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PRELIMS

POLITY & GOVERNANCE

STREET VENDORS (PROTECTION OF LIVELIHOOD AND REGULATION OF STREET VENDING) Act, 2014

Context: A decade has passed since the **Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act** came into effect on May 1, 2014.

Background:

- The Provisions of the act are aimed at creating **a conducive atmosphere where street vendors, are able to carry out their business in a fair and transparent manner**, without the fear of harassment and eviction.

Features of the Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act, 2014 :

- The Act provides for constitution of a **Town Vending Authority in each Local Authority**, for implementing the provisions of the Bill.
- In order to ensure participatory decision making for aspects relating to street vending activities like determination of natural market, identification of vending zones, preparation of street vending plan, survey of street vendors etc. the **TVC (Town Vending Committees) is required to have representation of officials and non-officials and street vendors, including women vendors with due representation from SC, ST, OBC, Minorities and persons with disabilities.**
- It mandates the States/ULB's conduct surveys to identify SV's atleat once in **5yrs.**
- The TVC are formed at municipal level to facilitate implementation of Act's provisions. TVC's are responsible for identifying vending zones, issuing vending certificates and addressing grievances of street vendors.
- Those street vendors who have been issued a certificate of vending/license etc. before the commencement of this Act, they will be deemed to be a street vendor for that category and for the period for which he/she has been issued such certificate of vending/license.
- It has been provided that no street vendor will be evicted until the survey has been completed and certificate of vending issued to the street vendors.
- There is a provision for establishment of an independent **dispute redressal mechanism** under the chairmanship of retired judicial officers to maintain **impartiality towards grievance redressal** of street vendors.

Source: The Hindu

RIGHT TO PEACEFUL DEATH

Context: A recent article contrasts end-of-life care in India and the West, emphasizing the need for a dignified passing for all.

Background:

- It calls for a rational system to ensure peaceful deaths and highlights the Right to Peaceful Death.

Key takeaways

- It refers to the **right of individuals to pass away with dignity and without unnecessary suffering**.

- Although it is not explicitly mentioned in the Indian Constitution, the Supreme Court of India has recognized the right to die with dignity as a fundamental aspect of the right to life under **Article 21.**
- This recognition came in the landmark case of **Common Cause (A Regd. Society) v. Union of India (2018),** where the court upheld the legality of passive euthanasia and the right to make advance directives for medical treatment.

Euthanasia in India

- The law in India distinguishes between active and passive euthanasia.
- **Active Euthanasia**: This involves intentionally administering lethal compounds to end a person's life. It remains illegal in India.
- **Passive Euthanasia**: This refers to withholding or withdrawing life support, allowing a terminally ill or vegetative patient to pass away naturally. Passive euthanasia has been legalized in specific circumstances.

Source: The Hindu

ARTICLE 31C

Context: During the ongoing hearing in the Supreme Court to decide whether the government can acquire and redistribute private property, a nine-judge Constitution Bench led by Chief Justice of India D Y Chandrachud asked a question of "radical constitutional consequence": Does Article 31C still exist?

Background:-

- Bench stated that the question of whether **Article 31C still lives following the Minerva Mills decision has to be decided to avoid "constitutional uncertainty".**

Key Takeaways

- **Article 31C was introduced by The Constitution (Twenty-fifth) Amendment Act, 1971.**
- **Article 31C** protects laws enacted to ensure the **"material resources of the community" are distributed to serve the common good (Article 39(b)) and that wealth and the means of production are not "concentrated" to the "common detriment" (Article 39(c)).**
- **Article 39** of the Constitution lists certain directive principles of state policy, which are meant to be guiding principles for the enactment of laws, but are not directly enforceable in any court of law.
- As per Article 31C, these particular directive principles (Articles 39(b) and 39(c)) cannot be challenged by invoking the right to equality (Article 14) or the rights under Article 19 (freedom of speech, right to assemble peacefully, etc).
- The 25th amendment was challenged in the seminal Kesavananda Bharati case (1973) in which 13 judges held by a narrow 7-6 majority that the Constitution has a "basic structure" that cannot be altered, even by a constitutional amendment.
- As a part of this verdict, the court struck down the last portion of Article 31C, i.e., the part that states "…and **no law containing a declaration that it is for giving effect to such policy shall be called in question in any court on the ground that it does not give effect to such policy".**
- In 1976, Parliament enacted The Constitution (Forty-second) Amendment Act, which expanded the **protection under Article 31C to "all or any of the principles laid down in Part IV of the Constitution", under clause 4. As a result, every single directive principle (Articles 36-51) was protected from challenges under Articles 14 and 19 of the Constitution.**

- The Statement of Objects of Reasons for the amendment stated that it was meant to give precedence to the directive principles "over those fundamental rights which have been allowed to be relied upon to frustrate socio-economic reforms for implementing the directive principles".
- **In 1980, in its judgment in Minerva Mills v. Union of India**, the SC struck down clauses 4 and 5 of the amendment. The five-judge Bench held that Parliament's power to amend the Constitution was limited, and it could not be used to remove these limitations and grant itself "unlimited" and "absolute" powers of amendment.
- However, this ruling birthed a conundrum that the apex court must now address. By striking down part of the 25th amendment, did the court strike down Article 31C as a whole, or did it restore the post-Kesavananda Bharati position wherein Articles 39(b) and (c) remained protected?

Source:Indian Express

CENTRAL DRUGS STANDARD CONTROL ORGANISATION (CDSO)

Context: Starting from May 15, 2024, the Central Drugs Standard Control Organisation (CDSCO) will be the sole authority responsible for granting no-objection certificates (NOCs) to pharmaceutical companies that manufacture drugs.

Background:

- These NOCs will specifically apply to drugs that are unapproved, new, or banned, but intended solely for export purposes. This move centralizes the approval process and removes the power previously held by state governments in this regard.

About CENTRAL DRUGS STANDARD CONTROL ORGANISATION (CDSCO) :

- It plays a crucial role in **regulating pharmaceuticals and ensuring their safety and quality in India.**
- The CDSCO is the **Central Drug Authority** responsible for discharging functions assigned to the Central Government under the Drugs and Cosmetics Act of 1940.
- It **operates under the Ministry of Health & Family Welfare and serves as the National Regulatory Authority (NRA)** for India.

CDSCO's primary responsibilities include:

- **Approval of Drugs:** It approves new drugs for use in India.
- **Clinical Trials:** It oversees and regulates clinical trials conducted within the country.
- **Setting Standards for Drugs:** CDSCO lays down quality standards for pharmaceuticals.
- **Quality Control of Imported Drugs**: It ensures the quality of imported drugs.
- **Coordination with State Drug Control Organizations**: CDSCO collaborates with state regulators.
- **Specialized Licenses**: Along with state regulators, CDSCO grants licenses for critical drugs such as vaccines and sera.
- **Medical Devices Oversight:** The Indian government plans to subject all medical devices (including implants and contraception) to CDSCO scrutiny.

Drugs Controller General of India (DCGI):

- The DCGI is the **head of the CDSCO department.**
- DCGI approves **licenses for specific categories of drugs,** including blood products, IV fluids, vaccines, and sera in India.

- Additionally, DCGI sets standards for drug manufacturing, sales, import, and distribution within the country.

Source: Livemint

POLLING STATION

Context: During the ongoing Lok Sabha elections, 96.8 crore electors are enrolled to vote at 10.5 lakh polling stations spread out nationwide.

Background:-

- The list of polling stations undergoes verification and updation during annual electoral roll revisions.

Key Takeaways

- **Under Section 25 of the Representation of the People Act, 1951 (RPA), the district election officer (DEO**) is assigned the responsibility to set up and publish the list of polling stations in their district.

Setting up polling stations: norms and principles

- These include ensuring that **voters do not have to travel more than two km to cast their vote, a station has a minimum area of 20 sq m, a station serves at most 1,500 electors, and a village with over 300 voters is provided with a polling station.**
- Polling stations **can even cater to less than 300 voters in case access is a problem, say in rugged terrain**. For instance, Arunachal Pradesh's Malogam polling station caters to only a single voter. Conversely, in cases where the number of electors exceeds 1,500, auxiliary polling stations are set up, preferably in the same building.
- **In urban areas, one building can have a maximum of four polling stations, whereas in rural areas, it can have at most two.**
- Locations for polling stations are chosen carefully, preferably in government or semi-government institutions.
- Private buildings are avoided unless absolutely necessary — they can be requisitioned with the written consent of the owner, or forcefully under Section 160 of the RPA.

Layout and some facilities

- There are separate entrance and exit points for voters, even if only one door is available. This is done with temporary fencing using bamboo and ropes.
- The voting compartment, constructed from opaque corrugated plastic sheet or steel grey flex board, measures at least 24 x 24 x 30 inches (length × width × height), and is positioned away from windows or doors.

People inside a polling station, voting day restrictions

- Only specific individuals are permitted access inside a polling station. These include registered electors, polling officers, candidates, one polling agent per candidate at a time, authorised media personnel, public servants on election duty, ECI-appointed observers, micro observers, videographers, photographers, and webcasting staff for critical/sensitive polling stations, a child in arms accompanying an elector, a person assisting a blind or infirm voter, and any others admitted by the Presiding Officer for voter identification or assistance purposes, whenever required.
- A polling party comprises a Presiding Officer and three Polling Officers. The first Polling Officer verifies the elector's identity, the second applies indelible ink on their left index finger, maintains the voters' register, and issues voter slips. The third Polling Officer takes back the voter slip issued

by the second polling officer, manages the EVM's control unit, and ensures that an elector has been properly inked before being allowed to vote.

- There is also a prohibition on any political propaganda, use of cameras, mobile phones, and canvassing within a 100 metre radius of the polling station, with a penalty of arrest without warrant under Section 130 of the RPA.
- Candidates can set up their booths beyond 200 m of the polling station, and there are also restriction on the use of loudspeakers.

Source:Indian Express

CITIZENSHIP AMENDMENT ACT

Context: Two months after the Union Ministry of Home Affairs (MHA) notified the rules for implementation of the Citizenship (Amendment) Act, which was enacted by Parliament in December 2019, Union Home Secretary Ajay Bhalla on Wednesday physically handed over the first set of citizenship certificates to 14 applicants.

Background:

- Despite its enactment four years ago, the CAA could not be implemented because the rules were not notified untill recently.Centre notified CAA rules ahead of Lok Sabha polls schedule announcement.

About CITIZENSHIP AMENDMENT ACT:

- Under the Citizenship Act, 1955, an illegal migrant is a foreigner who enters the country without valid travel documents like a passport and visa, or enters with valid documents, but stays beyond the permitted time period.
- The Citizenship Amendment Act, 2019 (CAA) was passed by the Parliament of India on 11 December 2019. This act amended the Citizenship Act, 1955.
- The CAA intends to make it easier for persecuted people from neighbouring countries **(covers migrants from Afghanistan, Bangladesh, and Pakistan.Specifically, it benefits those of Hindu, Sikh, Buddhist, Jain, Parsi, and Christian faiths)** to become Indian citizens. It's for **illegal migrants who entered India before December 31, 2014.**
- CAA does not grant such eligibility to Muslims from these countries
- The Amendment exempts the members of the six communities from any criminal case under the Foreigners Act, 1946 and the Passport Act, 1920. The two Acts (Foreigners Act, 1946 and the Passport Act, 1920) specify punishment for entering the country illegally and staying here on expired visas and permits.
- Under The Citizenship Act, 1955, one of the requirements for citizenship by naturalisation is that the applicant must have resided in India during the last 12 months, as well as for 11 of the previous 14 years.
- The amendment relaxes the second requirement from 11 years to 5 years as a specific condition for applicants belonging to these six religions (Hindu, Sikh, Buddhist, Jain, Parsi, and Christian) and the aforementioned three countries (Afghanistan, Bangladesh, and Pakistan).

Source:Indian Express

RAJYA SABHA NOMINATION

Context: Recently, Sudha Murthy was nominated as a Member of Rajya Sabha.

Background:

- Sudha Murthy, who is married to the co-founder of Infosys, N. R. Narayana Murty, has made significant contributions to diverse fields including social work, philanthropy, and education. Last year, she was honoured with the **Padma Bhushan**, India's third-highest civilian award, in recognition of her outstanding social work.

Composition of the Rajya Sabha:

- It consists of a total of 245 members, including 233 elected members and 12 nominated members.
- The constitutional limit for the Upper House's strength is 250 members. **The number of elected seats varies based on** states' population, which can change due to mergers, bifurcations, or the creation of new states.
- The nominated members are chosen by the President of India and come from fields such as art, literature, science, and social service.

Tenure and Permanent House:

- The Rajya Sabha is a permanent body and not subject to dissolution. However, like the Lok Sabha, it can be prorogued by the President.
- **Every RS MP has a tenure of six years,** and elections to one-third seats are held every two years.
- The Vice-President is the ex-officio chairperson of the RS.
- The Deputy Chairman, who is elected from amongst the house's members, takes care of the day-to-day matters of the house in the absence of the Chairman.

Election Process:

- It's members are **elected indirectly by the people through the MLAs.**
- MLA's vote in the **RS elections in proportional representation with the single transferable vote (STV) system. Each MLA's vote is counted only once.**
- To win a RS seat, a candidate should get a required number of votes.
- That number is found out using the formula:

Required vote = Total number of votes / (Number of Rajya Sabha seats + 1) + 1

Nominated Members:

- The **President of India nominates these 12 members** from individuals who have special knowledge or practical experience in art, literature, science, and social service.
- This principle of nomination is adopted to provide distinguished persons a place in the RS without going through the process of election.

Source: Livemint

CONSUMER PROTECTION ACT 2019

Context: The Supreme Court ruled on Tuesday that advocates cannot be held liable under the Consumer Protection Act of 1986 for deficiency in service. The court ruled that a complaint to this effect against an advocate was not maintainable before a consumer forum.

Background:

- The Supreme Court overturned a 2007 ruling by the National Consumer Disputes Redressal Commission (NCDRC) which said services provided by lawyers fell under the Consumer Protection Act.

About CONSUMER PROTECTION ACT 2019:

- The Act replaced **the previous Consumer Protection Act, 1986.**
- It was introduced in response to the evolving needs of consumers in the digital age, and aims to provide timely and effective administration and settlement of consumer disputes.

Salient Features

- As compared to CPA 1986, the definition of "Consumer" has been broadened under CPA 2019 .E-commerce has been **specifically mentioned and is defined as buying and selling of goods/services/digital products online.**
- Unfair contracts: to protect the interests of these consumers, **"unfair contracts" has been added as a ground for filing complaints in the present Act.**
- **Place of filing complaints:** CPA 1986 required consumers to file complaints (in Forums) only at places where the opposite party had its office. However, CPA 2019 allows complainant to file a complaint in the area where he/she resides/ works.
- **Renaming of the Forums**: Earlier, the consumer courts at district level were termed as "forums" and at the state and national level as "commissions". Giving two different names created confusion.Thus, to eliminate the ambiguity around the terms, the district forums have been renamed as district commissions under CPA 2019.
- **Central Consumer Protection Authority (CCPA):** CPA 2019 establishes a Central Consumer Protection Authority (to be termed as Central Authority) that will promote, protect and enforce the rights of consumers as a class.
- **Misleading Advertisements**: A provision, authorising the CCPA to issue directions and levy penalties against deceptive advertisements under CPA 2019 (Section 21), is an important one.CCPA can impose a penalty of up to Rs. 10 lakhs, which may extend up to Rs. 50 lakhs in case of subsequent violations.
- **Pecuniary Jurisdiction**: According to CPA 1986, consumers seeking compensation of Rs. 20 lakhs and less were to approach the district forums, from Rs. 20 lakhs to Rs. 1 crore the state commissions, and above Rs. 1 crore, the national commission. However, under the present Act, in order to facilitate easy access for consumers from districts and to ensure quick disposal of cases, the pecuniary jurisdiction has been increased to upto Rs. 1 crore for district commissions, from Rs. 1 crore to Rs. 10 crores for state commissions and above Rs. 10 crores for national commission.
- **Product Liability**: refers to the responsibility of a product manufacturer/seller of any product or service to compensate for any harm caused to the consumer by such defective product manufactured/sold or due to deficiency in services.

Source: Citizen Consumer And Civic Action Group

RULES ON CONTESTING SEATS

Context: Congress leader Rahul Gandhi is contesting in 2 seats. One in UP and other in Kerala. According to the Representation of the People Act (RPA), 1951, a candidate is permitted to contest an election from up to two constituencies, but he or she can hold only one seat at a time if elected from both.

Background:

- Opposition to a candidate contesting from many seats was basically due to the fact that many byelections had to be conducted after polls were over.

Rules on contesting elections:

- **A sub-section, 33 (7),** of the RPA, allowing a candidate to co**ntest from two seats, was introduced through an amendment in 1996,** prior to which there was no bar on the number of constituencies from which a candidate could contest.
- However**, Section 70 of the same Act stipulates that a candidate can hold only one seat at a time, regardless of whether he or she has been elected from more than one seat**. Thus, if a candidate wins from two seats, a byelection is necessary from the seat he or she vacates.

Additional Information

- The **minimum age for a person to contest Lok Sabha and Assembly polls is 25 years,** while one can become a member of the Rajya Sabha or the State Legislative Council only at 30 years.
- There **is no minimum educational qualification needed to contest general elections in India.** Candidates must be citizens of India, registered in some constituency of the country as a valid voter.

How can a candidate be disqualified?

- A person shall be disqualified from being chosen as or being a member of either House if he holds any office of profit under the Government of India or the Government of any State; if he or she is of unsound mind and stands so declared by a competent court; if he or she is an undischarged insolvent; if he or she is not a citizen of India or has voluntarily acquired the citizenship of a foreign state.
- Under the RPA Act, **if a person is convicted of any offence and sentenced to an imprisonment of two years or more, this will lead to his disqualification to contest elections.**
- Even if this person is out on bail, after the conviction and his appeal is pending for disposal, he or she is disqualified from contesting an election.
- For some **categories of serious offences, one may incur disqualification for any conviction, regardless of the quantum of punishment.**

Source:Hindu

COMPETITION COMMISSION OF INDIA

Context:15th Foundation Day of the Competition Commission of India was celebrated in New Delhi.The commission was formed on 14 October 2003. However, It became fully functional in 2009 with Dhanendra Kumar as its first chairman

Background:

- CCI got its enforcement and regulatory powers after the substantive provisions of the Competition Act relating to anti-trust enforcement and regulation of combinations came into force on May 20, 2009 and June 1, 2011, respectively.

About Competition Commission of India (CCI):

- **The chief national competition regulator in India. As a statutory body within the Ministry of Corporate Affairs,** its primary responsibility is to enforce the Competition Act, 2002.
- It should be noted that on the recommendations of Raghavan committee, the **Monopolies and Restrictive Trade Practices Act, 1969 (MRTP Act)** was repealed and replaced by the Competition Act, 2002.
- The Commission consists of a Chairperson and **not more than 6 Members appointed by the Central Government.**

- **It is a quasi-judicial body.**

Objective and Vision:

- Aim: promote and sustain an **enabling competition culture through engagement and enforcement.**
- Vision: inspiring businesses **to be fair, competitive, and innovative, enhancing consumer welfare, and supporting economic growth**.

Functions and Activities:

- **Antitrust Regulation**: The CCI ensures fair competition by preventing anti-competitive practices such as cartels, abuse of dominant position, and anti-competitive agreements.
- **Regulation of Combinations**: It reviews mergers, acquisitions, and other combinations to assess their impact on competition.
- **Competition Economics**: The CCI analyzes market dynamics and economic factors related to competition.
- **Advocacy:** It promotes competition awareness and educates stakeholders.
- **International Cooperation:** The CCI collaborates with global competition authorities.
- **Anti-Profiteering:** It investigates cases of undue profiteering by businesses.

Source: Indian Express

CONTEMPT OF COURTS ACT 1971

Context: The Allahabad High Court recently observed that orders issued by the **Central Administrative Tribunal (CAT) under the Contempt of Courts Act, 1971,** are subject to appeal solely to the Supreme Court and not the High Court.

Background:

- While the High Courts retain jurisdiction under **Article 226/227 for orders issued under Section 14(1) of the Administrative Tribunals Act,** orders falling under **Section 17 of the Act** are appealable solely to the Supreme Court.

About CONTEMPT OF COURTS ACT I971

- It's an **important statute in India that defines and limits the powers of courts in punishing contempts of courts.**
- It also **regulates their procedure in relation to contempt proceedings**.
- The act defines contempt of court as any act that **scandalizes or tends to scandalize, prejudices, or interferes with the administration of justice.**
- The Act **distinguishes between civil and criminal contempt.**
- **Civil contempt**: refers to willful disobedience to any judgment, decree, direction, order, writ, or other process of a court, or willful breach of an undertaking given to a court.
- **Criminal contempt:** includes any act that scandalizes or tends to scandalize, or lowers the authority of the court, prejudices, or interferes with judicial proceedings, or obstructs the administration of justice.
- The Act provides **protection for the innocent publication and distribution of matter not considered contempt, ensuring freedom of expression while respecting the course of justice.**

- It allows for **fair and accurate reporting of judicial proceedings and fair criticism of judicial acts** without constituting contempt of court.
- High Courts have the **power to punish contempts of subordinate courts and to try offences committed outside their jurisdiction.**
- The Act outlines the **punishment for contempt of court**, which can include imprisonment or fine.

Source: Livelaw

THE ELECTION COMMISSION OF INDIA'S OPTIONS WHEN POLLING PROCESS IS DISRUPTED

Context: The Election Commission of India (EC), under various Sections of the Representation of People Act, 1951 (RPA), declared void the poll on April 19, in 11 polling stations of Manipur, and 8 polling stations of Arunachal Pradesh. Repolls were conducted on April 22 and April 24 respectively. Elections were also adjourned in Madhya Pradesh's Betul Lok Sabha constituency due to the death of a candidate on April 9. The polling, originally scheduled on April 26, will now be held on May 7.

Background:

- India's election laws **provide a framework for handling situations where the normal polling process is disrupted for any reason, including damage to EVMs, booth-capturing, natural disasters, or a candidate's death.** The provisions for repolls, adjournments, and the voiding of polls ensure that the democratic process remains fair, transparent, and uninterrupted.

Circumstances when the normal polling process is disrupted — and the options before the EC in each case:

1. Intentional destruction, taking away of EVMs:

- **Under Section 58 of the RPA ('Fresh poll in the case of destruction, etc., of ballot boxes'),** the EC can declare the poll at a polling station to be void if:
  - an unauthorised person has unlawfully taken away any EVM;
  - any EVM has been accidentally or intentionally destroyed, or lost, or damaged, or tampered with; or
  - a mechanical failure develops in any EVM during the recording of votes.
- In such cases, the **Returning Officer (RO) immediately informs the EC and the Chief Electoral Officer of the state about the relevant facts and material circumstances,** after considering which, the EC can declare the poll void and formally fix the date and time for a new poll.
- All electors will be allowed to vote at the fresh poll. During the repoll, the voters' left middle fingers are inked to distinguish between the mark made during the original poll (on their left forefinger).

2. Booth capturing

- Booth-capturing, defined in **Section135A of the RPA,** includes all or any of the following activities by any person or persons:
  - seizure of a polling station, **affecting the conduct of elections;**
  - taking possession of a polling station, **allowing only his or their supporters to vote;**
  - intimidating or threatening any elector and preventing him from going to the polling station;
- Booth capturing is **punishable for a term of not less than one year,** which may extend to three years for lay people, and not less than three years, **extending to five years for government servants.**
- Under **Section 58A ('Adjournment of poll or countermanding of election on the ground of booth capturing')** the Presiding Officer of a polling station immediately closes the Control Unit of EVM

and detaches the Ballot Unit(s) from the Control Unit under Rule 49X of the Conduct of Election Rules, 1961.

- **He/She then informs the RO, who reports the full facts to the EC through the fastest means of communication. The EC, based on the material facts, may:** (a) declare the poll at that polling station to be void and direct a fresh poll on a new date; or (b) countermand the election in the constituency in case booth capturing has taken place in a large number of polling stations, or if it has affected the counting of votes.

3. Natural disasters, other disruptions to polling

- The Presiding Officer of a polling station can adjourn the poll at a polling station under section 57(1) of the Representation of the People Act, 1951, in case of:
  - a natural calamity like a flood, a severe storm;
  - non-receipt or loss or damage to essential polling materials like EVM, electoral roll etc;
  - interruption or obstruction due to any riot or, open violence;
  - non-arrival of the polling party due to obstruction or any other serious difficulty; or
  - non-commencement of the poll within two hours from the scheduled time due to malfunctioning of EVM, or any other reason.
- After seeking the EC's approval on the date and hours, the adjourned poll will recommence from the stage at which it was left immediately before the adjournment. Contesting candidates or their agents are informed, and only electors who have not already voted before the poll was adjourned are permitted to vote.

4. Death of a candidate

- As per Section 52 of RPA,the poll shall be adjourned only in case of **the death of a recognised political party's candidate.**
- The above provision applies if the candidate with a valid nomination dies at any time after 11.00 a.m. on the last date for making nominations, until the commencement of the poll. The RO reports the fact to the EC and orders the adjournment of the poll to a date to be notified later by the Commission.
- The EC then calls upon the concerned political party to nominate another candidate for the said election in place of the deceased candidate. The political party must make the nomination within seven days.
- In Betul, the candidate's death occurred one day after the last day for withdrawing candidature. Therefore, the polls were adjourned. However, in the Moradabad Lok Sabha constituency, the candidate died after voting, in which case a by-election will be held if he emerges as the winner of the seat after counting.

Source: Indian Express



INTERNATIONAL RELATIONS



GROUP OF SEVEN (G7)

Context: Energy ministers from the Group of Seven (G7) reached a deal to shut down their coal-fired power plants by 2035.

Background:

- The agreement on coal marks a significant step in the direction indicated last year by the COP28 United Nations climate summit to phase out fossil fuels, of which coal is the most polluting.

About G7 :

- It's an **intergovernmental political and economic forum** that brings together seven of the world's advanced economies.
- **These countries include** Canada, France, Germany, Italy, Japan, the United Kingdom, and the United States.
- Additionally, the **European Union (EU) is considered a "non-enumerated member" of the G7.**

Key points :

- **Origins**: The G7 originated from an ad hoc gathering of finance ministers in 1973. Over time, it has evolved into a formal and high-profile venue for discussing and coordinating solutions to major global issues. These issues span areas such as trade, security, economics, and climate change.
- Annual Summit: Each member's head of government or state, along with the EU's Commission President and European Council President, meet annually at the G7 Summit. Throughout the year, other high-ranking officials from the G7 and the EU engage in discussions and initiatives.
- Russia's Membership: Russia was a formal member of the G7 (as part of the G8) from 1997 until its expulsion in 2014 due to the annexation of Crimea.
- Rotating Presidency: The G7 operates without a legal or institutional basis. It is organized through a rotating presidency among the member states. The presiding state sets the group's priorities and hosts the annual summit. For 2024, Italy holds the presidency.

Source: Times of India

INDIA NEPAL BORDER ISSUES

Context: Nepal's cabinet last week decided to put a map on its Rs 100 currency note showing certain areas administered by India in Uttarakhand as part of its territory, provoking External Affairs Minister S Jaishankar to say that such "unilateral measures" by Kathmandu would not change the reality on the ground.

Background:-

- The map was adopted by consensus in Nepal's Parliament four years ago.Unlike then, when the new map was adopted, the May 2 decision to put it on the currency note has met with scepticism and criticism in Nepal.

About the issue

- The territorial dispute is about a **372-sq-km area that includes Limpiadhura, Lipulekh, and Kalapani at the India-Nepal-China trijunction in Uttarakhand's Pithoragarh district.** Nepal has claimed for long that these areas belong to it both historically and evidently.

- **The Treaty of Sugauli at the end of the Anglo-Nepalese War of 1814-16** resulted in Nepal losing a chunk of territory to the East India Company. Article 5 of the treaty took away the jurisdiction of Nepal's rulers over the land to the east of the Kali River.
- Modi and his Nepali counterpart, Sushil Prasad Koirala, agreed to set up a boundary working group for speedy settlement of the border issue in Kalapani and Susta, a 145-sq-km area that had fallen on the Indian side after the River Gandak changed course.

Frictions in bilateral ties



- The goodwill of the 2005-14 period when India mediated the transition of the Hindu Kingdom into a secular federal republic, evaporated in 2015 after the Maoists rejected outright the suggestion of New Delhi, conveyed through then Foreign Secretary Jaishankar, that Nepal's new constitution should be delayed until the concerns of the Terai parties were addressed.
- The 134-day blockade of Nepal that began in September 2015 created significant distrust against India, and K P Sharma Oli, who took over as Prime Minister that October, quickly moved to sign a trade and transit agreement with China in order to create a fallback source for the supply of essential goods.
- In February 2018, Oli, chairman of the Communist Party of Nepal (Unified Marxist Leninist), returned as Prime Minister with a massive mandate in the first election held under the new constitution.
- In 2020, he took the lead to build the consensus in Parliament for Nepal's new map that formally included the 372 sq km in Uttarakhand, and pledged to bring it back.
- India decried Nepal's "cartographic aggression" as unacceptable, but said the issue would have to be sorted out through diplomatic channels, on the basis of evidence. It is noteworthy that the cabinet decision on printing the new Rs 100 notes has come less than two months after Oli's party joined the ruling coalition in Nepal.
- Nepal sorted out its boundary issues with China in the early 1960s through bilateral meetings of the boundary commission. Nepal's former Prime Minister Kirtinidhi Bista had once quoted his Chinese counterpart Chou En-lai as saying that border issues, if left unresolved, turn into problems for the future generations that are difficult to address.

Source:Indian Express

INTERNATIONAL SOLAR ALLIANCE (ISA)

Context: Spain has become the 99th member of the International Solar Alliance.

Background:

- **ISA is a collaborative platform for increased deployment of solar energy technologies as a means for bringing** energy access, ensuring energy security, and driving energy transition in its member countries.

About INTERNATIONAL SOLAR ALLIANCE (ISA):

- The ISA was conceived as a joint effort by **India and France to mobilize efforts against climate change through deployment of solar energy solutions.**
- It was conceptualized on the side-lines of the **21st Conference of Parties (COP21) to the United Nations Framework Convention on Climate Change (UNFCCC) held in Paris in 2015.**
- **Vision**: Let us together make the sun brighter.
- **Mission**: Every home no matter how far away, will have a light at home
- The Headquarters is in India with its **Interim Secretariat being set up in Gurugram.**
- With the amendment of its **Framework Agreement in 2020**, all member states of the United Nations are now eligible to join the ISA.
- This would help mitigate **global solar emissions** to the tune of 1,000 million tonnes of CO2 every year.
- ISA is **headed by the Director General**, who leads the operations and carries out the functions of the ISA Secretariat and is responsible to the ISA Assembly.
- **One Sun One World One Grid (OSOWOG):** The OSOWOG **focuses on a framework for facilitating global cooperation**, building a global ecosystem of interconnected renewable energy resources (mainly solar energy) that can be seamlessly shared.

Source: newsonair

ARAB LEAGUE

Context: Arab League **welcomed a decision made by Norway, Ireland and Spain to recognize the Palestinian state**

Background:

- Arab League, regional organization of Arab states in the Middle East and parts of Africa, formed in Cairo on March 22, 1945, as an outgrowth of Pan-Arabism.

About ARAB LEAGUE:

- **It's formally the League of Arab States**, is a regional organization in the Arab world with 22 members. **Each member has one vote on the League Council,** decisions being binding only on those states that have voted for them.
- **It was formed in Cairo on 22 March 1945, initially with six members:** Egypt, Iraq, Transjordan (now Jordan), Lebanon, Saudi Arabia, and Syria.Yemen joined as a member on 5 May 1945.
- **Goal:** "draw closer the relations between member states and co-ordinate collaboration between them, to safeguard their independence and sovereignty, and to consider in a general way the affairs and interests of the Arab countries".
- After Egypt signed a peace treaty with Israel on March 26, 1979, the other members of the Arab League voted to suspend Egypt's membership and to transfer the league's headquarters from Cairo to Tunis. Egypt was reinstated as a member of the Arab League in 1989, and the league's headquarters returned to Cairo in 1990.

Source: The Hindu

INTERNATIONAL COURT OF JUSTICE (ICJ)

Context: International Court of Justice **ordered Israel to halt its offensive in the southern Gaza city of Rafah and withdraw from the enclave,** in a case brought by South Africa accusing Israel of genocide, citing "immense risk" to the Palestinian population.

Background:

- Friday's decision marked the third time this year the 15-judge panel has issued preliminary orders seeking to rein in the death toll and alleviate humanitarian suffering in Gaza. While orders are legally binding, the court has no police to enforce them.

About International Court of Justice (ICJ)

- The ICJ is the principal judicial organ of the United Nations (UN), established in June 1945 by the Charter of the United Nations and began work in April 1946. It is also known as the World Court.
- The **court is the successor to the Permanent Court of International Justice (PCIJ)**, which was brought into being through, and by, the League of Nations, 1922.
- Like the PCIJ, the ICJ is **based at the Peace Palace in The Hague**.
- After World War II, **the League of Nations and PCIJ were replaced by the United Nations and ICJ respectively.**

Seat and role

- **Role:** "to settle, in accordance with international law, legal disputes submitted to it by States and to give advisory opinions on legal questions referred to it by authorized United Nations organs and specialized agencies".
- English and French are the ICJ's official languages.
- All members of the UN are automatically parties to the ICJ statute, but this does not automatically give the ICJ jurisdiction over disputes involving them. The ICJ gets jurisdiction only if both parties consent to it.
- The judgment of the ICJ is final and technically binding on the parties to a case. There is no provision of appeal; it can at the most, be subject to interpretation or, upon the discovery of a new fact, revision.
- However, the ICJ **has no way to ensure compliance of its orders, and its authority is derived from the willingness of countries to abide by them.**

Judges of the court

- The ICJ has **15 judges who are elected to nine-year terms by the UN General Assembly and Security Council, which vote simultaneously but separately.**
- To be elected, a **candidate must receive a majority of the votes in both bodies, a requirement that sometimes necessitates multiple rounds of voting.** Elections are held at the UNHQ in New York during the annual UNGA meeting.
- **The president and vice-president of the court are elected for three-year terms by secret ballot.** Judges are eligible for re-election.
- Four Indians have been members of the ICJ so far.

India at the ICJ

- India has been a party to a case at the ICJ on six occasions, four of which have involved Pakistan. They are:

1. Right of Passage over Indian Territory (Portugal v. India, culminated 1960);
2. Appeal Relating to the Jurisdiction of the ICAO Council (India v. Pakistan, culminated 1972);
3. Trial of Pakistani Prisoners of War (Pakistan v. India, culminated 1973);
4. Aerial Incident of 10 August 1999 (Pakistan v. India, culminated 2000);
5. Obligations concerning Negotiations relating to Cessation of the Nuclear Arms Race and to Nuclear Disarmament (Marshall Islands v. India, culminated 2016);
6. (Kulbhushan) Jadhav (India v. Pakistan, culminated 2019).

Source:Al Jazeera

DIPLOMATIC PASSPORT

Context: Nearly a month after a Special Investigation Team (SIT) began probing allegations of sexual assault and abuse against Hassan MP Prajwal Revanna, sources in the Ministry of External Affairs (MEA) have confirmed that the Karnataka government's request to revoke the diplomatic passport of the suspended Janata Dal (Secular) leader is "being processed."

Background:

- Members of Parliament , when using diplomatic passports, are required to **apply for prior political clearance directly to the MEA. This is also applicable for private visits**. For a private visit to a foreign country requiring a visa, the MEA issues visa notes after the member submits a specific request with the necessary political clearance.
- **Diplomatic passport holders do not need a visa note from the MEA when travelling to any of the 34 countries with which India has mutual visa waiver agreements for diplomatic passports.** Under this exemption, the permitted period of stay ranges between 30 and 90 days.
- Germany, where Mr. Revanna allegedly fled to in April, is one of the countries that has an operational 'Visa Exemption Agreement 'for diplomatic passport holders. A person travelling to Germany on a diplomatic passport can stay in the country without a visa for up to 90 days.
- Notably, the MEA has clarified that Mr. Revanna did not seek the necessary political clearances before his travel, nor did the Ministry issue any visa note for his travel to Germany.

What is a diplomatic passport?

- Under the **Passport Act of 1967, the Central government issues three classes of passports — ordinary, official and diplomatic** — and travel documents such as emergency certificate and certificate of identity, to those who wish to leave the Indian mainland by land, water or air.
- **The deep blue-coloured 'ordinary passports'** are issued for personal travel for adults with a validity of 10 years for adults and five years for minors or till they attain the age of 18.
- **Official passports with white jackets** are issued to designated Government officials and other individuals specifically authorised by the Centre to work abroad on official assignments.
- **A diplomatic passport, or Type 'D' passport,** is for designated members authorised by the Centre to represent the country. These include senior government officials, Members of Parliament and their spouses. Such a passport has a **maroon jacket** and is valid for five years or less, depending on the position of the holder and the nature of their assignment and visit.

- While ordinary passports are issued through a network of passport offices across the country, the Consular, Passport & Visa (CPV) Division of the MEA exclusively handles matters related to the issuance of diplomatic and official passports.

Are MPs entitled to diplomatic passports?

- According to the MEA, diplomatic passports are issued to individuals with diplomatic status, those on diplomatic assignments abroad, and those in "certain positions" as determined by the Central Government.
- Diplomatic passport holders travelling abroad for official purposes typically do not pay visa fees. They have immunity from arrest under international law and expedited immigration.
- Besides officials working at Indian missions, Members of Parliaments and their spouses are entitled to diplomatic passports along with ordinary passports.

What are the conditions under which a diplomatic passport can be revoked?

- The passport authority can impound or revoke a passport if the holder has provided wrong information, has been convicted in a criminal case, or there are pending criminal proceedings in an Indian court, or if it is in the overall national interest.
- **Section 10(3) of the Passport Act of 1967**, which deals with 'variation, impounding and revocation' of passports and travel documents mentions the circumstances in detail
    - As per the Act, a passport can be impounded or revoked if the holder is in wrongful possession or has provided wrong information to the authority; or if the authority believes that impounding or revoking the passport will be in the interests of the country's sovereignty, integrity, security, maintaining friendly relations with a foreign country, in the interests of the general public; or any of the conditions of the passport have been unfollowed.
- **The passport of a person convicted by an Indian court of an offence involving** "moral turpitude" and sentenced to at least two years in prison after the passport was issued can also be confiscated.
- **Further, a passport can also be impounded or revoked if there are** pending proceedings against the holder in a criminal court, or if a court has **issued a warrant or summons for the holder's appearance or arrest**, or if there is a court order prohibiting their departure from India, as in the case of Prajwal Revanna.
- Last week, a **Special Court for Elected Representatives issued an arrest warrant against the absconding MP in the sexual assault case against him on a plea by the SIT investigating the case.**

Source:Hindu

COALITION FOR DISASTER RESILIENT INFRASTRUCTURE (CDRI)

Context: At the UN 4th International Conference on SIDS in Antigua and Barbuda, the Coalition for Disaster Resilient Infrastructure (CDRI)published a Call for Proposals for funding to improve infrastructure resilience in **Small Island Developing States (SIDS).**

Background:-

- The financing appeal, totaling $8 million, was revealed during the **SIDS4Conference in Antigua and Barbuda as a component of CDRI's Infrastructure for Resilient Island States Programme (IRIS).**

About Coalition for Disaster Resilient Infrastructure (CDRI):

- It's a **multi-stakeholder globalpartnership of national governments, UN agencies and programmes, multilateral development banks and financing mechanisms, the private sector, and**

knowledge institutions that aims to promote the resilience of new and existing infrastructure systems to climate and disaster risks in support of sustainable development.

- It was launched by the Indian Prime Minister Narendra Modi at the 2019 UN Climate Action Summit in September 2019.
- **It's not an intergovernmental organization**, which are ordinarily treaty-based organizations. As of 2023, the CDRI has 39 members, including 31 national governments.
- The CDRI Secretariat is based in New Delhi, India.
- **National governments that endorse the CDRI Charter and become a members have akey role in setting its substantive agenda as well as in its governance.**
- It may be noted that the policies, standards and other outputs of CDRI would not be binding on its members.
- **The following are CDRI's strategic priorities:** Technical Support and Capacity-building, Research and Knowledge Management, Advocacy and Partnerships
- **Promote:** research and knowledge sharing in the fields of infrastructure risk management, standards, financing, and recovery mechanisms.
- CDRI's initial focus is on developing disaster-resilience in ecological(natural waterways, waste management, etc.), social(schools, hospitals, etc.), and economic infrastructure(energy, telecommunication, roads, railways, airports, etc.).
- It aims to achieve substantial changes in member countries' policy frameworks and future infrastructure investments, along with a major decrease in the economic losses suffered due to disasters.

Source:CDRI

INTERNATIONAL CRIMINAL COURT AND RECENT DEVELOPMENTS

Context: In a noteworthy international development, the International Criminal Court (ICC)'s prosecutor, Karim Khan, has requested the court to issue arrest warrants against leaders of Israel — Prime Minister Benjamin Netanyahu and Defence Minister Yoav Gallant — and leaders of Hamas.

Background:

- The arrest warrants are sought on the charges of war crimes and crimes against humanity over the October 7 attack on Israel and the subsequent war in Gaza.

About INTERNATIONAL CRIMINAL COURT:

- It is an **intergovernmental organization and international tribunal that sits in The Hague, Netherlands.**
- **It has jurisdiction to prosecute individuals for the international crimes of:** genocide, crimes against humanity, war crimes, crime of aggression.
- It is intended to complement existing national judicial systems.
- It may exercise its jurisdiction only **when national courts are unwilling or unable to prosecute criminals.**
- It may only investigate and prosecute crimes committed within member states, crimes committed by nationals of member states, or crimes in situations referred to the Court by the United Nations Security Council. The ICC lacks universal territorial jurisdiction.

About the development

- The ICC, in recent times, has come under scathing criticism for targeting leaders of the third world, and not Western powers and their allies.
- The application to issue arrest warrants against the leaders of Israel, thus, is a first and quite significant one. It can be argued that by charging both the Hamas leadership and the Israeli leaders, the prosecutor is seemingly giving an impression of being balanced and even-handed.
- Under international law, heads of state are immune from criminal proceedings at any foreign or international court. At the ICC, however, no head of state can claim immunity.
- The basis of ICC's jurisdiction in this case is that Palestine is a state party to the court. Thus, the ICC can exercise what is known as territorial jurisdiction. This means that if a crime is committed on the territory of a state party to the ICC, the Court can exercise jurisdiction on that crime, even if it is committed by people belonging to a state that isn't its member. Hence, the crimes committed by Israeli soldiers in Gaza fall within the Court's jurisdiction.

Source: Indian Express



Economy



GOODS AND SERVICE TAX

Context: The Goods and Services Tax (GST) collections as a percentage of Gross Domestic Product (GDP) have reached pre-GST levels, Finance Minister Nirmala Sitharaman said.The monthly gross GST collections have been picking pace, recently crossing a landmark of Rs 2 lakh crore, as per the monthly data for April released on May 1 by the Finance Ministry.

Background:

- In India, GST is a successor to VAT used on the supply of goods and service. It is a comprehensive, multistage, destination-based tax. It has subsumed almost all the indirect taxes except a few state taxes. GST replaced many indirect taxes in India such as the excise duty, VAT, services tax, etc. The Goods and Service Tax Act was passed in the Parliament on 29th March 2017 and came into effect on 1st July 2017.

About Goods and Service Tax (GST):

- GST is one **indirect tax for the whole nation**, **is a single tax on the supply of goods and services,** right from the manufacturer to the consumer.
- In the earlier indirect tax regime, the **Centre could tax goods up to the production or manufacturing stage, while States collected taxes on the sale or distribution of goods.** The right to tax services was vested with the Centre alone.
- Under the GST, **both the Centre and the States can tax the entire supply chain** in both goods as well as services – right from production to distribution.
- The final consumer will thus bear only the **GST charged by the last dealer in the supply chain,** with set-off benefits at all the previous stages.

Benefits of GST

For business and industry

- **Easy compliance**: A robust and comprehensive IT system would be the foundation of the GST regime in India. Therefore, all tax payer services such as registrations, returns, payments, etc. would be available to the taxpayers online, which would make compliance easy and transparent.
- **Uniformity of tax rates and structures:** GST will ensure that indirect tax rates and structures are common across the country, thereby increasing certainty and ease of doing business. In other words, GST would make doing business in the country tax neutral, irrespective of the choice of place of doing business.
- **Removal of cascading**: A system of seamless tax-credits throughout the value-chain, and across boundaries of States, would ensure that there is minimal cascading of This would reduce hidden costs of doing business.

For Central and State Governments

- **Simple and easy to administer**: Multiple indirect taxes at the Central and State levels are being replaced by GST. Backed with a robust end-to-end IT system, GST would be simpler and easier to administer than all other indirect taxes of the Centre and State levied so far.
- **Better controls on leakage**: GST will result in better tax compliance due to a robust IT infrastructure. Due to the seamless transfer of input tax credit from one stage to another in the

chain of value addition, there is an in-built mechanism in the design of GST that would incentivize tax compliance by traders.

- **Higher revenue efficiency**: GST is expected to decrease the cost of collection of tax revenues of the Government, and will therefore, lead to higher revenue efficiency.

For the consumer:

- **Single and transparent tax proportionate to the value of goods and services:** Due to multiple indirect taxes being levied by the Centre and State, with incomplete or no input tax credits available at progressive stages of value addition, the cost of most goods and services in the country today are laden with many hidden taxes. Under GST, there would be only one tax from the manufacturer to the consumer, leading to transparency of taxes paid to the final consumer.
- **Relief in overall tax burden:** Because of efficiency gains and prevention of leakages, the overall tax burden on most commodities will come down, which will benefit consumers.

Source: Indian Express

DRIP PRICING

Context: The Centre recently warned about "drip pricing", saying it can surprise consumers with "hidden charges".

Background:

- **The Department of Consumer Affairs asked consumers to reach out to NCH 1915** for assistance or via WhatsApp at 8800001915 if they need help with 'drip pricing'.

About DRIP PRICING

- It's a pricing technique used by firms where they i**nitially advertise only part of a product's price (referred to as the "headline price").**
- As the customer proceeds through the buying process, additional charges are gradually revealed.
- This technique can lead to "**hidden charges**" that surprise consumer.

Key Points:

- **Initial Advertisement:** Firms may advertise a base price that does not include mandatory fees such as booking, service, or credit card fees.
- **Incremental Disclosure:** Additional costs, which may be unavoidable, are disclosed one by one or "dripped" to the buyer at the point of purchase.
- **Common Usage:** This strategy is prevalent in the hospitality and travel industries, as well as for other online payments.
- **Consumer Impact:** It can make comparison shopping difficult and may cause frustration for consumers who expect to know the full cost upfront.
- **Example: An airline ticket advertised without including baggage fees is a form of drip pricing**

Source: Hindustan Times

INDIA VOLATALITY INDEX (IVI)

Context: India VIX, which is an indicator of the market's expectation of volatility over the near term, surged past the 21 mark on Tuesday (May 14).

Background:-

- The rise shows that fear among traders or market participants on the expected volatility is more now, as compared to 15 days earlier. Currently, the fear among the market players is coming from the outcome of the ongoing Lok Sabha elections.

About INDIA VOLATILITY INDEX (VIX)

- It's the '**rate and magnitude of changes in prices' and in finance often referred to as risk**. Usually, during periods of market volatility, the market moves steeply up or down and the volatility index tends to rise. **As volatility subsides, the Volatility Index declines.**
- Also known as **Fear Index, is a measure of the market's expectation of volatility over the near term.**
- **Measure:** of the amount by which an **underlying index is expected to fluctuate in the near term,** (calculated as annualised volatility, denoted in percentage e.g. 20 per cent) based on the order book of the underlying index options.
- India VIX is a volatility index computed by the NSE based on the order book of NIFTY Options. For this, the best bid-ask quotes of near and next-month NIFTY options contracts, which are traded on the F&O segment of NSE are used.
- **India VIX indicates the investor's perception of the market's volatility** in the near term i.e. it depicts the expected market volatility over the next 30 calendar days.
- The higher the India VIX values, the higher the expected volatility and vice versa, as per NSE.

Additional Information

- **The Chicago Board of Options Exchange (CBOE)** was the first to introduce the volatility index for the US markets in 1993 based on S&P 100 Index option prices. In 2003, the methodology was revised and the new volatility index was based on S&P 500 Index options.
- **'VIX' is a trademark of the CBOE,** and Standard & Poor's has granted a license to NSE, with permission from the CBOE, to use such a mark in the name of the India VIX and for purposes relating to the India VIX

Source:Indian Express

FUTURE & OPTIONS TRADING

Context: A frenzied growth in India's equity options market, which clocked a record 85 billion trades last year and accounted for 84 percent of all equity option contracts traded in 2023, is unnerving policymakers and regulators.

Background:-

- Retail investors now make up over 35 percent of options trades, undeterred by the fact that 9 out of 10 individual traders in the equity futures and options (F&O) segment are recorded to have incurred losses.

About FUTURES & OPTIONS TRADING

- **Futures and options are two types of financial derivatives** that derive their value from an underlying asset such as shares, stock market indices, commodities, ETFs, and more.

Futures:

- **Futures are contracts that obligate the buyer to purchase a specific asset, and the seller to sell and deliver that asset, at a specific future date.** They are typically used to hedge against risk or for speculation.

Key Features:

- **Standardization**: Futures contracts are standardized in terms of contract size, expiration date, and other details, facilitating easy trading on exchanges.
- **Obligation**: Both parties in a futures contract are obligated to fulfill the contract at expiration. This means the buyer must purchase, and the seller must sell the asset at the agreed-upon price.
- **Margin Requirements**: Traders must maintain a margin account, which requires an initial margin (a percentage of the contract's value) and maintenance margin (minimum balance to keep the position open).

Options:

- **Options, on the other hand, give the buyer (of an options) the right, but not the obligation, to buy (or sell) an asset at a specific price at any time during the life of the contract.** They are often used for hedging, speculation, or arbitrage.

Key Features:

- **Types:** There are two types of options: call and put. A call option confers the right to buy a stock at the strike price before the agreement expires. A put option gives the holder the right to sell a stock at a specific price.
- **Right, Not Obligation:** The buyer has the right but is not obligated to exercise the option.The seller, however, has the obligation to fulfill the contract if the buyer exercises the option.
- **Premium:** The buyer of an option pays a premium to the seller for this right. The premium is the price of the option.
- **Strike Price:** This is the price at which the option holder can buy (call) or sell (put) the underlying asset.
- **Expiration Date**: Options have a finite life and must be exercised on or before their expiration date.

Here are some key differences between futures and options:

- **Obligation**: A futures contract requires a buyer to purchase the underlying security or commodity—and a seller to sell it—on a specific future date, unless the holder's position is closed earlier. An options contract, however, gives an investor the right, but not the obligation, to buy (or sell) shares at a specified price at any time before the contract's expiration.
- **Risk**: Futures contracts carry a high level of risk because they require the holder to buy or sell the asset in the future, regardless of its current market price. Options contracts, on the other hand, limit potential loss to the cost of the option premium.
- **Cost**: Futures require margin and can involve significant capital commitment.Options require the payment of a premium, which is generally lower than the margin for futures.

Source: Indian Express

MEDIUM, SMALL AND MICRO ENTERPRISES (MSME's)

Context: The government is considering options to relax the 45-day payment rule to MSMEs that has come into effect from April 1, 2023, as the rule may disrupt business practices due to its severity and lack of clarity.

Background:

- According to **Section 43B(h) of the Income Tax Act**, **introduced through the Finance Act 2023,** if a larger company does not pay an MSME on time — within 45 days in case of written agreements — it cannot deduct that expense from its taxable income, leading to potentially higher taxes.MSMEs fear that due to this provision, large buyers could cold-shoulder MSME suppliers and start buying either from those MSMEs that are not registered with Udyam or from non-MSMEs.

About MSMEs:

- They silently operate across various areas in India, contributing significantly to the country's GDP.
- **More than 6 crore MSMEs serve as the backbone of our economy,** accounting for approximately 30% of the GDP, 45% of manufacturing output, and around 48% of exports.
- Remarkably, over **11 crore people find employment within the MSME sector,** making it an essential driver of economic growth.

Recent Changes:

- **As of July 1, 2020, the definition of MSMEs** was revised to align with market conditions and ease of doing business.
- **The new criteria are based on investment amount and turnover:**
    - Micro: Investment up to ₹1 crore and turnover up to ₹5 crore.
    - Small: Investment up to ₹10 crore and turnover up to ₹50 crore.
    - Medium: Investment up to ₹50 crore and turnover up to ₹250 crore.
- Importantly, exports turnover is excluded from the turnover calculation, encouraging MSMEs to

Issues associated with the MSMEs:

- Lack of skilled labour, Bureaucratic red tape, Competition from larger companies, Access to finance,

  Lack of infrastructure, Lack of technological know-how, Issues with supply chain and logistics, Lack of formalization, Lack of marketing and networking opportunities are some of the issues associated with MSME's.

Source:Business Today

PRODUCTION LINKED INCENTIVE (PLI) SCHEME

Context: Smartphone PLI (production-linked incentive), the most successful of all 14 such schemes, may be extended by a couple of years beyond 2025-26 when it officially ends.

Background:-

- The scheme aimed to attract significant foreign investment in the sector while encouraging domestic mobile phone makers to expand their units and presence in India.

Key takeaways

- PLI Scheme is an **initiative by the Government of India** to boost domestic manufacturing and reduce imports.
- The scheme **provides companies with incentives on incremental sales** from products manufactured in domestic units.

Here are some key details about the PLI Scheme:

- The scheme was announced with an outlay of INR 1.97 Lakh Crores across 14 key sectors to create national manufacturing champions, create 60 lakh new jobs, and an additional production of 30 lakh crore during the next 5 years.
- **Aim**: attract foreign companies to set up units in India while encouraging local companies to expand their manufacturing units.
- **Targets**: labor-intensive sectors in the hope to create new jobs for the ballooning employable workforce of India.
- **The 14 sectors are:** (i) Mobile Manufacturing and Specified Electronic Components, (ii) Critical Key Starting Materials/Drug Intermediaries & Active Pharmaceutical Ingredients, (iii) Manufacturing of Medical Devices (iv) Automobiles and Auto Components, (v) Pharmaceuticals Drugs, (vi) Specialty Steel, (vii) Telecom & Networking Products, (viii) Electronic/Technology Products, (ix) White Goods (ACs and LEDs), (x) Food Products, (xi) Textile Products: MMF segment and technical textiles, (xii) High efficiency solar PV modules, (xiii) Advanced Chemistry Cell (ACC) Battery, and (xiv) Drones and Drone Components.
- PLI Schemes for all 14 Sectors have been notified by the concerned Ministries/ Departments after due approval. These Schemes are in various stages of implementation by the implementing Ministries/ Departments.

Source:Financial Express

WORLD INEQUALITY LAB (WIL)

Context: A tax package for the ultra-wealthy in India including an annual wealth tax and an inheritance tax for those with net wealth exceeding Rs 10 crore accompanied by redistributive policies has been proposed by a report of the World Inequality Lab.

Background:-

- Recently, the debate on inheritance tax had picked up after Sam Pitroda, a former adviser to Rajiv Gandhi and an associate of Rahul Gandhi, described the inheritance tax in the United States as an “interesting law”.

ABOUT WORLD INEQUALITY LAB (WIL)

- It’s a **research organization that aims to promote understanding of global inequality dynamics.**
- The lab **gathers social scientists committed to helping everyone understand the drivers of inequality worldwide through evidence-based research.**
- The WIL hosts and maintains the World Inequality Database, the most comprehensive **open-access database on global inequality dynamics.**
- **Mission:** maintain and expand the World Inequality Database.
- It works in close coordination with a large international network of researchers covering nearly seventy countries.
- **Projects involved:** the Political Cleavages and Social Inequalities project, the Climate Inequality Report, and the World Inequality Report.

- The WIL is **based at the Paris School of Economics**. It is committed to empowering civil society and reinforcing democracy, powered with data.

Key takeaways from the Report and associated note released by the researchers:

- **India's top 1 per cent income and wealth shares have reached historical highs** and are among the very highest in the world.
- **The report proposes an annual wealth tax and an inheritance tax** for those with net wealth exceeding Rs 10 crore, equivalent to the top 0.04 per cent of the adult population (~370,000 adults), who currently hold over a quarter of the total wealth.
- Outlining a baseline scenario, the report said, a 2 per cent annual tax on net wealth exceeding Rs 10 crore and a 33 per cent inheritance tax on estates exceeding Rs 10 crore in valuation would generate a 2.73 per cent of Gross Domestic Product (GDP) in revenues.
- However, the report noted that since the inheritance tax comes into force when wealth is bequeathed (i.e. upon the death of the dynast), hence, it delivers relatively small tax revenues on an annual basis.
- **On the other hand, a small wealth tax** would deliver significant annual revenues (given the larger base) while also dampening accumulation at the very top..
- **The tax package needs to be accompanied by redistributive policies to support the poor, lower castes, and middle classes.** The report cited an example of how the baseline scenario would allow doubling the current public spending on education, which has stagnated at 2.9 per cent of GDP over the past 15 years.

Source: Indian Express

ZIMBABWE GOLD OR ZiG GOLD

Context: Zimbabwe has launched the world's newest currency – the ZiG. The new gold-backed currency called the ZiG, or Zimbabwe Gold, has come into effect from April 5, 2024 and will co-circulate with other foreign currencies in the economy.

Background:

- The introduction of ZiG is an attempt to mitigate the currency instability and hyperinflation that has plagued the country for decades.

Key Takeaways

- **The ZiG stands out as a new currency backed by gold reserves, ensuring its value is supported by the physical gold held by the government.** ZiG notes and coins will be available and issued in denominations: 1ZiG, 2ZiG, 5ZiG, 10ZiG, 20Zig, 50ZiG, 100ZiG, and 200ZiG, with the gold backing aims to provide stability and prevent currency devaluation.

Transition from Zimbabwean Dollars to ZiG Banknotes and Coins:

- **Zimbabwe has grappled with high inflation, with rates surpassing 500% in recent years**. Following this, the country relied on various currencies, primarily the US dollar, leading to limited control over its economy.
- **The collapse of the Zimbabwean dollar in 2009, with hyperinflation peaking at 5 billion per cent, marked one of the most severe currency crashes in history.** The ZiG, now the sixth currency used by Zimbabwe, has been in circulation since April 5, 2024.

- **Banks in Zimbabwe have been converting the previous national currency,** the Zimbabwe dollar, into ZiGs, aiming to promote simplicity, certainty, and predictability in monetary and financial matters, Additionally, will circulate alongside other foreign currencies in the economy.
- Zimbabwe central bank governor has committed that the amount of currency in circulation will be backed by equivalent value in gold (or foreign exchange) in order to prevent the currency losing value like its predecessors.

Source:Indian Express

SENSEX & NIFTY

Context: Adani Ports and Special Economic Zone Ltd (APSEZ) will become the first Adani Group company to be included in the Sensex at the Bombay Stock Exchange, replacing Wipro on June 24.

Background:-

- APSEZ and Adani Enterprises, the flagship of the Adani Group, are already part of the Nifty at the National Stock Exchange, the other stock market index that tracks the performance of a set of large and actively traded companies in the country.

ABOUT STOCK MARKET AND INDICES

- **The stock market refers to public markets that exist for issuing, buying, and selling stocks.** Stocks, also known as equities, represent fractional ownership in a company, and the stock market is a place where investors can buy and sell ownership of such investible assets.
- **The stock market serves two very important purposes**.
    - The first is to provide capital to companies that they can use to fund and expand their businesses. By offering stock shares instead of borrowing the capital needed for expansion, the company avoids incurring debt and paying interest charges on that debt.
    - The secondary purpose the stock market serves is to give investors – those who purchase stocks – the opportunity to share in the profits of publicly traded companies. Investors can profit from stock buying in one of two ways. Some stocks pay regular dividends (a given amount of money per share of stock someone owns). The other way investors can profit from buying stocks is by selling their stock for a profit if the stock price increases from their purchase price.
- **Most of the trading in the Indian stock market takes place on its two stock exchanges**: the Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE). The BSE was established in 1875. The NSE was founded in 1992 and started trading in 1994.
    - Both the Sensex and Nifty are tools/index by which traders and market participants measure the domestic market's performance.
- **Launched in 1986, the Sensex is the country's oldest and most tracked index.** It is designed to measure the performance of the 30 largest, most liquid, and financially sound companies across key sectors of the Indian economy that are listed at BSE Ltd.
- **Among the companies that are part of the Sensex are Reliance Industries, ICICI Bank, and ITC Ltd.**These companies are selected to represent the broader Indian equity marketplace. As such, even though the Sensex is composed of only 30 stocks, investors make decisions to buy or sell based on the movement of the Sensex.
- The Sensex is reconstituted biannually, in June and December of every year.

How is Nifty different from Sensex?

- While the Sensex constitutes 30 companies trading on the BSE, the Nifty 50 is a broad-based index consisting of 50 blue chip large and liquid stocks traded on the National Stock Exchange of India (NSE).
- The Nifty 50 was started in November 1995.

Source: Indian Express

TRADE DEFICIT

Context: India has recorded a trade deficit, the difference between imports and exports, with nine of its top 10 trading partners, including China, Russia, Singapore, and Korea, in 2023-24, according to official data.

Background:

- The data also showed that the deficit with China, Russia, Korea, and Hong Kong increased in the last fiscal compared to 2022-23, while the trade gap with the UAE, Saudi Arabia, Russia, Indonesia, and Iraq narrowed.

Key Takeaways

- Trade deficit **arises in the course of international trade when the payments for imports exceed the receipts from export trade. A trade deficit is also referred to as a negative balance of trade.**
- According to trade experts, **a deficit is not always bad, if a country is importing raw materials or intermediary products to boost manufacturing and exports. However, it puts pressure on the domestic currency.**
- A rising trade deficit, even from importing raw materials and intermediates, can cause the country's currency to depreciate because more foreign currency is needed for imports. This depreciation makes imports more expensive, worsening the deficit.
- **Economic think tank Global Trade Research Initiative (GTRI)** said that a bilateral trade deficit with a country isn't a major issue unless it makes us overly reliant on that country's critical supplies. However, a rising overall trade deficit is harmful to the economy.
- To cover the growing deficit, the country might need to borrow more from foreign lenders, increasing external debt and this can deplete foreign exchange reserves and signal economic instability to investors, leading to reduced foreign investment.
- **Reducing a trade deficit involves a combination of increasing exports and decreasing imports.**

Some strategies that could be used to reduce trade deficit:

- **Improve Infrastructure:** Improving infrastructure can help facilitate trade and make it easier for businesses to export goods.
- **Diversify Trade Relations:** By diversifying trade relations, India can reduce its dependence on any single trading partner and potentially increase exports.
- **Quality Control Orders (QCOs):** These can be used to cut down on cheap and non-essential imports.
- **Devalue Domestic Currency:** This can make exports cheaper and imports more expensive, thereby reducing the trade deficit.
- **Attract Foreign Direct Investment (FDI):** FDI can bring in more funds and technology, which can boost domestic production and exports.

- **Energy Diversification**: Reducing reliance on imported crude oil by diversifying energy sources can help in reducing the trade deficit.
- **India's total trade deficit in the last fiscal narrowed to** $238.3 billion as against $264.9 billion in the previous fiscal.

Source:Hindu

NAVRATNA STATUS

Context: Recently, the Indian Renewable Energy Development Agency, a public sector undertaking company was granted 'Navratna status' from the department of public enterprises.

Background:

- IREDA provides funding and other services to projects focused on renewable energy and energy efficiency.

About NAVRATNA STATUS

- The Indian government **classifies Public Sector Undertakings (PSUs) into three categories:** Maharatna, Navratna, and Miniratna.
- Navratna companies are a group of Central Public Sector Enterprises (CPSEs) in India that enjoy a certain degree of financial autonomy from the government.
- These companies have specific privileges and can invest up to Rs 1,000 crore without explicit government approval.

Eligibility Criteria for Navratna Status:

- A company **must first be a Miniratna** and **have four independent directors** on its board before it can be considered for Navratna status.
- The company **must score at least 60 out of 100 based on specific parameters**, including:PBDIT (Profit Before Depreciation, Interest, and Taxes), Total Manpower Cost, Cost of Services, Capital Employed, Net Worth, Net Profit.
- Additionally, the company must **demonstrate significant global presence or international operations.**

Benefits of Navratna Status:

- Navratna companies **can invest up to Rs 1,000 crore or 15% of their net worth in a single project, whichever is lower.**
- They can also **invest up to 30% of their net worth over the entire year, again not exceeding Rs 1,000 crore.**
- These companies have **greater financial flexibility** and can undertake strategic projects without seeking explicit government approval.

Indian Renewable Energy Development Agency (IREDA)

- **It's a 'Navratna' Government of India** Enterprise under the administrative control of the Ministry of New and Renewable Energy (MNRE).
- **Established in 1987, IREDA serves as a Non-Banking Financial Institution (NBFC)** dedicated to promoting, developing, and extending financial assistance for projects related to renewable sources of energy and energy efficiency/conservation.

Source: Livemint

AJRAKH

Context: Ajrakh from Kutch gets GI tag

Background:

- Ajrakh's GI tag was granted by the Geographical Indications Registry after a long process of documentation and verification of its origin and production techniques.

About AJRAKH:



- The Ajrakh fabric is a traditional hand-block printing technique that **originated in Kutch, Gujarat.**
- It uses natural dyes and intricate patterns to create beautiful textiles, which are then used to make **sarees, dupattas, stoles, and other garments**.
- The unique feature of Ajrakh is its use of **geometric patterns and rich earthy colours like indigo, madder**, and mustard.
- This fabric has been an integral part of **the Kutchi culture** for centuries and has now gained recognition on a global level with its GI tag.

What is a GI tag?

- A **Geographical Indication** (GI) tag is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin.
- It serves as an intellectual property right, attributing a product's quality, reputation, or other characteristic to its geographical origin.
- In India, GI tags are issued by the Geographical Indication Registry under the Department of Industry Promotion and Internal Trade, Ministry of Commerce and Industry.
- The GI tag helps to ensure that only those registered as authorized users are allowed to use the popular product name.
- This system of tagging is beneficial as it prevents unauthorized use of a registered GI by others, promotes economic prosperity of producers of goods produced in a geographical territory, and helps consumers to get quality products of desired traits.

Some other Indian textiles that have received similar recognition.

- **Banarasi Silk**: It is a fine variant of silk originating from the city of Varanasi in Uttar Pradesh, India.
- **Chanderi Fabric**: Chanderi, a small town in Madhya Pradesh, is known for its delicate and lightweight fabric.
- **Kanjeevaram Silk**: The Kanjeevaram sarees from Tamil Nadu are famous for their vibrant colours, fine silk, and intricate zari work.
- **Kota Doria**: One of the many varieties of sari clothing produced in Muhammadabad Gohna, Mau in Uttar Pradesh and the surrounding area, as well as in Kota, Rajasthan, is kota doria.
- **Odisha Ikat**: One type of ikat, which is a resist dyeing method, comes from Odisha and is called Orissa Ikat.

Source: Times of India

REMITTANCES

Context: India has become the first country to receive over $100 billion in remittances in 2022, according to the United Nations migration agency.

Background:

- The International Organisation for Migration (IOM) revealed in its World Migration Report 2024 that India's remittances surged to $111 billion in 2022.

Top countries receiving remittances							
2010		2015		2020		2022	
India	53.48	India	68.91	India	83.15	India	111.22
China	52.46	China	63.94	China	59.51	Mexico	61.10
Mexico	22.08	Philippines (the)	29.80	Mexico	42.88	China	51.00
Philippines (the)	21.56	Mexico	26.23	Philippines (the)	34.88	Philippines (the)	38.05
France	19.90	France	24.07	Egypt	29.60	France	30.04
Nigeria	19.75	Nigeria	20.63	France	28.82	Pakistan	29.87
Germany	12.79	Pakistan	19.31	Pakistan	26.09	Egypt	28.33
Egypt	12.45	Egypt	18.33	Bangladesh	21.75	Bangladesh	21.50
Belgium	10.99	Germany	15.57	Germany	19.32	Nigeria	20.13
Bangladesh	10.85	Bangladesh	15.30	Nigeria	17.21	Germany	19.29

About REMITTANCES

- Foreign remittances refer to funds sent from one country to another, often by migrant workers to their families back home.
- These remittances play a crucial role in the economies of many countries, including India.

Top Sources of Remittances to India:

- **United Arab Emirates (UAE):** Migrant workers from India contribute significantly to the UAE's economy, and their remittances form a substantial portion of India's inflow.
- **United States**: Indian diaspora in the US sends substantial remittances back home.
- **Saudi Arabia:** Another major destination for Indian migrant workers, contributing to remittance inflows.
- **Oman and Kuwait:** These Gulf States also play a significant role in India's remittance landscape.

India's Role as a Source of Migrants:

- India has the largest number of international migrants, with nearly 18 million people living abroad.
- Major diasporas from India are located in the United Arab Emirates, the United States, and Saudi Arabia.
- India is also the 13th most popular destination for immigrants, with people coming from other countries to live in India.

Challenges and Risks:

- While remittances provide vital support to families and communities, migrant workers face numerous risks:
- **Financial Exploitation:** Some workers may be exploited financially.
- **Migration Costs:** Excessive financial debt due to migration costs.

- Xenophobia and Workplace Abuses: Migrants encounter challenges related to xenophobia and workplace mistreatment.
- The Gulf States, common destinations for Indian migrant workers, have faced criticism for rights violations.

Source: Times of India

GEOGRAPHY

RAFAH

Context: Israel's military has ordered tens of thousands of Palestinians to leave Rafah, a city in the southern Gaza Strip.

Background:

- This move comes amid growing fears of a ground assault on the area. Approximately 100,000 people in eastern Rafah have been directed to evacuate to "an expanded humanitarian area" along the coast.

About Rafah :



- **Rafah is a city in the southern part of the Gaza Strip**, near the border with Egypt.
- It is the capital of the Rafah Governorate of the State of Palestine, located approximately 30 kilometers southwest of Gaza City.

Importance:

- Historical Significance: **Rafah has been a site of human settlement since ancient times, known by various names throughout history.**
- Geopolitical Relevance: The city's proximity to Egypt makes it a strategic location for trade and cross-border movement. It is also the site of the Rafah Border Crossing, the **sole crossing point between Egypt and the Gaza Strip.**
- Humanitarian Aspect: Rafah has been a refuge for Palestinians, especially during conflicts. As of April 2024, it is reported that about **1.5 million people are believed to be sheltering in Rafah** due to massive bombardment and ground assaults in other parts of the Gaza Strip.

- Military Operations: **The city has been a focal point in the ongoing conflict between Israel and Hamas,** with recent military operations targeting the area due to the presence of Hamas operative.

Source: The Hindu

CHABAHAR PORT

Context: India and Iran signed a 10-year contract on Monday for the operation of a terminal at the strategically important Chabahar port in Iran.

Background:-

- The long-term bilateral contract was signed between Indian Ports Global Limited (IPGL) and the Port & Maritime Organisation (PMO) of Iran, enabling the operation of the Shahid Beheshti terminal at the Chabahar Port for 10 years.The pact replaces one-year contracts that were being signed to keep the port operational until now.

About CHABAHAR PORT

- Chabahar is a deep water port in Iran's Sistan-Baluchistan province.
- It is the Iranian port that is the closest to India, and is located in the open sea, providing easy and secure access for large cargo ships.

Chabahar Port History and Progress

• **Modern Chabahar was established in the 1970s**, recognizing Tehran's strategic importance during the Iran-Iraq war.
• In 2003, President Khatami and Prime Minister Atal Bihari Vajpayee signed a strategic cooperation roadmap, including Chabahar.
• The project, which could link South Asia, was halted by India's growing relationship with the US.
• The US's labeling of Iran as an **"axis of evil"** led to India abandoning its strategic relationship with Tehran.
• Despite spending $100 million on a 218-km road, **the port project progressed slowly.**
• IPGL operates Chabahar port through its subsidiary, **India Ports Global Chabahar Free Zone.**
• The **port handles over 90,000 TEUs** of container traffic and 8.4 MMT of bulk and general cargo.

Chabahar and INSTC

- To better realise Chabahar ports potential, the development of the port must be integrated with the larger connectivity project of the International **North South Transport Corridor (INSTC).**
- The INSTC, which was initiated by **Russia, India, and Iran**, is a multi-modal transportation route envisaged to link the Indian Ocean and Persian Gulf to the Caspian Sea via Iran, and onward to northern Europe via St Petersburg in Russia.

Source:Indian Express

SLOVAKIA

Context: Slovak Prime Minister Robert Fico was shot multiple times on Wednesday after a cabinet meeting in the central town of Handlova.

Background:

- The former Communist Party member took power for the fourth time last October and has turned the country's foreign policy towards more pro-Russian views.During his current term, Robert Fico has garnered worldwide attention after a series of inflammatory comments about Ukraine, calling for Kyiv to cede territory to Moscow to end the war.

ABOUT SLOVAKIA :



- **Officially the Slovak Republic , is a landlocked country in Central Europe, it's bordered by Poland to the north, Ukraine to the east, Hungary to the south, Austria to the west, and the Czech Republic to the northwest.**
- The capital and largest city is Bratislava, while the second largest city is Košice.
- Slovakia became an independent state on 1 January 1993.
- **Slovakia is a developed country with an advanced high-income economy.**
- **Slovakia is a member of the European Union,** the Eurozone, the Schengen Area, the United Nations, NATO, CERN, the OECD, the WTO, the Council of Europe.
- The world's largest per-capita car producer, Slovakia manufactured a total of 1.1 million cars in 2019, representing 43% of its total industrial output.

Source: NDTV

KYRGYZSTAN

Context: The government on Saturday urged Indian citizens in Kyrgyzstan to stay indoors amid reports of attacks on foreign students and protests by local residents in Bishkek on Friday night over concerns about "growing numbers of migrants from Pakistan, Bangladesh, India, Egypt and other countries".

Background:

- Around 15,000 Indian students are estimated to be in Kyrgyzstan, mostly enrolled in medical colleges and universities.

About KYRGYZSTAN :

- Kyrgyzstan, officially the Kyrgyz Republic, is a landlocked country in Central Asia, lying in the Tian Shan and Pamir mountain ranges.



- Kyrgyzstan is bordered by Kazakhstan to the north, Uzbekistan to the west, Tajikistan to the south, and China to the east and southeast.
- Ethnic Kyrgyz make up the majority of the country's 7 million people, followed by significant minorities of Uzbeks and Russians.
- On 31 August 1991, Kyrgyzstan declared independence from the USSR and a democratic government was established. Kyrgyzstan attained sovereignty as a nation state after the breakup of the Soviet Union in 1991.
- After independence, Kyrgyzstan was officially a unitary presidential republic; after the Tulip Revolution it became a unitary parliamentary republic, although it gradually developed an

executive president and was governed as a semi-presidential republic before reverting to a presidential system in 2021.

Source: Indian Express

BALTIC SEA

Context: A Russian defence ministry proposal to revise Russia's maritime border in the eastern Baltic Sea was deleted on Wednesday from an official portal after creating confusion and concern among NATO members such as Finland, Sweden, Lithuania and Estonia.

Background:

- Finland and the Baltic states are all members of the EU and Nato and the military alliance is committed to defending their borders.



About Baltic Sea

- **The Baltic Sea is enclosed** by Denmark, Estonia, Finland, Germany, Latvia, Lithuania, Poland, Russia, Sweden, and the North and Central European Plain.
- It is a shelf sea and marginal sea of the Atlantic with limited water exchange between the two, making it an inland sea.
- The Baltic Sea drains through the Danish Straits into the Kattegat by way of the Øresund, Great Belt and Little Belt. It includes the Gulf of Bothnia (divided into the Bothnian Bay and the Bothnian Sea), the Gulf of Finland, the Gulf of Riga and the Bay of Gdańsk.

Additional information

- The Baltic states or the Baltic countries is a geopolitical term encompassing Estonia, Latvia, and Lithuania.
- All three countries are members of NATO, the European Union, the Eurozone, and the OECD. The three sovereign states on the eastern coast of the Baltic Sea are sometimes referred to as the "Baltic nations".

Source: Wikipedia

TROPICAL CYCLONES

Context: At least 36 people were killed in heavy rains and landslides in four northeastern states on Tuesday, May 28, under the impact of Cyclone Remal.

Background:

- Tropical cyclones develop in tropical oceans at least 5° - 30° latitude north or south of the equator, where the sea temperature is at least 27 °C. They are important on Earth for transferring heat and energy between the equator and the poles.

About TROPICAL CYCLONES :

- These are formed in the **region between the Tropics of Capricorn and Cancer.**
- They're large-scale weather systems that form over tropical or subtropical oceans and coalesce into surface wind circulation.
- Tropical cyclones are **one of the world's most destructive natural disasters**.

Favourable Conditions for the Formation of Tropical Cyclone

- A large area of water surface with a **temperature above 27° C**
- The **Coriolis force is strong enough to form a cyclonic vortex.**
- Variations in the vertical wind speed are minor.This allows the storm clouds to rise vertically to high levels.
- A weak low-pressure area or low-level cyclonic circulation already exists.
- Above the sea level system, there should be a higher divergence.

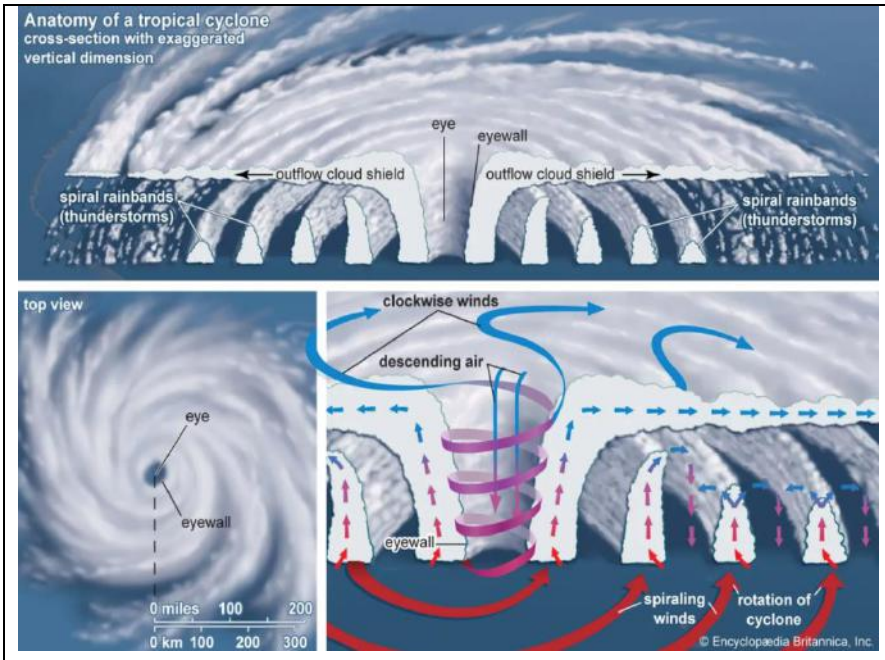
Origin of Tropical Cyclones

- Tropical cyclones **arise over tropical oceans in late summers** and have a thermal origin (August to mid-November).
- Because of the Coriolis effect, **the powerful local convectional currents** take on a whirling motion at these regions.

Developmental Stages of Tropical Cyclone

Tropical cyclones can be classified into three stages throughout their development:

- **Formation and Initial Development Stage**
    - The transport of water vapor and heat from the warm ocean to the overlying air, largely through evaporation from the sea surface, is crucial to the creation and early development of a cyclonic storm.
    - Convection with condensation of rising air above the ocean surface stimulates the creation of huge vertical cumulus clouds.
- **Mature Stage**



o The air rises in powerful thunderstorms as a tropical storm intensifies, and it tends to spread out horizontally at the tropopause level.

o When air spreads out, a positive pressure is created at high elevations, speeding up the downward migration of air due to convection.

o When subsidence is induced, the air warms up due to compression, resulting in a warm 'Eye' (low-pressure center).

o A mature tropical cyclone in the Indian Ocean has a concentric pattern of highly turbulent large cumulus thundercloud bands as its principal physical feature.

- **Modification and Decay**
  - o As soon as its source of warm moist air begins to ebb or is abruptly cut off, a tropical cyclone begins to weaken in terms of core low pressure, internal warmth, and extremely fast speeds.
  - o This occurs after it reaches land or passes across chilly water.

Source:Bureau of Meteorology

SHARAVATI RIVER

Context: The National Green Tribunal (NGT) has directed Karnataka government to stop any unlawful/illegal sand mining in Sharavathi river.

Background:

- The NGT's southern zone bench was hearing an application regarding damage caused to flora and fauna due to illegal sand mining.

About Sharavathi River

- Sharavati is a river which originates and flows entirely within the state of Karnataka in India.
- **It is one of the few westward flowing rivers** of India and a major part of the river basin lies in the Western Ghats.
- The total length of the river is around 128 km and it joins the Arabian Sea at Honnavar in Uttara Kannada district.
- On its way, the Sharavati forms the Jog Falls where the river falls from a height of 253 m.
- Jog Falls is the highest waterfall in India if the single drop water fall and the volume of water are considered for height.Otherwise, it is the third highest waterfall in India (after Kunchikal Falls and Barkana Falls); all three are located in Shivamogga district.

Source:Deccan Herald

RUSSIA'S DISPUTE WITH ESTONIA OVER NARVA RIVER

Context: Recently, a new dispute has emerged between Russia and Estonia, centered around the removal of navigation markers on the Narva River, which forms part of the border between the two countries.

Background:

- The conflict between Ukraine and Russia has not only strained relations between the two countries but has also aggravated tensions with Russia's neighbouring nations.

About ESTONIA :

- Estonia, officially the Republic of Estonia, is a country by the Baltic Sea in Northern Europe.
- It is bordered to the north by the Gulf of Finland across from Finland, to the west by the sea across from Sweden, to the south by Latvia, and to the east by Lake Peipus and Russia.
- Tallinn, the capital city, and Tartu are the two largest urban areas of the country.
- Estonia is a developed country, with a high-income advanced economy, ranking 31st (out of 191) in the Human Development Index.

NARVA RIVER

- The Narvaflows 77 kilometresnorth into the Baltic Sea and is the largest Estonian river by discharge. A similar length of land far to the south, together with it and a much longer intermediate lake, Lake Peipus, all together nowadays form the international border between Estonia and Russia.
- The Narva River is a vital shipping route for Estonia, and any disruption due to territorial disputes could have significant economic repercussions. Russia's opposition to floating markers on the river raises concerns about its intentions and could be seen as a strategic move to intimidate Estonia and its NATO allies.

Source:Financial Express

LA CUMBRE VOLCANO

Context: On March 2, La Cumbre erupted for the first time since 2020, NASA's Earth Observatory reported, when lava began slowly seeping out of a fissure near the summit of the volcano's 4,850-foot-tall (1,480 meters) southeast flank and dribbling down the mountain's tree-covered slopes. Since then, lava has continually poured out of the volcano, and in early April, the river of molten rock reached the island's coastline around 6 miles (10 km) from La Cumbre

Background:-

- Fernandina, the youngest of the Galápagos islands, is also the most volcanically active. The island's La Cumbre volcano lies directly atop the mantle plume, or hot spot, that produced all of the Galápagos islands.

About La Cumbre volcano

- La Cumbre is a **shield volcano** on Fernandina Island in the Galápagos Islands.
- Volcanic emissions from Galápagos volcanoes are **typically rich in sulfur dioxide and have little ash.**
- Since Fernandina – the most volcanically active island among the Galápagos Islands – is uninhabited, there is no risk to people or infrastructure, but the lava flow reportedly has proven to be a spectacle for people on passing ships.

- Most of Fernandina is rocky and inhospitable to vegetation due to recent lava flows, but a ring of vegetation grows on the volcano's upper slopes. Vegetation covers more of the other Galápagos islands, where eruptions are less frequent.
- The island is also home to a large population of rare land iguanas which nest and lay eggs on the rim of the volcano's large crater or caldera and deep within it.

About Galápagos Islands

- The Galápagos Islandsare an archipelago of volcanic islands in the Eastern Pacific, located around the Equator 900 km (560 mi) west of South America. They form the Galápagos Province of the Republic of Ecuador.
- All of these islands are protected as part of Ecuador's Galápagos National Park and Marine Reserve.

Source: Nasa

GREAT RIFT VALLEY

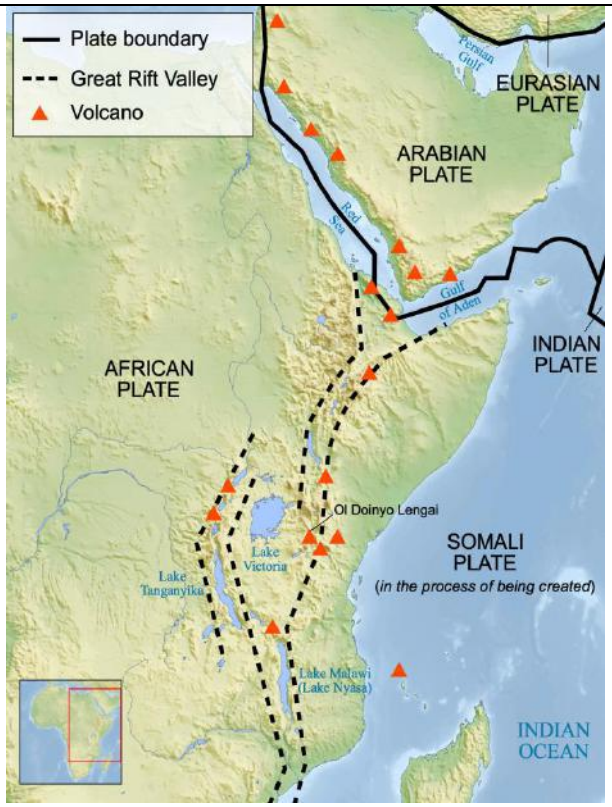
Context: At least 42 people died when a dam burst its banks near a town in Kenya's Rift Valley, as heavy rains and floods battered the country recently.

Background:

- A rift valley is a lowland region that forms where Earth's tectonic plates move apart, or rift. Rift valleys are found on land and at the bottom of the ocean, where they are created by the process of seafloor spreading. Rift valleys differ from river valleys and glacial valleys in that they are created by tectonic activity and not the process of erosion.

About the Great Rift Valley

- It's a series of **contiguous geographic trenches**, approximately 7,000 kilometres (4,300 mi) in total length, that runs from Lebanon in Asia to Mozambique in Southeast Africa.
- **The northern part of the system is the Jordan Rift Valley.** The Jordan Rift Valley stretches from the Golan Heights, near Israel's border with Syria and Lebanon, to the Dead Sea, to the Gulf of Aqaba—an inlet of the Red Sea that separates the Sinai Peninsula from the Arabian Peninsula.
- Associated with the Jordan Rift Valley to the south is the Red Sea Rift. Millions of years ago, the Arabian Peninsula was connected to Africa. Seafloor spreading caused the Arabian and African plates to rift apart. The Indian Ocean flooded the rift valley between the continents, creating the Red Sea.
- Today, Africa and Asia are connected by the triangle of the Sinai Peninsula. Eventually, the Red Sea Rift will separate Africa and Asia entirely and connect the Mediterranean and Red Seas.
- South of the Red Sea Rift lies the massive, complex East African Rift. Throughout the East African Rift, the continent of Africa is splitting in two. The African plate, sometimes called the Nubian plate, carries most of the continent, while the smaller Somali plate carries Horn of Africa.



- Several deep, elongated lakes, called ribbon lakes, exist on the floor of this rift valley: Lake Malawi and Lake Tanganyika are examples of such lakes. The region has a unique ecosystem and contains a number of Africa's wildlife parks.

Source:National Geographic

EL NINO, ANTI CYCLONE SYSTEMS: WHY LARGE PARTS OF INDIA EXPERIENCED HEATWAVES IN APRIL

Context: April has been abnormally hot. In the first 26 days of the month, either a small pocket or a considerably large geographical area in India experienced heatwave conditions, according to the India Meteorological Department (IMD).

Background:-

- Intergovernmental Panel on Climate Change (IPCC) reports and analysis of weather models clearly indicate that heatwaves in India shall no longer remain contained over only those regions traditionally believed to be prone. Newer areas, especially from the southern peninsular India, are already experiencing heatwaves.

Which areas in India are prone to heatwaves?

- The Core Heatwave Zone (CHZ) spanning central, north, and peninsular India between Gujarat and West Bengal is prone to heatwave conditions every year, during the summer season March to June and occasionally in July.
- Rajasthan, Punjab, Haryana, Chandigarh, Delhi, West Madhya Pradesh, Uttar Pradesh, Chhattisgarh, Odisha, Vidarbha in Maharashtra, parts of Gangetic West Bengal, coastal Andhra Pradesh, and Telangana are the most heatwave-prone states or regions.

When does IMD declare a heatwave?

- IMD declares a heatwave when the normal maximum temperature recorded over at least two localities in plains touches 40 degree Celsius or exceeds 4.5 degree Celsius from normal. Heatwave is declared in hilly and coastal regions when temperatures cross 30 degree Celsius and 37 degree Celsius respectively.
- A severe heatwave is declared if the temperature departure exceeds 6 degree Celsius from normal.

Why has April been so hot?

- **One, 2024 is a year that began in an El Niño state**. El Niño,is a climate pattern that describes the unusual warming of surface waters in the eastern Pacific Ocean, which leads to extreme heat in many parts of the world and the ocean. It developed in June 2023 and generally, the years which

begin in an El Niño state, experience extreme temperatures, harsh, multiple and extended heatwave spells, and lack of pre-monsoon rainfall.

- **Two, the persistent presence of anticyclone systems over southern peninsular and southeastern coastal areas is also partly responsible for such a hot April**. These high-pressure systems, which exist at about the altitude of 3 km and extend between 1,000 and 2,000 km in length, push the air underneath them towards the Earth, in a process called air subsidence. As a result, the forcefully sunk air generates more heat on the surface closer to the Earth.
- The presence of anticyclone systems also leads to wind flow from land towards the sea and prevents the incoming cooler sea breeze, which is otherwise responsible for cooling land from time to time.
- **El Niño and anticyclone systems collectively created sweltering hot conditions and heatwaves during April, especially over Gangetic West Bengal, Odisha, Andhra Pradesh, Telangana, Tamil Nadu, Kerala, Karnataka, and Maharashtra.**

What has been the extent of heatwaves this April?

- Except for four days (April 1, 10, 11 and 12), a small pocket or a considerably large geographical area of the country experienced either heatwave or severe heatwave conditions, according to the IMD.
- **The worst affected were southern peninsular India and the southeastern coast areas, including Odisha, Andhra Pradesh, and Tamil Nadu along with large areas of Karnataka, Kerala, Sikkim, Bihar, Jharkhand, and Gangetic West Bengal.**
- The inclusion of Kerala and Sikkim to the list of heatwave-affected areas indicates that the temperatures are rapidly rising during summers over regions outside that of the CHZ.

Source: Indian Express

SEA OF JAPAN

Context: North Korea recently fired a ballistic missile into the Sea of Japan, also known as the East Sea.

Background:

- These missile tests are closely monitored by neighboring countries and the international community due to their potential implications for regional security.

About SEA OF JAPAN:

- The Sea of Japan is a **marginal sea located between the Japanese archipelago, Sakhalin, the Korean Peninsula, and the mainland of the Russian Far East.**
- The Japanese archipelago separates the Sea of Japan from the Pacific Ocean.

The Sea of Japan is bordered by the following countries:

- Japan: To the east and south, with major islands like Hokkaidō, Honshū, and Kyūshū.
- Russia: To the north, including the mainland and Sakhalin Island.
- North Korea: To the west, across the sea from Japan.
- South Korea: Also, to the west, separated from Japan by the Korea Strait.

Source: NDTV

TORNADO

Context: Recently, a deadly tornado struck in West Bengal and killed five people and injured over a 100.

Background:

- In recent years, an increasing number of tornadoes have formed across India and caused destruction. A warming Bay of Bengal and land, along with anomalous wind patterns could be a reason behind this.

About TORNADO

- It's a **violently rotating column of air that extends from a cumulonimbus cloud to the Earth's surface, and** forms during severe thunderstorms.
- Unlike cyclones (which are large-scale weather systems), tornadoes are local and short-lived.
- **The formation of a tornado involves a complex interaction between a supercell (a long-lived thunderstorm with a deep, persistently rotating updraft) and the environment.**
- Tornadoes are more likely to occur in regions where warm, moist air near the surface interacts with cooler, drier air aloft.

Composition and Impact:

- Tornadoes consist of air, water droplets, dust, and debris.
- **The visible funnel is primarily made of** water droplets formed from condensation, along with dust and debris picked up from the ground.
- The core of the storm contains the lowest pressure and highest wind speeds, causing the most significant damage.
- Tornadoes can destroy buildings, homes, bridges, vehicles, uproot trees, and alter landscapes.

Source: Down To Earth

INDIAN OCEAN DIPOLE (IOD)

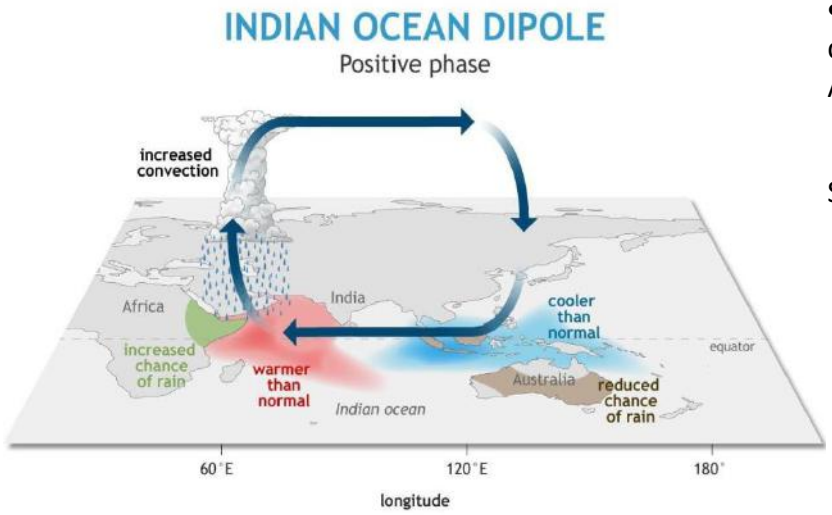
Context: According to recent reports, Positive Indian Ocean Dipole (IOD) or Indian Nino may re-emerge for the second consecutive year in the latter half of 2024.

Background:

- It is for the first time since 1960 that the event is recurring for a second consecutive year since record-keeping began.

About INDIAN OCEAN DIPOLE (IOD) :

- It's also known as the **Indian Niño, is an irregular oscillation of sea surface temperatures** in which the western Indian Ocean **becomes alternately warmer (positive phase) and then colder (negative phase) than the eastern part of the ocean.**
- **During a positive phase**, warm waters are pushed to the Western part of the Indian Ocean, while cold deep waters are brought up to the surface in the Eastern Indian Ocean. This pattern is reversed during the negative phase of the IOD.
- **The IOD can modulate the impact of El Niño on the Indian monsoon.** When a positive IOD coincides with an El Niño event, it can partially offset the negative effects of El Niño.



- The IOD influences local weather, causing heavy rains and/or droughts in Africa and Australia.

Source: Hindu Businessline



Environment and Ecology



MAHARASHTRA GEARS UP FOR TIGER RELOCATION

Context: The Maharashtra forest department is gearing up for translocation of a few tigers from the Tadoba-Andhari Tiger Reserve (TATR) in Chandrapur to Sahyadri, the lone tiger reserve in the state's western region.

Background:

- The translocation project is an important step in conservation of tigers in the forests of the northern Western Ghats, which form a key wildlife corridor between Maharashtra and Karnataka.

Key Takeaways

- **The Sahyadri Tiger Reserve (STR) is one of only five tiger reserves** in the country — Kaval in Telangana, Kamlang in Arunachal Pradesh, Dampa in Mizoram and Satkosia in Odisha being the other four — with zero tigers within the reserve and the translocation is part of a long-term plan to revive the population of the big cats in the northern Western Ghats forests.
- **STR straddles Kolhapur, Satara, Sangli and Ratnagiri districts in western Maharashtra. Spread over 1,165 sq km, the reserve was notified in 2010 by amalgamating the Chandoli national park and Koyna wildlife sanctuary.**
- The Maharashtra forest department is awaiting the final go-ahead from the Union Environment Ministry. **The NTCA (National Tiger Conservation Authority) had cleared the translocation plan in October 2023. In the initial phase, a male tiger or a pair of male and female tigers would be translocated from TATR to STR.**
- **The 2023 tiger population estimation report highlighted the threats posed to the wildlife corridor between the STR and Karnataka's forests from mining activities, road projects and human settlements.**
- The translocation of tigers from TATR will be followed by their translocation from the Pench Tiger Reserve landscape, which is spread over Maharashtra and Madhya Pradesh. In all, the plan involves translocation of eight tigers — three males and five females.
- The translocation is part of the phase-II of the tiger recovery project; phase-I involves preparing the habitat for the big cats' reintroduction, augmentation of prey, improving forest protection and building of a temporary enclosure for their soft release.

Source: Indian Express

CARBON FARMING

Context: Some techniques within carbon farming can reduce greenhouse gas emissions.

Background:

- By adopting carbon-smart farming methods, we can simultaneously meet food demands and combat climate change.

About CARBON FARMING:

- It's also known as carbon sequestration refers to a set of practices aimed at reducing greenhouse gas emissions from agriculture and land use.
- It aimed at storing carbon in the soil, crop roots, wood, and leaves.

Key Techniques in Carbon Farming:

- **Rotational Grazing**: This involves moving livestock between pastures to allow vegetation to recover, enhancing soil carbon storage.
- **Agroforestry**: Incorporating trees and shrubs into farming systems to sequester carbon in vegetation.
- **Conservation Agriculture:** Practices like zero tillage, crop rotation, and cover cropping to minimize soil disturbance and improve organic content.
- **Integrated Nutrient Management:** Using organic fertilizers and compost to promote soil fertility and reduce emissions.
- **Livestock Management**: Optimizing feed quality and managing animal waste to lower methane emissions.

Challenges in Implementation:

- Geographical Factors: Effectiveness varies based on location, soil type, crop selection, and water availability.
- Biodiversity: Practices must be balanced with maintaining ecosystem services to avoid negative impacts like biodiversity loss.
- Policy and Community Engagement: Requires sufficient policy support and community involvement for successful adoption.

Source: The Hindu

KAZIRANGA NATIONAL PARK

Context: Assam's Kaziranga National Park collects highest revenue in its 50-year history.The revenue collected by the authorities of Kaziranga National Park, from the tourists, touches ₹8.8 crore.

Background:

- Kaziranga National park is sprinkled with elephant-grass meadows, swampy lagoons, and dense forests is home to more than 2200 Indian one-horned rhinoceros, approximately 2/3rd of their total world population.

About Kaziranga National Park:

- It's located in the **Golaghat and Nagaon districts of the state of Assam**, India. Located on the edge of the Eastern Himalaya biodiversity hotspot, the park combines high species diversity and visibility.
- **The park is a UNESCO World Heritage Site.** It's renowned for its population of the Indian one-horned rhinoceros, which is one of the main attractions for visitors.
- **Kaziranga is also recognized as an Important Bird Area** by BirdLife International for the conservation of avifaunal species.The Endangered Ganges dolphin is also found in some of the closed oxbow lakes.Over the time, the tiger population has also increased in Kaziranga, and that's the reason why Kaziranga was declared as Tiger Reserve in 2006.
- Due to the difference in altitude between the eastern and western areas of the park, one can see mainly four types of vegetation like alluvial inundated grasslands, alluvial savanna woodlands, tropical moist mixed deciduous forests, and tropical semi-evergreen forests.
- Kumbhi, Indian gooseberry, the cotton tree, and elephant Apple are amongst the famous trees that can be seen in the park.It is criss-crossed by four major rivers, including the Brahmaputra.

Source: Hindu

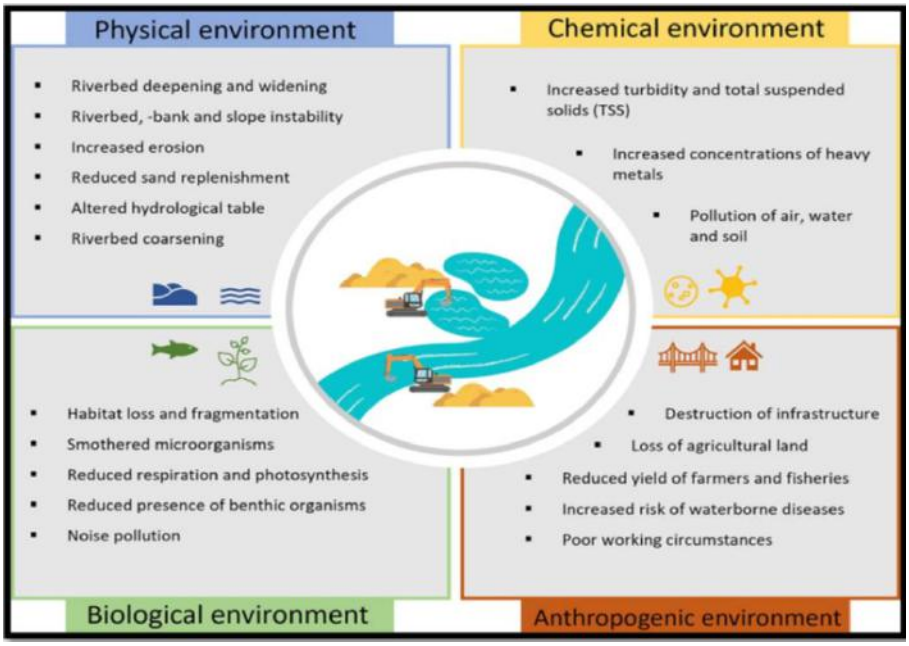
SAND MINING

Context: The Supreme Court sought a report from the Directorate of Enforcement (ED) related to the illegal sand mining case in Tamil Nadu.

Background:

- The Enforcement Directorate (ED) has provisionally attached assets worth Rs 130.60 crore in connection with an illegal sand mining case in Tamil Nadu.

Key takeaways:



- Sand mining involves the extraction of sand from various sources, including rivers, beaches, and seabeds.
- Sand is a valuable resource used in construction, manufacturing, and other industries.
- Approximately 40-50 billion tonnes of sand are extracted globally each year, but the management and regulation of sand mining are often inadequate in many countries.

Illegal Sand Mining in India

- Illegal sand mining in India is a crime under Sections 378 and 379 of the Indian Penal Code, 1860.
- Natural resources are considered public property, and the state acts as their trustee.
- Violating this trust by engaging in illegal sand mining can have severe consequences.

Some of the social and economic Impacts:

- Displacement: Communities that depend on riverbanks for their livelihoods, such as fishing and agriculture, face displacement due to sand mining.
- Environmental Degradation: Excessive sand mining disrupts river ecosystems, alters river channels, and contributes to erosion. For instance, in the Papagani catchment area in Karnataka, illegal sand mining has led to groundwater depletion and environmental degradation in communities along the river banks in both Andhra Pradesh and Karnataka.
- Groundwater Depletion: Unregulated extraction impacts groundwater levels, affecting water availability for communities.

Source: The Hindu

DEMISE OF RANGELANDS

Context: About half of the world's rangelands are degraded and need policy interventions, and communities depending on them need focused support, according to a new report of the United Nations Convention on Combating Desertification (UNCCD).

Background:

- Rangelands cover 80 million sq km, which is 54 per cent of the earth's land surface. They are characterised by low vegetation and comprise grasslands, shrublands, wetlands, desert, semi-arid land, mountain pastures, plateaus and tundra.
- In India, rangelands occupy about 1.21 million sq km, from the Thar Desert to Himalayan meadows, as per the UNCCD report.

What are rangelands?

- **The UNCCD report defines rangelands as natural or semi-natural ecosystems that are grazed by livestock or wild animals. Rangelands contain vegetation such as grasses, shrubs, bushes, open forests, and agroforestry systems (land which contains trees and crops or pastures). The exact nature of rangelands' vegetation is influenced by rainfall, temperature, and other climate phenomena.**
- Currently, rangelands cover 80 million sq km of Earth's terrestrial surface area (over half of Earth's land), and are thus the largest land cover or land use type in the world, the UNCCD report said.
- **They act as carbon sinks (which absorb more carbon from the atmosphere than they release), storehouses of freshwater, and prevent desertification of land. Millions of people worldwide depend on rangelands for food security, and livelihoods.**

What does the UNCCD report say?

- The UNCCD report found that **nearly 50% of the world's rangelands can be considered "degraded" and are facing a "silent demise".**
- Climate change, unsustainable land and livestock management practices, biodiversity loss, and the conversion of rangelands to farmlands are some of the primary drivers of rangeland degradation. Uncertainty over land rights among pastoralist communitiesalso leads to their degradation.
- This, in turn, severely affects the communities dependent on rangelands as their deterioration impacts soil fertility and biodiversity, leading to a dip in incomes and rise in conflicts with authorities over grazing rights.

Who are pastoralists?

- Pastoralism is a livelihood system based on livestock production. **This includes livestock rearing, dairy production, meat production, wool production, and leather production.**
- The communities and groups, both indigenous and non-indigenous, who are involved in livestock production are known as pastoralists. These communities rear sheeps, goats, cattle, horses, donkeys, camels, yaks, llamas, alpacas, and pigs. Some pastoralist communities also rear ducks and chickens.
- Their livelihood is highly dependent on the quality of pasture (or rangelands) they have access to, and their rights over them.

What are pastoralists' economic contributions in India?

- India is home to 20% of the world's livestock population. Around 77% of these animals are reared in pastoralist systems, meaning they are either herded or left to range on common lands, according to the Accounting for pastoralists in India (2020) report.

- Pastoralists also protect indigenous livestock breeds, and protect traditional knowledge about animal rearing.
- Currently, India is the largest producer of milk in the world, accounting for 23% of global dairy production. It is also the largest producer of buffalo meat and largest exporter of sheep and goat meat, according to the Department of Animal Husbandry and Dairying.

Source: Indian Express

MERCURY POLLUTION

Context: The Governments of Albania, Burkina Faso, India, Montenegro and Uganda have united to combat chemical pollution, launching a $134-million "Phasing out mercury measuring devices in healthcare project" to eliminate the use of mercury in medical devices.

Background:

- Mercury is one of the top 10 chemicals of major public health concern, according to WHO. Released to the environment from a variety of human activities, it is transported around the globe and bioaccumulates in the food chain.People are exposed to mercury primarily through eating contaminated fish or breathing mercury vapour directly.Children, infants and fetuses are at the highest risk because of their developing nervous systems.

About Mercury

- Mercury is a chemical element symbolized by Hg on the periodic table. It's the only common metal that is liquid at ordinary temperatures.It's heavy, silvery-white, and slowly tarnishes in moist air.

Mercury Pollution

- Mercury occurs naturally in the earth's crust, but human activities, such as mining and fossil fuel combustion, have led to widespread global mercury pollution.
- Mercury is a persistent, bioaccumulative, toxic pollutant. When released into the environment, it accumulates in water laid sediments where it converts into toxic methylmercury and enters the food chain. Mercury contamination is a significant public health and environmental problem because methylmercury easily enters the bloodstream and affects the brain.
- Mercury is introduced into the environment in three ways. First, mercury is emitted into the air naturally from volcanoes, the weathering of rocks, forest fires, and soils. Second, mercury is emitted into the air from the burning of fossil fuels and municipal or medical waste. Lastly, mercury can be re-introduced into the environment through natural processes such as evaporation of ocean water.
- Human exposure to mercury occurs from eating fish and shellfish contaminated with methylmercury.
- Mercury can also be found in beauty products, particularly skin-lightening creams, but also eye make-up and eye-cleansing products6.
- Globally, artisanal and small-scale gold mining is the largest source of anthropogenic mercury emissions (37.7%), followed by stationary combustion of coal (21%)

Phasing out mercury measuring devices in healthcare project

- **Led by the United Nations Environment Programme (UNEP),** with funding from the Global Environment Facility (GEF) and executed by the World Health Organization (WHO), the Phasing out mercury measuring devices in healthcare project will develop and implement nation-wide strategies to stop the import, export and manufacture of thermometers and sphygmomanometers, encourage the adoption of accurate, affordable and safe mercury-free alternatives, while improving the management of mercury-containing medical waste

Source:UNEP

CHIR PINE

Context: As forest fires rage through Uttarakhand, the chir pine with its inflammable leaves or needles has been in the news for less benign reasons.

Background:

- After a government ban in 1981 on felling of trees over 1,000 mt above sea level, the chir got an opportunity to spread lower in the foothills where the sal grows and above too, among the broad-leaved trees because it's the nature of the chir to colonise.

About CHIR PINE

- **A conifer that can grow up to a height of about 30 metres, the chir pine covers almost 16.5 per cent of Uttarakhand's total forested area.**
- Native to the Himalayas, chir pine or the longleaf Indian pine, was not introduced to the Himalayas by the British, as is often believed, though they certainly gave it preference for its commercial use, mainly lumber and resin.
- Found all over the Western Himalayas at an elevation between 1,000 and 2,000 metres, it covers large parts of Garhwal and Kumaon where it can be seen growing on steep slopes and can be identified by its three needles and its round cone (unlike the elongated cones of other pines).
- Chir Pine does not allow any vegetation to grow around it as it makes the soil more acidic through its fallen needles. The accumulating carpet of needles on the forest floor under these trees makes conditions unfavourable for many common plants and trees to grow
- For the short-term and long-term management of the forests in Uttarakhand, the dense chir pine forest, more like monocultures, must be converted into mixed forests. More broad-leaf species should be promoted and the dense chir pine forests should be effectively managed and their wood used. The state could effectively manage the chir forests and earn some revenue as well.
- While its needles tend to catch fire easily because of their high resin content, the bark of the tree doesn't catch fire easily since it has a very high ignition temperature — which is why blacksmiths use the bark of the chir pine in their furnaces to melt metal.

Source: Indian Express

VENEZUELA LOST ITS GLACIER

Context: Venezuelahas likely become the first country — it certainly won't be the last — in modern history to lose all its glaciers. This comes after scientists reclassified the Humboldt glacier, Venezuela's last remaining glacier, as an ice field earlier this month.

Background:-

- Scientists expected the Humboldt glacier to last another decade. However, it melted at a faster rate than expected.

Key takeaways

- Venezuela used to be home to six glaciers, located at about 5,000 metres above sea level in the Andes mountains.
- By 2011, five of them had vanished. Scientists expected the Humboldt glacier to last another decade. But it melted at a faster rate than expected, and has shrunk to an area of less than 2 hectares, leading to its downgrade from a glacier to an ice field.

What are glaciers?

- **Glaciers are essentially large and thick masses of ice that are formed on land due to the accumulation of snow over centuries**.
- According to the United States Geological Survey (USGS), they usually exist and form in areas where mean annual temperatures reach near the freezing point; winter precipitation leads to significant snow accumulations; and temperatures throughout the remaining year do not cause the complete loss of the previous winter's snow accumulation.

Why are glaciers disappearing?

- Since the Industrial Revolution kicked off in the 18th Century, human activities such as burning fossil fuels have been releasing GHGs like carbon dioxide and methane into the atmosphere. These invisible gases trap heat — they allow sunlight to pass through the atmosphere but prevent the heat that sunlight brings from returning into space — leading to a rise in global temperatures.
- In recent decades, the emission of GHGs has skyrocketed, which has resulted in the global average temperature to increase by at least 1.1 degree Celsius since 1880. The temperature rise may seem small but it has had devastating consequences.
- The Andes — a mountain range running through parts of Argentina, Bolivia, Chile, Colombia, Ecuador, Peru, and Venezuela — has witnessed a temperature increase of a high rate of 0.10 degree Celsius in the past seven decades. That is one of the major reasons why Venezuela has lost all of its glaciers.
- In the case of the Humboldt glacier, the melting was accelerated by El Niño, which developed in July 2023. El Niño refers to an abnormal warming of surface waters in the equatorial Pacific Ocean and leads to warmer temperatures.

What are the impacts of glacier loss?

- Glaciers are a crucial source of freshwater, especially during hot, dry periods, for local communities, plants, and animals. Their disappearance would mean that one would have to be entirely dependent on spot rainfall for freshwater.
- The cold water that runs off glaciers keeps downstream water temperatures cooler. This is crucial for many aquatic species in the region as they need cold water temperatures to survive. Glacier loss directly impacts such species.
- Melting glaciers can also contribute to the rise in sea level. The Greenland and Antarctic ice sheets — they are also considered as glaciers — are the largest contributors to global sea level rise.

Source: Indian Express

ANTACRTICA PARLIAMENT

Context: India is hosting the 46th Antarctic Treaty Consultative Meeting (ATCM 46), also known as the Antarctic Parliament, from May 20-30 in Kochi.

Background:

- The National Centre for Polar and Ocean Research, Goa, through the Ministry of Earth Sciences (MoES), has organised the meeting, which will be attended by the 56 member countries of the Antarctic Treaty. India had last hosted the ATCM in New Delhi in 2007.

The Antarctic Treaty

- Twelve countries — Argentina, Australia, Belgium, Chile, France, Japan, New Zealand, Norway, South Africa, the USSR, the United Kingdom, and the United States — were the original signatories to the Antarctic Treaty, which was signed on December 1, 1959.
- The treaty came into force in 1961, and a total of 56 countries — including India, in 1983 — have joined it since then.
- The Antarctic Treaty, which was signed during the Cold War, effectively designated Antarctica as a "no man's land", outside the bounds of international geopolitical competition.

Few key features of the treaty are:

- Antarctica shall be used only for peaceful purposes, and no militarisation or fortification shall be allowed.
- All signatories will have the freedom to carry out scientific investigations, and should share plans for scientific programmes, extend required cooperation, and freely make available the data gathered.
- Nuclear testing or disposal of radioactive waste materials shall be prohibited anywhere in Antarctica.
- Today, the treaty forms the basis of all governance and activities in Antarctica, the fifth-largest continent on the planet.

India in Antarctica

- Since 1983, India has been a consultative party to the Antarctic Treaty. In this capacity, India votes and participates in all key decision-making processes regarding Antarctica. Of the 56 nations that are part of the Antarctic Treaty, 29 have consultative party status.
- India has undertaken scientific research in Antarctica since 1981. The first Indian Antarctica research station, Dakshin Gangotri, was set up in 1983, some 2,500 km from the South Pole in Queen Maud Land. The station operated till 1990.
- In 1989, India set up its second Antarctica research station, named Maitri, in the Schirmacher Oasis, a 3-km wide ice-free plateau with over 100 freshwater lakes. It is still operational.
- In 2012, India inaugurated Bharati, its third Antarctica research station, located around 3,000 km east of Maitri, on the Prydz Bay coast. Although the station focuses on oceanographic and geologic study, the Indian Space Research Organisation (ISRO) utilises it for receiving Indian Remote sensing Satellite (IRS) data.
- India plans to open a new station, Maitri II, a few kilometres from the ageing Maitri station. Operations are set to begin by 2029.
- In 2022, India enacted the Antarctic Act, reaffirming its commitment to the Antarctic Treaty.

Agenda before Antarctic Treaty Consultative Meeting (ATCM)

- The ATCM is meant to facilitate a global dialogue on law, logistics, governance, science, tourism, and other aspects of the southern continent.
- During the conference, India will seek to promote the idea of peaceful governance in Antarctica, and underline that geopolitical tensions elsewhere in the world should not interfere with the protection of the continent and its resources.
- **During the Kochi meet, India will also officially table its plan to construct Maitri II before the members. Any new construction or initiative in Antarctica requires the ATCM's approval.**

Source:Indian Express

SWEET SORGHUM

Context: The southern African region is battling with drought at present. The drought has hit the region's agricultural productivity hard. Malawi, Zambia and Zimbabwe have declared a state of disaster with respect to their current agricultural outputs.Sweet sorghum is a hardy, nutritious, biofuel crop that offers solutions in drought-hit southern Africa.

Background:

- Governments should be using their agriculture extension services to raise awareness among farmers, consumers about benefits & practical applications of sweet sorghum in people's diet.Sweet sorghum is viewed as a preferred alternative as it can provide both food and biofuel from the same land.

ABOUT SWEET SORGHUM

- **Sorghum (Jowar) is a large stout grass.Sweet sorghum is like any ordinary sorghum but with a high content of sweet juice in its stalks. While the juice is used to produce ethanol biofuel, the grains can be used as food or feed**
- The grains from sweet sorghum are prepared as steamed bread or porridge malt for traditional beer, as well as in commercial beer production.
- **It produces a lot of plant material (biomass) as it grows, which is left over after harvest. That's why it's useful as animal feed too.**
- Animal feed is made from what remains once the sweet sorghum crop has been harvested and its grains and stem juice stripped off. The residue is high in nutritional content, which can improve the quality of diets of animals, including cattle. The grains can also be used for animal feed.

Resilient

- One of the key characteristics of sweet sorghum varieties is their drought resistance. It allows them to enter a dormant state during extended periods of dryness and resume growth afterwards.
- Sweet sorghum's ability to withstand low water and nitrogen inputs, as well as its tolerance for salinity and drought stress, makes it an ideal crop for farmers in arid regions.

Indian Context

- The sweet sorghum varieties were first introduced in India from the US in the 1970s.
- India has achieved more than 10% blending of petrol with ethanol and the next target is 20% blending by 2025. The fact remains that ethanol production from sugarcane molasses alone does not ensure optimum supply levels needed to meet the demand at any given time owing to reasons such as the cyclical nature of sugarcane cultivation, difficulty in increasing sugarcane area due to high water intensiveness of the crop, erratic monsoon and power supply.
- Increasing the area under sugarcane at the cost of diverting land from other staple food crops is undesirable. Sweet sorghum, a widely adapted sugar crop with high potential for bioenergy and ethanol production and which produces higher biomass yield with fewer inputs is listed as a candidate crop for biofuel production in our National Policy on Biofuels 2018.

Source:Down To Earth

ALASKAN RIVERS TURN ORANGE

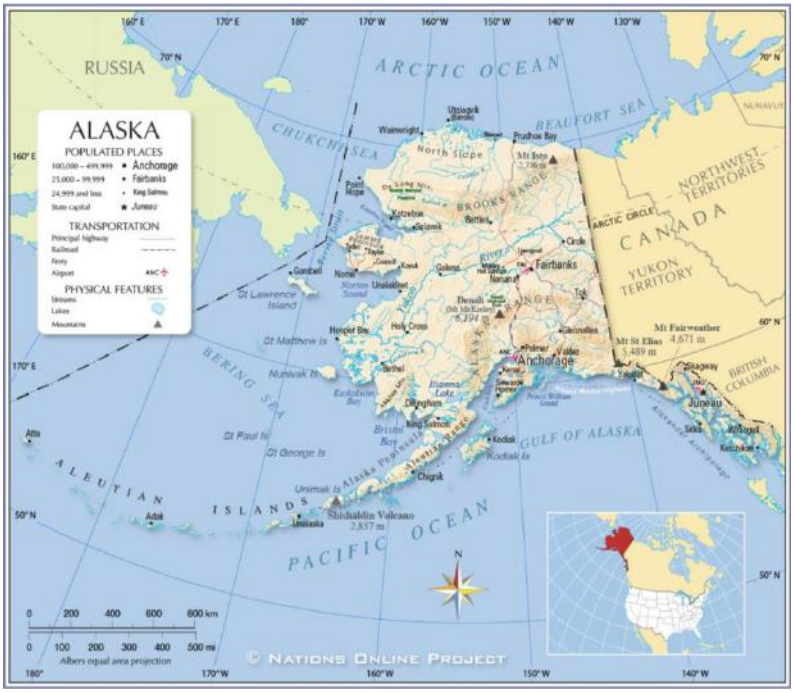
Context: Rivers and streams in Alaska are changing color – from a clean, clear blue to a rusty orange – because of the toxic metals released by thawing permafrost.

Background:

- The Arctic is warming four times faster than the rest of the world resulting in the thawing of permafrost.

Key Takeaways

- The discoloration and cloudiness are being caused by metals such as iron, zinc, copper, nickel and lead– some of which are toxic to the river and stream ecosystems – as permafrost thaws and exposes the waterways to minerals locked away underground for thousands of years.



- Arctic soils naturally contain organic carbon, nutrients and metals, such as mercury, within their permafrost. High temperatures have caused these minerals and the water sources around them to meet as permafrost melts.

Permafrost:

- Permafrost is soil or underwater sediment which continuously remains below 0 °C (32 °F) for two years or more: the oldest permafrost had been continuously frozen for around 700,000 years.While the shallowest permafrost has a vertical extent of below a meter (3 ft), the deepest is greater than 1,500 m (4,900 ft).

- Around 15% of the Northern Hemisphere or 11% of the global surface is underlain by permafrost. This includes large areas of Alaska, Canada, Greenland, and Siberia. It is also located in high mountain regions, with the Tibetan Plateau a prominent example.
- Only a minority of permafrost exists in the Southern Hemisphere, where it is consigned to mountain slopes like in the Andes of Patagonia, the Southern Alps of New Zealand, or the highest mountains of Antarctica

Alaska

- Alaska lies at the extreme northwest of the North American continent, and the Alaska Peninsula is the largest peninsula in the Western Hemisphere. Because the 180th meridian passes through the state's Aleutian Islands, Alaska's westernmost portion is in the Eastern Hemisphere. Thus, technically, Alaska is in both hemispheres.
- Alaska is bounded by the Beaufort Sea and the Arctic Ocean to the north, Canada's Yukon territory and British Columbia province to the east, the Gulf of Alaska and the Pacific Ocean to the south, the Bering Strait and the Bering Sea to the west, and the Chukchi Sea to the northwest.
- Alaska, constituent state of the United States of America. It was admitted to the union as the 49th state on January 3, 1959. The capital is Juneau, which lies in the southeast, in the panhandle region.

Source:CNN

EUCALYPTUS

Context: The Kerala government issued an order allowing the Kerala Forest Development Corporation (KFDC) to plant eucalyptus trees for its financial sustenance in 2024-2025. Environmentalists soon protested the decision saying the move would adversely affect forests and heighten human-animal conflicts in future.

Background:

- In 2021, the State government had published an eco-restoration policy. Among other things, it sought to address what it called the "proliferation of invasive species that are not suitable for our environment" and the resulting "depletion of natural forests".Environmental activists alleged following the State's order — permitting the KDFC to plant eucalyptus trees — contravened the policy's aspirations.

Key Takeaways

- **Eucalyptus is a genus of more than 700 species of flowering plants in the family Myrtaceae. Most species of Eucalyptus are native to Australia.**
- About three-quarters of Australian forests are eucalypt forests. Many eucalypt species have adapted to wildfire, are able to resprout after fire, or have seeds that survive fire.
- Eucalyptus trees are a source of timber, pulpwood, and firewood. They grow rapidly and can be harvested in short rotation cycles.
- **Introduction**: Eucalyptus was first introduced in India around 1790 by Tipu Sultan, the ruler of Mysore. He planted it in his palace garden on Nandi hills near Bangalore. According to one version, he received seed from Australia and introduced about 16 species.
- **Expansion**: Initially, eucalyptus was planted as an ornamental tree to enhance the beauty of gardens. However, its plantation on a larger scale began only in 1952.
- **Industrial Use**: In the 1950s, eucalyptus and acacia trees were planted extensively throughout the Nilgiris to supply pulp wood.Later, eucalyptus was planted extensively with financial assistance from the World Bank.

Controversies:

- Eucalyptus was targeted in the 1980s when water crisis first hit the district, though this tree was not the only causative factor. The state government decided not to go for fresh plantations in 1988 due to a Save Nilgiris Campaign.
- A study by the Kerala State Forest Protective Staff Organisation found that replacing exotic plants in forested areas with the corresponding natural species could help ensure food for wild elephants. Some of the the prime elephant habitat of the state like Chinnakanal in Munnar is filled with eucalyptus trees.

Source:Hindu

FOREST FIRE

Context: At least five people have lost their lives in the fires that have now spread over 1,000 hectares of forests in much of Uttarakhand. There is a danger of the blaze fanning out to heavily populated urban centres including Nainital city. Poor visibility has constrained the Air Force's firefighting efforts.

Background:-

- Higher forest fire incidents are reported in India during March, April and May, due to the ample availability of dry biomass following the end of winter and the prevailing summer season.

Key takeaways

- According to experts, **three factors cause the spread of forest fires** — fuel load, oxygen and temperature. Dry leaves are fuel for forest fires. **The Forest Survey of India (FSI) website states that nearly 36 per cent of India's forests are prone to frequent fires.**
- Experts believe that **burning forests intensifies heat and leads to black carbon emissions, adversely affecting water systems and air quality.**However, some of the combustion is necessary for the forest to regenerate — the burning of litter promotes the growth of fresh grasses.
- **The state government has blamed the fires on human activities and banned people from setting fire to fodder for a week**. Urban bodies have been asked to stop people from burning solid waste in and around forests. These are much-needed emergency measures.
- Severe fires occur in many forest types particularly dry deciduous forest, while evergreen, semi-evergreen and montane temperate forests are comparatively less prone.
- The monsoon was deficient in large parts of the state last year. Uttarakhand has witnessed its driest April in five years. In such moisture-deficient conditions, fire spreads fast, especially in oxygen-rich environs such as forests.

How are forest fires prevented and doused?

- **The Ministry of Environment, Forest and Climate Change (MoEFCC)** lists the following methods to prevent and control a forest fire: construction of watch towers for early detection; deployment of fire watchers; involvement of local communities, and the creation and maintenance of fire lines.
- **According to the National Disaster Management Authority (NDMA) website**, two types of fire lines are in practice – Kachha or covered fire lines and Pucca or open fire lines. In Kaccha fire lines, the undergrowth and shrubs are removed while trees are retained to decrease the fuel load.
- The Pucca fire lines are clear felled areas separating a forest/compartment/block from another to control the spread of potential fires.

Source: Indian Express

SCIENCE & TECHNOLOGY

ARTIFICIAL GENERAL INTELLIGENCE (AGI)

Context: In a recent interview, Sam Altman, CEO of OpenAI, expressed his commitment to invest billions of dollars towards the development of Artificial General Intelligence (AGI). But even as Altman continues to champion what is considered to be the pinnacle of AI development, many in the global tech community are very apprehensive.

Background:-

- AGI aims to emulate human cognitive abilities such that it allows it to do unfamiliar tasks, learn from new experiences, and apply its knowledge in new ways.

About Artificial General Intelligence (AGI)

- AGI refers to a **machine or a software that can perform any intellectual task that a human can do.** AGI aims to **emulate human cognitive abilities such that it allows it to do unfamiliar tasks, learn from new experiences, and apply its knowledge in new ways.**
- The main difference between AGI and the more common form of AI, also known as narrow AI, lies in their scope and capabilities.
- Narrow **AI is designed to perform specific tasks** such as image recognition, translation, or even playing games like chess—at which it can outdo humans, but it remains limited to its set parameters.
- On the other hand, **AGI envisions a broader, more generalised form of intelligence, not confined to any particular task (like humans).** This is what puts AGI at the summit of all developments in artificial intelligence.
- The idea of AGI first emerged in the 20th century with a paper written by Alan Turing, widely considered to be the father of theoretical computer science and artificial intelligence.
- In 'Computing Machinery and Intelligence' (1950), he introduced what is now known as the Turing test, a benchmark for machine intelligence. Simply put, if a machine can engage in a conversation with a human without being detected as a machine, according to the Turing test, it has demonstrated human intelligence.
- When Turing wrote this influential paper, humans were nowhere close to developing artificial intelligence — even computers were in their nascency. Yet, his work led to wide-ranging discussions about the possibility of such machines, as well as their potential benefits and risks.
- In theory, **AGI has innumerable positive implications in various fields such as healthcare, education, finance, and business.**
- Despite the promise AGI holds, it continues to fuel widespread apprehensions, due to a number of reasons. For instance, the humongous amount of computational power required to develop AGI systems raises concerns about its impact on the environment, both due to the energy consumption and generation of e-waste.
- AGI could also lead to a significant loss of employment, and widespread socio-economic disparity.
- It could introduce new security vulnerabilities, and its development could outrun the ability of governments and international bodies to come up with suitable regulations. And if humans were to become dependent on AGI, it might even lead to the loss of basic human skills and capabilities.

But the most serious fear regarding AGI is that its abilities can outpace human beings', making its actions difficult to understand and predict.

Source:Indian Express

OLEANDER FLOWERS

Context: Two Kerala government-controlled temple boards, which together manage 2,500-odd temples in the state, have banned use of oleander flowers (locally known as arali) in temple offerings after a 24-year old woman died after accidentally chewing some oleander leaves.

Background:-

- Oleander's toxicity has been recognised across the world.

About Oleander

- Nerium oleander, commonly known as oleander or rosebay, **is a plant cultivated worldwide in tropical, subtropical, and temperate regions.**
- Known for its **drought tolerance,** the shrub is often used for ornamental and landscaping purposes.
- In Kerala, the plant is **known by the names of arali and kanaveeram and is grown along highways and beaches as a natural, green fencing.** There are different varieties of oleander, each with a flower of a different colour

References in Ayurveda :

- **The Ayurvedic Pharmacopoeia of India (API),** a government document that describes the quality, purity, and strength of drugs used in Ayurveda, mentions oleander. According to API, **an oil prepared from the root bark can be used to treat skin diseases.**
- **Charka [Charak Samhita]** has prescribed the **leaves of white-flowered variety externally in chronic and obstinate skin diseases of serious nature including leprosy.**
- **Bhavaprakasha has described Karvira [another name of the plant] as a visha (poison**) and indicated it in treatment of vrana (infected wounds), kustha (skin diseases including leprosy), krimi (microbes and parasites), kandu (itching) etc
- **Ingestion or inhalation of smoke from burning oleander can be intoxicating.** This is due to the properties of cardiac glycosides (a type of chemical) including oleandrin, folinerin, and digitoxigenin, which are present in all parts of the plant.
- **Cardiac glycosides are steroidal compounds capable of exerting pharmacological effects on cardiac muscle.** The primary therapeutic value of these glycosides lies in their ability to exert profound tonic effects on the heart [stronger and faster heart contractions].

Source:Indian Express

BIO HACKING

Context: Recently, the Biohacking India 2.0 event took place in New Delhi.

Background:

- This event provided a platform for discussions, knowledge sharing, and innovative ideas related to enhancing human health and performance through cutting-edge technology and lifestyle modifications.

About BIOHACKING

- Biohacking **refers to the practice of using science and technology to optimize and enhance the human body's performance and well-being.**

- It is a broad term that **encompasses a variety of practices,** from simple lifestyle and dietary changes to the use of advanced technology and biometrics.
- **It includes practices aimed at improving physical and mental performance,** longevity, and overall well-being.
- Currently, there are no laws in India that specifically address biohacking.

Techniques:

- **Dietary Modifications**: Implementing diets like ketogenic or intermittent fasting to boost energy and health.
- **Supplementation**: Using vitamins, minerals, or nootropics to enhance cognitive function and physical performance.
- **Physical Training**: Customized exercise routines to build strength and endurance.
- Sleep Optimization: Techniques to improve sleep quality and duration.
- **Mindfulness Practices:** Meditation and deep breathing exercises for mental well-being.
- **Biometric Monitoring**: Using devices to track health metrics for insights into health and performance.
- **Biofeedback Devices**: Devices to train the body's physiological responses.

Source: Livemint

OXYTOCIN

Context: The Delhi High Court has taken a strong stance against the spurious use of oxytocin on cattle in dairy colonies within the capital.

Background:

- The court has directed the Delhi government's Department of Drugs Control to conduct weekly inspections and register cases against the administration of oxytocin.

About OXYTOCIN:

- Oxytocin, often referred to as the **"love hormone," is a crucial hormone secreted by the pituitary glands in mammals during various physiological processes.**
- Oxytocin is a hormone **produced in the brain's hypothalamus and secreted by the pituitary gland.**
- It plays dual roles as **both a hormone and a neurotransmitter.**

Functions:

- **Childbirth**: Oxytocin helps contract the uterus during labour, facilitating childbirth.
- **Breastfeeding**: It promotes the release of breast milk.
- **Social Bonding**: Oxytocin is associated with social interactions, trust, and emotional bonding.

Controversies and Bans:

- **Misuse in Dairy Industry:** Oxytocin misuse occurs in the dairy sector, where livestock (milch cattle) is injected with the hormone to increase milk production.
- **Human Health Concerns**: The same hormone, when consumed through milk, can affect human health.

- **Government Ban**: In April 2018, the Indian government banned the use of oxytocin on milch cattle due to these concerns. The administration of oxytocin is considered as an "animal cruelty" and a cognizable offense under Section 12 of the Prevention of Cruelty to Animals Act, 1960.

Alternative Solution:

- The World Health Organization (WHO) **recommends carbetocin** as a safe and effective alternative to oxytocin, because carbetocin does not require refrigeration and remains effective even at higher temperatures.
- It **retains efficacy for at least three years** when stored at 30°C and 75% relative humidity.

Source: Times of India

55 CANCRI E

Context:A thick atmosphere has been detected around a planet that's twice as big as Earth in a nearby solar system, researchers reported Wednesday.The so-called super Earth — known as 55 Cancri e — is among the few rocky planets outside our solar system with a significant atmosphere, wrapped a blanket of carbon dioxide and carbon monoxide.

Background:

- Super Earth refers to a planet's size — bigger than Earth but smaller than Neptune. The boiling temperatures on this planet — which can reach as hot as 4,200 degrees Fahrenheit – mean that it is unlikely to host life.

About 55 Cancri e:

- **55 Cancri e, also known as Janssen**, is an exoplanet that orbits a Sun-like host star, 55 Cancri A.
- The planet is located in our Milky Way galaxy about 41 light-years from Earth, in the constellation Cancer. Four other planets, all gas giants, are known to orbit its host star.
- It was discovered on August 30, 2004**. It was the first super-Earth discovered around a main sequence star.**
- The mass of the exoplanet is about **eight Earth masses and its diameter is about twice that of the Earth.**
- Temperature: Due to its proximity to its star, 55 Cancri e is extremely hot, with temperatures on the day side exceeding 3,000 Kelvin.
- Infrared observations using two instruments aboard the James Webb Space Telescope indicated the presence of a substantial - if inhospitable - atmosphere, perhaps continuously replenished by gases released from a vast ocean of magma.
- Being so close to its star, any atmosphere should be stripped away by stellar irradiation and winds. But gases dissolved in the vast lava ocean thought to cover the planet may keep bubbling up to replenish the atmosphere
- All of the previous exoplanets found to have atmospheres were gaseous planets, not rocky ones. As Webb pushes the frontiers of exoplanet exploration, the discovery of a rocky one with an atmosphere represents progress.

What is an exoplanet?

- An exoplanet is any **planet beyond our solar system.** Most orbit other stars, but free-floating exoplanets, called rogue planets, orbit the galactic center and are untethered to any star.

Source: Hindu

INTERACTIVE VOICE RESPONSE SYSTEM (IVRS)

Context: Recently, voters are receiving Interactive Voice Response System (IVRS) calls every day and they want the Election Commission of India (ECI) should order an inquiry as to how the service providers of various telecom firms shared our mobile phone numbers with the political parties.

Background:

- The calls violate the Telecom Regulatory Authority of India (TRAI) Act, 1997. Lakhs of voters are affected daily by these IVRS calls. It's essential for the ECI to address this issue promptly to protect voters' privacy and ensure a fair electoral process.

About INTERACTIVE VOICE RESPONSE SYSTEM (IVRS):

- **An IVRS is an automated telephony technology** that allows a computer to interact with humans through the use of voice and DTMF tones input via a keypad.
- IVRS is commonly **used in call centers to manage large volumes of call**s and provide callers with self-service options.
- **Benefits:** improved customer service, reduced wait times, and lower operational costs for businesses.
- Modern IVRS can integrate with various networks, including public switched telephone networks (PSTN) and voice over IP (VoIP), and support multiple applications written in VoiceXML.

Political Campaigns and IVRS:

- During elections, political parties use IVRS to reach out to voters.
- Voters receive pre-recorded messages from candidates or party leaders, urging them to vote or highlighting campaign promises.
- These messages can be informative or persuasive, depending on the party's strategy.

Source: Hindu

FLiRT

Context: The new coronavirus variant called KP.2 — nicknamed FLiRT — that has been linked to rising cases of Covid-19 in the United States, United Kingdom, and South Korea, has been in circulation in India since November 2023, genomic surveillance data show.

Background:

- About 250 KP.2 sequences have been reported so far by INSACOG, the country's genome sequencing consortium.

Key Takeaways

- KP.2 is a descendant of the JN.1 variant of the virus. It is a sub-variant of the Omicron lineage with new mutations.
- **FLiRT, the nickname of KP.2, is based on the letters representing two immune escape mutations that allow the virus to evade antibodies.**
- FLiRT is characterised by its **ability to evade immunity from vaccines and previous infections.** Its symptoms are similar to those of earlier variants, including fever, cough, fatigue, and digestive issues.
- **The US Center for Disease Control and Prevention (CDC)** notes that there are currently no indicators suggesting that KP.2 would cause more severe illness than other strains.

- However, **FLiRT has a heightened transmission rate and, like its parent JN.1, it is likely to drive a wave of infections. Also, the infections are likely to spread silently — because without severe symptoms, most people are unlikely to get themselves tested.**

India's Case

- India has been reporting the highest proportion of KP.2 sequences in the world, global data show.
- KP.2 sequences made up 29% of Covid-19 sequences uploaded by India to the Global Initiative on Sharing All Influenza Data (GISAID), **the world's largest repository of these sequences, over the last 60 days.**
- However, JN.1 continues to be the dominant variant of SARS-CoV-2 in the country.

Source:Indian Express

MALARIA VACCINE

Context: Serum Institute of India (SII), the world's largest manufacturer of vaccines by number of doses, marked a significant milestone on Monday with the shipment of the first set of R21/Matrix-M malaria vaccine to seven-eight countries in Africa.

Background:

- Close to half-a-million children die of malaria each year in the African region, according to the World Health Organisation (WHO).

About MALARIA:

- Malaria is a **mosquito-borne infectious disease caused by parasites of the genus Plasmodium.** Five species of Plasmodium commonly infect humans. The three species associated with more severe cases are P. falciparum (which is responsible for the vast majority of malaria deaths), P. vivax, and P. Knowlesi.
- The disease is widespread in the tropical and subtropical regions that exist in a broad band around the equator. This includes much of sub-Saharan Africa, Asia, and Latin America.
- **Transmission:** It is spread exclusively through bites of infected female Anopheles mosquitoes. When an infected mosquito bites a person, it injects the malaria parasites into their bloodstream.
- **Symptoms:** Common symptoms include fever, chills, headache, muscle aches, and fatigue. Severe cases can lead to organ failure, coma, and death.

Prevention:

- **Antimalarial Medications:** Prophylactic antimalarial drugs can be taken by travelers to endemic areas.
- **Vaccines:** Recently, the WHO recommended the R21/Matrix-M vaccine for children, in addition to the RTS,S/AS01 vaccine. These vaccines aim to reduce malaria cases and deaths.
- **Treatment:** Antimalarial medications, such as artemisinin-based combination therapies (ACTs), are used to treat malaria. Early diagnosis and prompt treatment are crucial.

R21/Matrix-M malaria vaccine

- The R21/Matrix-M vaccine is a promising addition to malaria prevention efforts. It's designed to protect against the Plasmodium falciparum parasite, which causes the most severe form of malaria.
- Composition: The R21 vaccine contains a protein from the malaria parasite (R21) combined with an adjuvant called Matrix-M. The adjuvant enhances the immune response.
- Age Group: The vaccine is recommended for children aged 5 months and older.

- Implementation: The WHO's recommendation aims to accelerate the vaccine's deployment in malaria-endemic regions, particularly sub-Saharan Africa.
- The low-cost, high-efficacy R21/Matrix-M malaria vaccine was developed by the University of Oxford and SII leveraging Novavax's adjuvant technology.
- R21//Matrix-M vaccine is the second malaria vaccine to be authorised for use in children in malaria-endemic regions.

Source: Indian Express

PREFIRE

Context: On May 25, the National Aeronautics and Space Administration (NASA) launched one of the two climate satellites, which would study heat emissions at Earth's poles. The second satellite will be launched in the following days.

Background:-

- The two shoebox-sized cube satellites, or CubeSats, will measure how much heat the Arctic and Antarctica — two of the coldest regions on the Earth — radiate into space and how this influences the planet's climate. **The mission has been named PREFIRE (Polar Radiant Energy in the Far-InfraRed Experiment)** and was jointly developed by NASA and the University of Wisconsin-Madison (US).

what are CubeSats?

- **CubeSats are essentially miniature satellites** whose basic design is a 10 cm x 10 cm x 10 cm (which makes up for "one unit" or "1U") cube — just a little bigger than a Rubik's cube — and weight not more than 1.33 kg. Depending on the CubeSat's mission, the number of units can be 1.5, 2, 3, 6, and 12U, according to NASA.
- Each of the PREFIRE satellites is a 6U CubeSat. They measure around 90 cm in height and nearly 120 cm in width when the solar panels, which will power the satellite, are deployed. The two satellites will be placed in a near-polar orbit (a type of low Earth orbit) at an altitude of about 525 kilometres.

Why do researchers want to measure heat emissions at Earth's poles?

- It has to do with the Earth's energy budget, which is the balance between the amount of heat incoming to Earth from the Sun and the amount of heat outgoing from Earth into space. The difference between the two determines the planet's temperature and climate.
- A large amount of the heat radiated from the Arctic and Antarctica is emitted as far-infrared radiation — wavelengths of 3 µm to 1,000 µm within the infrared range of electromagnetic radiation. However, there is currently no way to measure this type of energy. As a result, there is a gap in knowledge about the planet's energy budget.

What is the PREFIRE (Polar Radiant Energy in the Far-InfraRed Experiment) mission?

- **The two CubeSats can study far-infrared radiation from the Earth's pole** and the data collected by them would help scientists better understand the energy budget of the planet.
- The observations will **help understand the fundamentals of Earth's heat balance**, allowing to better predict how our ice, seas, and weather will change in the face of global warming.
- Each of the **PREFIRE CubeSat is equipped with a thermal infrared spectrometer — known as Thermal Infrared Spectrometer (TIRS**) — to measure the amount of infrared and far-infrared radiation from the Arctic and Antarctica. The spectrometer features specially shaped-mirrors and detectors for splitting and measuring infrared light, according to NASA.

- The CubeSats will also measure the **amount of far-infrared radiation trapped by atmospheric water vapour and clouds at the poles and how this influences the greenhouse effect in the region**.

Source: Indian Express

CARBON FIBER

Context: Vice President recently visited National Aerospace Limited (NAL) facilities and laid the foundation stone of the Centre for Carbon Fiber and Prepregs in Bengaluru.

Background:

- Carbon fiber can have higher cost than other materials which has been one of the limiting factors of adoption.

Key Takeaways

- Carbon fiber, **also known as graphite fiber or carbon graphite**, is a material consisting of thin, strong crystalline filaments of carbon. These fibers are about 5 to 10 micrometers in diameter and composed mostly of carbon atoms.
- **The properties of carbon fibers include**: High stiffness, High tensile strength, High strength-to-weight ratio, High chemical resistance, High-temperature tolerance, Low thermal expansion
- **These properties make carbon fiber popular in various fields such as** aerospace, civil engineering, military, motorsports, and other competition sports. However, they are relatively expensive compared to similar fibers, such as glass fibers, basalt fibers, or plastic fibers.
- Carbon fibers are usually combined with other materials to form a composite. For example, when permeated with a plastic resin and baked, it forms carbon-fiber-reinforced polymer which has a very high strength-to-weight ratio and is extremely rigid although somewhat brittle. Carbon fibers are also composited with other materials, such as graphite, to form reinforced carbon-carbon composites, which have a very high heat tolerance.
- Carbon fiber-reinforced materials are used to make aircraft and spacecraft parts, racing car bodies, golf club shafts, bicycle frames, fishing rods, automobile springs, sailboat masts, and many other components where light weight and high strength are needed.

Source:Hindu

GOLDEN RICE

Context: A court in the Philippines recently revoked biosafety permits for commercial propagation of genetically modified golden rice and Bt eggplant.

Background:

- The decision is circulating in the international media, as proponents of GM crops have claimed the decision may have disastrous consequences for children with vitamin A deficiency, ignoring the safety violations noted by the court.

About Golden Rice

- Golden Rice is a **new type of rice that contains beta carotene (provitamin A**, a plant pigment that the body converts into vitamin A as needed). **This compound is what gives this grain its yellow-orange or golden color, hence its name.**
- Golden Rice is **developed through genetic engineering. While ordinary rice does produce beta carotene, it is not found in the grain.** Thus, scientists used genetic engineering to add the compound to the grain - a minor tweak that improved the grain's nutritive value.

- **The beta carotene in Golden Rice, which was made possible by the addition of two new enzymes, is identical to the beta-carotene found in green leafy and yellow-colored vegetables, orange-colored fruit, and even in many vitamin supplements and food ingredients.**
- Like ordinary rice, **Golden Rice does not require any special cultivation practices**, and generally has the same yield and agronomic performance.
- While vitamin A can be obtained from food products and supplements, challenges regarding their availability, accessibility, and affordability make it difficult to address the problem of vitamin A deficiency (VAD).

Source:International Rice Research Institute

DJIBOUTI RELEASES GM MOSQUITOES TO FIGHT MALARIA

Context: Genetically modified (GMO) mosquitoes were released in Djibouti, East Africa on May 23, 2024 to fight malaria.

Background:

- This is the first time a genetically engineered mosquito has been released in East Africa and the second time on the African continent. Burkina Faso in West Africa conducted the first ever release of genetically modified mosquitoes in Africa

Key Takeaways

- The release is part of the **'Djibouti Friendly Mosquito Programme'** started two years ago to stop the spread of Anopheles stephensi, an invasive species of mosquito.
- Unlike other malaria-carrying mosquitoes in Africa that primarily breed in rural areas, **Anopheles stephensi is well adapted to the urban environment.** This characteric is particularly dangerous for Djibouti, where 70 per cent of the population live in the capital city and are exposed to the malaria vector. It **bites both during the day and at night and is resistant to chemical insecticides**.
- To combat these urban invaders, Oxitec, a biotechnology company developed a method that uses mosquitoes to fight mosquitoes. They have released **genetically altered male mosquitoes carrying a special gene that prevents their female offspring from reaching adulthood**.
- The laboratory-produced mosquitoes carry a **"self-limiting" gene that prevents female mosquito offspring from surviving to adulthood when they mate.**
- Only their male offspring survive but would eventually die out, according to the scientists behind the project.
- The method targets female mosquitoes, which areresponsible for malaria transmission. By reducing the female mosquito population, scientists hope to significantly decrease the spread of the disease. Male mosquitoes do not bite and therefore cannot transmit malaria.

About Malaria

- **Malaria is a disease caused by a parasite.** The parasite is spread to humans through the bites of infected mosquitoes.
- **Anopheles mosquitoes are the type of mosquito that transmit malaria from one person to another.** Not all Anopheles mosquitoes have malaria, but if they bite a person with malaria, they can become infectious. Once they bite another person, this continues the cycle of spreading malaria from mosquito to people.
- There are 5 Plasmodium parasite species that cause malaria in humans and 2 of these species – P. falciparum and P. vivax – pose the greatest threat.

Source:Down To Earth

CONFLICT MINERALS

Context: Recently, the Democratic Republic of the Congo (DRC) has accused Apple of using conflict minerals.

Background:

- The DRC accuses Apple of purchasing minerals that have been smuggled out of the country into neighbouring Rwanda. These minerals are then integrated into Apple's global technology supply chain, where their origin is allegedly obscured.

About CONFLICT MINERALS:

- Conflict minerals**, also known as "blood minerals,"** are extracted from regions affected by armed conflict or human rights abuses and play a significant role in **funding violence and wars**.
- These minerals play a crucial role in various industries, including electronics, automotive, and renewable energy.
- The DRC is rich in untapped reserves of precious metals and minerals.
- The primary conflict minerals include: Tin , Tantalum, Tungsten, Gold. These minerals are **extracted from ores such as** cassiterite (tin), columbite-tantalite (tantalum), and wolframite (tungsten).

Impact of Conflict Minerals:In politically unstable regions, the trade in these minerals can:

- **Finance armed groups**: The proceeds from mineral sales may directly fund armed conflict.
- **Fuel forced labour**: Miners, including children, may be subjected to exploitative working conditions.
- **Support corruption and money laundering**: Illicit trade can perpetuate corruption and criminal activities.

Source: Down To Earth



HISTORY AND ART & CULTURE



MAY DAY OR LABOUR DAY

Context: Politicians, celebrities and members of civil society came together on Wednesday to celebrate Labour Day and recognise the contributions of workers.

Background:

- Also known as May Day, **May 1 is celebrated to honour the working class and commemorate their struggles for fair wages and better working conditions.**On the occasion, the International Labour Organisation (ILO) emphasised the importance of renewing commitment to social justice.

What is the history and significance of May Day?

- The origins of May Day can be traced back to the late 19th century, specifically to the 1886 Haymarket Massacre in Chicago.
- At the time, several countries were reeling under the heat of industrialisation. Crop failure in the 1840s had led to widespread anti-feudal upheavals collectively called 'The Revolutions of 1848,' with Karl Marx's The Communist Manifesto leaving a lasting impact on workers.
- The subsequent events resulted in the birth of an umbrella body for socialists and communist organisations in 1864, known as the First International.
- Thousands of blue-collar workers in the industrial city of Chicago in the United States went on a massive strike to demand better working conditions and a reduction in their work hours from 12-15 hours a day to eight hours daily. The movement spread throughout the country, rallying tens of thousands of workers.
- On May 4, 1886, a bomb went off at a protest rally at the Haymarket Square in Chicago and ended in a riot as workers clashed with police. At least 11 people were killed. The incident, which came to be known as the Haymarket massacre, is referred to as the origin of May 1 being celebrated as Workers' Day.
- The Second International that emerged in 1889 declared May 1 as International Workers' Day to commemorate the struggle for labour rights and as a tribute to those killed in Chicago.
- On June 28, 1894, U.S. President Grover Cleveland signed a bill into law declaring the first Monday in September as Labour Day and a national holiday.
- As for India, Labour Day was first celebrated on May 1, 1923, in Chennai (then known as Madras). Two meetings were held— one on the Triplicane Beach and another near the Madras High Court. These were organised by the Labour Kisan Party of Hindustan. The party was founded by Malayapuram Singaravelu Chettiar, who was one of the founders of the Communist Party of India (CPI).
- It was at these meetings that a resolution was passed urging the British government to declare May 1 as Labour Day and a government holiday.
- Interestingly, the red flag that is synonymous with the Left movement was used for the first time during the Chennai event.

Source: Hindu

CHAR DHAM YATRA

Context: With nearly twice the number of pilgrims embarking on the Char Dham yatra in Uttarakhand since it commenced on May 10, the state government has reported the death of 11 pilgrims in the first five days.

Background:

- The pilgrimage to Yamunotri, Gangotri, and Kedarnath Dham commenced on May 10, while the doors to Badrinath Dham opened on May 12. Over 1.55 lakh pilgrims have already journeyed to Kedarnath, over 70,000 to Yamunotri, and over 63,000 to Gangotri. Within just three days, Badrinath Dham has welcomed over 45,000 pilgrims.

About CHAR DHAM YATRA

- **The Chota Char Dham Yatra is a significant pilgrimage in India, particularly in the state of Uttarakhand, also known as Devbhoomi or the Land of Gods**. The term 'Char Dham' in Hindi translates to 'four religious destinations'.
- The pilgrimage sites are Yamunotri and Gangotri in Uttarkashi district, Kedarnath in Rudraprayag and Badrinath in Chamoli. While Kedarnath is the highest at around 11,700 ft, Gangotri is the lowest at around 10,200 ft.
    1. **Yamunotri**: The Yamunotri temple is lodged in a narrow gorge close to the source of River Yamuna, the second-most sacred Indian river after River Ganga, in Uttarkashi district. It is dedicated to Goddess Yamuna.
    2. **Gangotri**: Located in the Uttarkashi district, Gangotri is dedicated to Goddess Ganga, the most sacred of all Indian rivers.
    3. **Kedarnath**: Located in the Rudraprayag district, Kedarnath is dedicated to Lord Shiva.
    4. **Badrinath**: Home to the sacred Badrinarayan Temple, Badrinath is dedicated to Lord Vishnu.
- The Yatra is believed to be completed in a clockwise direction, starting from Yamunotri, proceeding towards Gangotri, onto Kedarnath, and finally ending at Badrinath.
- The high-altitude shrines remain shut for around six months every year, opening in summers (April or May) and closing with the onset of winter (October or November). Some devotees even do a Do Dham Yatra or a pilgrimage to two shrines - Kedarnath and Badrinath.

Additional Information:

- The Char Dham Yatra in India comprises Chota Char Dham and the Bada Char Dham. While the Chota Char Dham involves four sacred sites—Yamunotri, Gangotri, Kedarnath, and Badrinath—nestled in the picturesque Himalayan peaks, the Bada Char Dham consists of four dhams of utmost significance—Badrinath, Rameswaram, Puri, and Dwarka.

Source:Indian Express

HAMPI VIRUPAKSHA TEMPLE

Context: A portion of the Virupaksha temple in Karnataka collapsed following torrential rains. Located at the historical site of Hampi, the temple's pavilion or the saalu mantap was damaged, leading some conservationists to blame Archaeological Survey of India (ASI) officials for allegedly neglecting its protection and conservation.

Background:

- A nationally protected monument, the temple is believed to have its origins in the 7th century.

Key Takeaways

- Virupaksha Temple is located in Hampi in the Vijayanagara district of Karnataka, India. It is part of the Group of Monuments at Hampi, designated as a UNESCO World Heritage Site.
- The temple is dedicated to Lord Shiva, **known here as Virupaksha/Pampa pathi**, as the consort of the local goddess Pampadevi who is associated with the Tungabhadra River.
- Despite legends about its origins going further back, the **Virupaksha temple gained prominence and underwent extensive expansion in the 14th century during the Vijayanagara Empire (1336 to 1646).**
- **Founded by Harihara I of the Sangama dynasty,** the Vijayanagara empire expanded from a strategic position on the banks of the Tungabhadra river to become one of the most powerful kingdoms of its time.
- The temple flourished under the patronage of the Vijayanagara rulers, who were great builders and patrons of art. It became a vital centre for the religious and cultural activities of its time.
- It is a prime example of **Dravidian temple architecture**, characterised by its grand gopurams (towering gateways), the shikhara towering over the sanctum sanctorum, its intricate carvings and pillared halls. Richly adorned with carvings and sculptures, the gopuram depicts various deities, mythological scenes and animals.
- Virupaksha Temple complex is surrounded by three gopuras (towers). The main tower, on the east is an imposing structure, 9 stories, 50 meters tall, built in the fifteenth century. Eastern tower forms the main entrance to Virupaksha Temple. Eastern tower has extensive craftsmanship on each of its floors featuring hundreds of Hindu gods and goddesses. The inverted shadow of the main tower falls on a wall inside the temple.
- With several other temples and structures located there, Hampi was the empire's capital city. Hampi sits on the banks of the **Tungabhadra River**

Source: Indian Express

PATACHITRA PAINTING

Context: The first generation women patachitra artists of West Bengal sell their work online & are recognised world over, encouraging future generations to stay in the profession

Background:

- Patachitra or Pattachitra is a general term for traditional, cloth-**based scroll painting, based in the eastern Indian states of Odisha,West Bengal and parts of Bangladesh.** Patachitra artform is known for its intricate details as well as mythological narratives and folktales inscribed in it.

About Patachitra Painting :

- It dates back to the 12th century and its origins can be traced to the **modern-day state of Odisha**.
- Pattachitra is one of the **ancient artworks of Odisha, originally created for ritual use and as souvenirs for pilgrims to Puri, as well as other temples in Odisha**
- The word 'patachitra' is a **combination of two words in the Sanskrit language, where 'pata' means 'cloth' and 'chitra' means 'picture'.**
- It is a narrative scroll art and a form of storytelling, focusing on **themes of mythology and culture**.
- **It is difficult to determine the exact year when Patachitra as an art form reached Bengal but art historians believe that it occurred approximately three centuries ago.** When people migrated from Odisha to Bengal back then, they brought the tradition of Patachitra with them. In their native

homeland, they would worship Durga on Patachitra and carried this form of worship when they travelled to Bengal.

Source: Down To Earth

DEFENCE& SECURITY.

CHIEF OF DEFENCE STAFF (CDS)

Context: Media reports suggest that the complex and contentious policy issue of rewiring the Indian military into integrated theatre commands (ITC) is gaining traction and that the armed forces are looking at the appointment of a Vice Chief of Defence Staff and a Deputy Chief of Defence Staff.

Background:

- New posts at the higher level are being envisaged to enable the CDS to realise the larger objective of enhancing jointness (the sharing of domain expertise and assets) and improving composite combat capability.

About CDS:

- **It is the military head and permanent Chairman of the Chiefs of Staff Committee (COSC) of the Indian Armed Forces.**
- It is the **highest-ranking uniformed officer on active duty** in the Indian military and chief military adviser to the Minister of Defence.
- The Chief also **heads the Department of Military Affairs.**
- The first Chief of Defence Staff, General Bipin Rawat, took office on 1 January 2020
- The position was **created with the aim of improving coordination, tri-service effectiveness and overall integration of the combat capabilities of the Indian armed forces.**
- The government amended Service Rules of the Army, Navy and Air Force, allowing retired Service Chiefs and three-star officers eligible for consideration for the country's top military post.
- Age limit that the retired officer should not have attained 62 years on the date of appointment.

Source: Indian Express

PROJECT UDBHAV

Context: Army chief Gen Manoj Pande talked about Project Udbhav at a conference titled 'Historical Patterns in Indian Strategic Culture' held recently.

Background:

- Launched last year, this project aims to enrich the nation's outlook in the defence domain by integrating India's ancient strategic acumen into contemporary military practices, making the force "future-ready".

About Project UDBHAV

- 'Udbhav', which translates to 'origin' or 'genesis', acknowledges the vintage scriptures and writings of our Nation, that span centuries in the past and contain profound knowledge that can benefit modern military strategies.

- **Objective:** synthesize ancient wisdom with **contemporary military practices, forging a unique and holistic approach to address modern security challenges.**
- It is a visionary initiative by the Indian Army that **seeks to integrate age-old wisdom with contemporary military pedagogy.**
- Project UDBHAV will facilitate **in-depth understanding of our ancient knowledge systems and philosophies and also aim to comprehend their enduring connect, relevance and applicability in the modern day.**
- Literature like Chanakya's Arthashastra underscores the importance of strategic partnerships, alliances and diplomacy, aligning with modern military practices such as international cooperation and soft power projection. Chanakya's teachings on statecraft and warfare are studied by various institutions the world over.
- Similarly, the wisdom of Thirukkural, the classical Tamil text authored by Thiruvalluvar, the Tamil philosopher, advocates ethical conduct in all endeavours, including warfare. This aligns with modern military codes of ethics of just war and principles of Geneva Convention.
- There are examples of the Ahom Kingdom too, which successfully ruled for 600 years, repeatedly defeating the Mughals.
- The Naval Battle of Saraighat in 1671, led by Lachit Borphukan, stands as a stellar example of the use of clever diplomatic negotiations to buy time, employ psychological warfare, focus on military intelligence and exploiting the strategic weakness of the Mughals.
- **Project UDBHAV aims to effectively integrate this ancient wisdom with modern military pedagogy through interdisciplinary research, workshops and leadership seminars.** It will facilitate emergence of previously under-explored thoughts and theories related to strategic thinking, statecraft and warfare, foster deeper understanding and contribute to enriching military training curricula.

Source:PIB

EXERCISE SHAKTI

Context: The 7th edition of India- France Joint Military Exercise SHAKTI commenced recently at Umroi, in a fully developed and modern Foreign Training Node in Meghalaya.

Background:

- The Exercise is scheduled to be conducted from 13th to 26th May 2024.

ABOUT EXERCISE SHAKTI:

- Exercise Shakti is a **bilateral joint military exercise** between the armies **of India and France**. It is a **biennial training event** conducted alternatively in India and France. Last edition was conducted in France in November 2021.
- The exercise started in 2011, and since then, it has been undertaken to **promote defence cooperation and interoperability.**

7th edition of Exercise Shakti

- The latest**, 7th edition** of Exercise Shakti commenced on Monday, May 13, 2024, at Umroi, Meghalaya.
- The aim of this exercise is to **enhance the joint military capability of both sides to undertake multi-domain operations in a sub-conventional scenario.**

- The joint exercise focuses on **operations in semi-urban and mountainous terrain.**
- Objectives include achieving a high degree of physical fitness, rehearsing and refining drills for operations at a tactical level, and sharing best practices.
- Tactical drills practiced during the exercise include response to a terrorist action of capturing a defined territory, establishment of a Joint Command Post, establishment of an Intelligence & Surveillance Centre, securing of a helipad/ landing site, Small Team Insertion & Extraction, Special Heliborne Operations, Cordon & Search Operations, and employment of drones and counter-drone systems.
- Exercise Shakti **enables the two sides to share their best practices in Tactics, Techniques, and Procedures of conducting joint operations**. This joint exercise fosters inter-operability, bonhomie, and camaraderie between armed forces personnel of the two countries. It also enhances the level of defence cooperation, further fostering bilateral relations between the two friendly nations.

Source: Times of India

ASSAM RIFLES

Context: The Assam Rifles successfully intercepted a substantial cache of arms, ammunition, and other war-like supplies near the Indo-Myanmar border in Mon district, Nagaland.

Background:

- This significant recovery of heavy-caliber, military-grade weapons close to the border area is a commendable achievement for the ongoing border sealing operation by the Assam Rifles.

About ASSAM RIFLES :

- The Assam Rifles, **also known as the Sentinels of North East**, is the oldest paramilitary force in India.

Formation and Evolution:

- The Assam Rifles traces its **lineage back to the Cachar Levy**, which was formed in 1835 during British rule. Over time, it **underwent several name changes, including the Assam Frontier Police, Assam Military Police, and Eastern Bengal and Assam Military Police, before officially becoming the Assam Rifles in 1917.**
- It served in both World Wars and has a rich historical legacy.

Role and Responsibilities:

- **Counter-Insurgency Force**: The Assam Rifles plays a crucial role in counter-insurgency operations in the challenging terrain of the Northeastern states.
- **Law and Order**: It assists in maintaining law and order in the region, working alongside the Indian Army.
- **Indo-Myanmar Border Security**: The force guards the Indo-Myanmar border, ensuring security and preventing illegal activities.
- **Sentinels of the Northeast**: The Assam Rifles is aptly called the “Friends of the Hill People” due to its close interaction with the local communities.

Dual Control Structure:

- **The Assam Rifles is unique in having a dual control structure:**
    - Administrative Control: Lies with the Ministry of Home Affairs (MHA).
    - Operational Control: Rests with the Indian Army, which falls under the Ministry of Defence.

- This dual control arrangement ensures effective coordination and synergy between civil administration and military operations.

Source: PIB

GOVT. INITIATIVES, SCHEMES AND POLICIES, ORGANISATION

THE 'MUSLIM QUOTA' QUESTION

Context: In election season, India is debating fundamental constitutional questions around reservation. Can a secular country like India have religion-based reservation? Have Muslims ever been given reservation by reducing the quota for Scheduled Castes (SCs), Scheduled Tribes (STs), or Other Backward Classes (OBCs)?

Background:

- **The Justice Rajinder Sachar Committee (2006)** found that the Muslim community as a whole was almost as backward as SCs and STs, and more backward than non-Muslim OBCs. The Justice Ranganath Misra Committee (2007) suggested 15% reservation for minorities, including 10% for Muslims.

Key takeaways :

- **The Supreme Court has held that equality is** a dynamic concept with many aspects and dimensions, and it cannot be "cribbed, cabined and confined" within traditional and doctrinaire limits (E P Royappa vs State Of Tamil Nadu, 1973).
- **Formal equality is concerned with equality of treatment** — treating everyone the same, regardless of outcomes — which can at times lead to serious inequalities for historically disadvantaged groups. **Substantive equality, on the other hand,** is concerned with equality of outcomes. Affirmative action promotes this idea of substantive equality.
- **The first constitutional amendment inserted Article 15(4)**, which empowered the state to make "any special provision for the advancement of any socially and educationally backward classes of citizens or for the Scheduled Castes and the Scheduled Tribes".
- **Article 15 specifically prohibits the state from discriminating against citizens on grounds only** of both religion and caste (along with sex, race, and place of birth). After the Supreme Court's judgment in State of Kerala vs N M Thomas (1975), reservation is considered not an exception to the equality/ non-discrimination clauses of Articles 15(1) and 16(1), but as an extension of equality.
- **The crucial word in Articles 15 and 16 is 'only'** — which implies that if a religious, racial, or caste group constitutes a "weaker section" under Article 46, or constitutes a backward class, it would be entitled to special provisions for its advancement.
- Some Muslim castes were given reservation not because they were Muslims, but because these castes were included within the backward class, and reservation was given without reducing the quota for SCs, STs, and OBCs by creating a sub-quota within the OBCs.
- **The Mandal Commission**, following the example set by several states, included a number of Muslim castes in the list of OBCs.

- The Supreme Court in Indra Sawhney (1992) laid down that any social group, whatever its mark of identity, if found to be backward under the same criteria as others, will be entitled to be treated as a backward class.

Source: Indian Express

MISCELLANEOUS

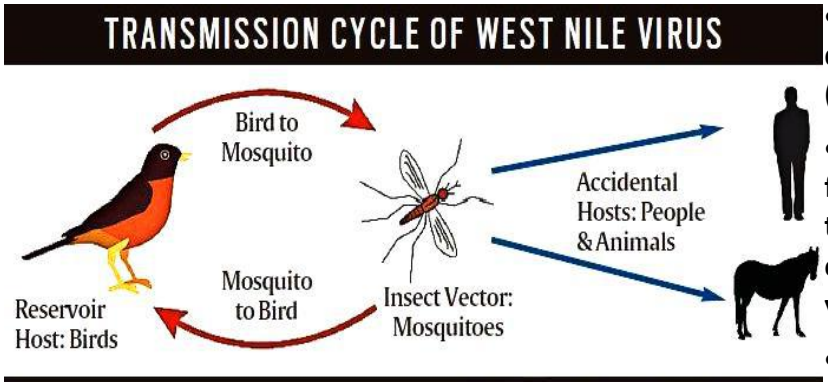
WEST NILE FEVER

Context: Multiple cases of West Nile fever reported in Kerala.

Background:

- The Kerala Health Department is emphasizing mosquito control and the elimination of breeding sites to prevent further spread of the disease.

About WEST NILE FEVER (WNF) :



- West Nile Fever is an **infectious disease caused by the West Nile virus (WNV).**
- West Nile virus belongs to the **flavivirus family and is related to viruses that cause other diseases such as St. Louis encephalitis, Japanese encephalitis, and yellow fever.**
- **It is a mosquito-borne, single-stranded RNA virus.**

Global Prevalence:

- WNV outbreaks occur along major bird migratory routes.
- The virus was **first identified in Uganda in 1937** and has since been found in various regions including Europe, Africa, Asia, Australia, and North America.
- The virus is **commonly found in regions including** Africa, Europe, the Middle East, North America, and West Asia.

Prevalence in India:

- In India, WNV activity has been **reported in southern, central, and western regions**.
- Antibodies against WNV **were first detected in humans in Mumbai in 1952.**
- WNV has been isolated from mosquitoes in Andhra Pradesh, Tamil Nadu, Maharashtra, Karnataka and Kerala.

Transmission Cycle:

- The principal vector for transmission is the Culex species of mosquitoes.
- Birds act as reservoir hosts for the virus.
- Infected mosquitoes transmit WNV between humans and animals, including birds.

Prevention and Control:

- There is currently no vaccine available for humans; therefore, prevention and supportive care are key in managing the risk and impact of this disease.
- Preventive measures include mosquito control, avoiding mosquito bites, and monitoring bird populations.

Source: The Hindu

ZERO DAY VULNERABILITY

Context: Google released a security update for Chrome to fix a zero-day vulnerability.

Background:

- The security vulnerability was being exploited in the wild, and could result in data leakage, code execution and crashes in the software.This is the fifth zero-day exploit Google fixed since the start of this year. Three zero-day exploits were discovered in March 2024.

Key Takeaways

- A Zero-Day Vulnerability is an **undiscovered flaw in an application or operating system**.
- The term "Zero day" refers to the fact that the **software or device vendor has zero days to fix the flaw because malicious actors can already use it to access vulnerable systems**. The unknown or unaddressed vulnerability is referred to as a zero-day vulnerability or zero-day threat.
- **A Zero-Day Exploit is a cyberattack vector that takes advantage of this unknown or unaddressed security flaw.** A zero-day attack is when a **malicious actor uses a zero-day exploit to plant malware, steal data, or otherwise cause damage to users, organizations, or systems**.
- Zero-day vulnerabilities—especially in **widely-used operating systems or computing devices—are a severe security risk.** They leave huge numbers of users or entire organizations wide **open to cybercrime until the vendor or the cybersecurity community identifies the problem and releases a solution.**
- A similar but separate concept, Zero-Day Malware**, is a virus or malware for which the signature is unknown or as yet unavailable, and therefore undetectable by many antivirus software solutions or other signature-based threat detection technologies**

Source: Hindu

SICKLE CELL DISEASE

Context: Recently, a 12-year-old boy from USA, became the first person in the world with sickle cell disease to begin a commercially approved gene therapy that may cure the condition.

Background:

- The U.S. Food and Drug Administration (FDA) has approved a gene therapy called Casgevy, which represents a significant milestone in the treatment of sickle cell disease.

About Sickle Cell Disease :

- Sickle cell disease **is a group of inherited disorders that affect the hemoglobin, the protein that carries oxygen through the body.**
- People with SCD **have a typical haemoglobin molecules called haemoglobin S, which can distort RBCs into a sickle or crescent shape**.

- This distortion leads to several complications, including chronic anaemia, acute painful episodes (known as sickle cell crises), organ infarction (tissue damage due to blocked blood flow), and chronic organ damage.
- Unfortunately, SCD **significantly reduces life expectancy for affected individuals.**

Treatment Options:

- **Blood Transfusions**: These helps relieve anaemia and reduce the risk of pain crises.
- **Hydroxyurea**: A medication that reduces the frequency of painful episodes and prevents long-term complications.
- **Bone Marrow or Stem Cell Transplantation**: In some cases, this approach can be used to replace the defective stem cells with healthy ones.

Government Initiatives in India:

- The National Health Mission in India supports states in preventing and managing Sickle Cell Disease (SCD).
- The government aims to **eliminate Sickle Cell Anaemia by 2047.**
- Technical operational guidelines have been **released for prevention and control of hemoglobinopathies, including sickle cell anaemia.**

Source: New York Times

Mpox

Context: A study revealed a novel adaptation mechanism of the Mpox virus enhancing its ability to infect humans amidst recent outbreaks. The name was changed from "monkeypox" to "mpox" to avoid stigma towards monkeys and reflect the virus's direct human infectivity.

Background:

- Mpox was first discovered in laboratory monkeys in the 1950s. The first human case was reported in 1970 in the DRC. Since then, outbreaks of mpox have been small and traceable to a single person – typically someone who had recently returned from a country where the virus is endemic – such as west and central Africa.

Mpox:

- It is **also known as monkeypox and is a DNA virus**. It belongs to the **family Poxviridae, which consists of large, double-stranded DNA viruses.**
- It is primarily **transmitted to humans from animals, particularly rodents and primates, through direct contact or contaminated objects.**
- Its infection in humans **typically presents with fever, headache, muscle aches, and a characteristic rash that progresses from macules to papules to vesicles and pustules**.
- While a vaccine for mpox exists, its availability and effectiveness are limited, highlighting the need for improved prevention and control measures.
- Mpox gained **global attention during a widespread outbreak that affected over 100,000 people in more than 118 countries in 2022-2023.**
- The outbreak was characterised by **high human-to-human transmission**, particularly through close contact and the sexual route.
- The World Health Organization (WHO) declared the **mpox outbreak a public health emergency, leading to coordinated efforts to control its spread.**

Genomic Features:

- Mpox genomes are divided into clades I and II, with evidence suggesting clade I has higher mortality. **The 2022 outbreak involved a new lineage, clade IIb, better adapted for human-to-human transmission.**
- Researchers found evidence of a distinct lineage of clade I associated with **human-to-human transmission, suggesting a recent zoonotic spillover event.**
- Mpox viruses can **undergo genomic accordion through gene duplication or deletion to adapt to different hosts and environments.**
- A study published in Nature Communications sequenced the genome of the mpox virus from the 2022 outbreak, **revealing that certain sections strongly influence human-to-human transmission and are the virus's genomic accordions.**

Source: The Conversation

LOKCBIT RANSOMEWARE

Context: The U.S. Department of Justice (DOJ) has recently announced a $10 million reward for information leading to the apprehension of Dimitry Yuryevich Khoroshev, an alleged creator of the LockBit ransomware group.

Background:

- FBI Director Christopher Wray emphasized the agency's commitment to disrupting ransomware organizations. The substantial reward underscores the severity of Khoroshev's alleged crimes and the urgency in bringing him to justice.

About LOCKBIT RANSOMWARE

- LockBit ransomware is a **malicious software designed to block user access to computer systems in exchange for a ransom payment.**
- LockBit automatically **vets valuable targets, spreads the infection, and encrypts all accessible computer systems on a network.**
- It primarily **targets enterprises and government organizations rather than individuals**.
- **LockBit operates as a Ransomware-as-a-Service (RaaS)** model. RaaS is a business model where cybercriminals create and distribute ransomware, allowing other individuals (affiliates) to use it for their attacks.

The threats posed by LockBit include:

- **Operations disruption**: Essential functions come to a sudden halt.
- **Extortion**: The hacker seeks financial gain.
- **Data theft and illegal publication**: Blackmail if the victim does not comply.

Origins and Notable Targets:

- Attacks using LockBit began around September 2019, **initially known as the "abcd virus**."
- Notable past targets include organizations in the United States, China, India, Indonesia, and Ukraine.
- LockBit has reportedly targeted India's **National Aerospace Laboratories (NAL).**
- LockBit **avoids attacking systems local to Russia or other countries within the Commonwealth of Independent States, presumably to avoid prosecution there.**

Source: The Hindu

SALMONELLA

Context: The United States has taken stringent measures in response to salmonella contamination in shipments from Mahashian Di Hatti Pvt Ltd (MDH).

Background:

- The situation underscores the importance of stringent quality control measures in food production and distribution. Authorities worldwide are closely monitoring such incidents to protect consumers and maintain food safety standards.

About SALMONELLA

- Salmonella is a **group of bacteria that can cause gastrointestinal illness and fever, a condition known as salmonellosis.**
- These bacteria naturally **reside in animals' intestines and can be found in their faeces**.
- Humans can contract salmonella **if they come into contact with infected animals or contaminated items in their environment.**
- This ubiquitous and hardy bacteria **can survive for several weeks in a dry environment and several months in water.**
- People infected with salmonella **might experience diarrhea, fever, and stomach cramps.**
- The United States **Centers for Disease Control and Prevention (CDC)** estimates that salmonella causes approximately 1.2 million illnesses, 23,000 hospitalizations, and about 450 deaths in the United States every year.

Source: Economic Times

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MAINS

PAPER 1

CARE ECONOMY IN INDIA

GS I –ROLE OF WOMEN

Context: The Supreme Court ruling declared the denial of Childcare Leave (CCL) to a government employee in Himachal Pradesh as a violation of workingwomen's Constitutional rights.

Key findings of the study:

Care Economy:

- **It** refers to the sector of economic activity that encompasses the provision of care and support services, particularly those related to health, education, childcare, eldercare, and other forms of social care.
- It encompasses **paid and unpaid care work** crucial for human survival, welfare, and labour force reproduction.
- It contributes to meeting **material, emotional, and developmental needs** but is often unrecognised or undervalued, leading to a **"hidden care economy".**
- It is different from the Monetized Economy, which is the **formal market-based system** where goods and services are bought and sold using money.
- It encompasses industries like **manufacturing, technology, healthcare (formal sector), and retail.**
- The value of work in the monetized economy is directly tied to its market price.
- Historically, feminist economists have critiqued the conventional definition of **"work"** for excluding **unpaid labour,** particularly the significant contributions made by women within households.This challenge led to the **Beijing Platform for Action in 1995**, which advocated for recognising and valuing women's roles in caregiving, domestic work, and volunteerism.

Challenges in India's Care Economy

• Limited coverage and applicability of policies like maternity benefits and childcare leave in small-scale enterprises and the informal sector.
• Unequal burden of care work hinders women's workforce participation and career advancement opportunities.
• Access to affordable and quality care services remains a challenge, especially for low-income households.
• Women's unpaid care and domestic work represent almost 15%-17% of India's GDP.
• Societal expectations and cultural norms reinforce the perception that caregiving is primarily a woman's responsibility, preventing men from actively participating in caregiving duties.

Promote **corporate social responsibility initiatives** that support the care economy. This could involve companies sponsoring childcare centers in low-income communities or offering flexible work arrangements for employees with caregiving responsibilities.

Note:

- **Paid Care Work**refers to care jobs in sectors like **health, education, personal care, and domestic work** that are compensated.Women are overrepresented in care roles like **nurses, domestic workers, personal carers, teachers,** and childcare assistants.
- **Unpaid Care and Domestic Work**include **domestic services** (cooking, cleaning), **caregiving** (children, elderly, sick), and community/voluntary services.Under this, direct care involves dependents, while indirect care includes domestic tasks, and multitasking blurs these boundaries.

Care Diamondrepresents the four main actors in care provision - State, markets, households, and communities.

CHOLA ADMINISTRATION

GS I –INDIAN CULTURE

Context: The Prime Minister (PM) placed the Sengol orsceptre, presented by the head of ThiruvaduthuraiAdheenam (monastery) to India's first PM, in the new Parliament building.

Chola Administration:

- King was the most powerful individual andhe had complete power, although he wasguided by a council of ministers.
- Top authorities were known as Peruntaram, while the inferior officials were known as Siruntaram.
- The Chola Empire was divided into mandalams or provinces and these, inturn, were divided into valanadu and nadu.
- Sometimes princes of the royal family were appointed governors of provinces.
- Chola administration encouraged local self-government in the villages all over their empire.
- There is a mention of two assemblies, Ur and the Sabha or Mahasabha, in many inscriptions(including the Uthiramerur Inscriptions).
  - Ur was a general assembly of the village.
  - Mahasabha was a gathering of adult men in Brahmana villages which were called agraharas.
- Villages enjoyed a large measure of autonomy. The affairs of villages were managed by an executive committee to which educated persons owning property were elected.
- Cholas paid attention to irrigation. River Kaveri and other rivers were used for this purpose. Many tanks and wells for irrigation were built.
- In addition to land tax, Chola rulers drew their income from tolls on trade, taxes on professions, etc.
- Chola kings gave some rich landowners titles like muvendavelan (a velan or peasant serving threekings), araiyar(chief), etc.
- Brahmanas often received land grants or brahmadeya.
  Cholas had an efficient army and navy. The Chola army consisted of elephants, cavalry, and infantry.Rajaraja's son Rajendra I developed a navy for his expeditions in Ganga Valley, Sri Lanka, and countries of SoutheastAsia.

Sengol:

- **On August 14, 1947, Pandit Jawaharlal Nehrureceived the 'Sengol' as a symbol of the transfer** ofpower to the government from the specially arrivedpriests from the ThiruvaduthuraiAdheenam in Tamil Nadu.
- The word 'sengol' is derived from the Tamil word 'semmai', whichmeans 'righteousness'. As theobserver of 'justice',the hand-engraved Nandi sits atop it, watching with his unflinching gaze.
- The Sengol reminds **the recipient that he has the "aanai" (order ordecree) to rule justly and fairly.**
- **Sengol had an important place in Tamil culture.**When a new king is crowned, he would be presented with a 'sengol' during the coronation by the high priest as asymbol of the transfer of power.
- In the Chola Empire, Sengol wasconsidered a symbol of the path of duty,the path of service, and the path of thenation.

Historians suggest that handingoverof a sceptre to denote the transfer ofpower has been in practice for nearly2,000 years since the Sangam Age andfinds mention in texts such asPurananooru, Kurunthogai,Perumpaanatrupadai, and Kalithogai.

EXPANSION OF HIMALAYAN GLACIERS

GS I –Important geophysical phenomena

Context: The satellite monitoring data by the Indian Space Research Organisation (ISRO) has shown a large expansion in glacial **lakes between 1984 and 2023** in the Himalayan region, which has posed an alarming situation for the downstream areas.

ISRO's Observation on Expansion of Himalayan Glaciers:

- Of the 2,431 lakes larger than 10 hectares identified during 2016-17, **676** glacial lakes have notably expanded since 1984.130 of these lakes are situated within India, with 65, 7, and 58 lakes located in

the **Indus, Ganga, and Brahmaputra** River basins, respectively.Of these lakes 601 lakes (89%) have expanded more than twice, 10 lakes have grown between 1.5 to 2 times, and 65 lakes at 1.5 times.

- **Elevation-based analysis reveals** that 314 lakes are located in the 4,000 to 5,000 m range and 296 lakes are above 5,000 m elevation.
- Long-term changes in the **Ghepang Ghat glacial lake (Indus River Basin)** at an elevation of 4,068 m in Himachal Pradesh, India, show a **178% increase** in size from 36.49 to 101.30 hectares between **1989 and 2022.**
- **Moraine-dammed (307)** are formed when piles of rocks and debris (moraines) left behind by retreating glaciers block valleys, creating natural dams that hold back meltwater.
- **Ice-dammed (8)**are formed when a glacier itself acts as a dam, blocking the flow of meltwater.
- **Erosion (265)**lakes occupy depressions carved directly into bedrock by glaciers.

Causes of Expanding Glacial Lakes in Himalayas

• Global Warming: Rising temperatures cause glacier melting, leading to lake expansion.
• New Glacial Lakes: Glaciers melt, exposing new land surfaces, allowing for new formations.
• Moraines: Shrinkage of glaciers weakens moraines, triggering Glacial Lake Outburst Floods.
• Changes in Precipitation: Increased rainfall and snowfall fill lakes, allowing more water to fill them.
• Permafrost: Warming temperatures thaw permafrost, creating depressions that collect water, contributing to lake expansion.
• Infrastructure Development: Roads and hydropower projects alter natural lake drainage patterns, leading to lake expansion.
• Activities: Mining and deforestation accelerate climate change.

Given the transboundary nature of many **glacial lakes** in the Himalayas, international cooperation is essential for effective management and risk reduction.Collaborative efforts among countries sharing glacier-fed river basins can facilitate information sharing, joint monitoring, and coordinated action to address common challenges.

STRATEGIC IMPORTANCE OF THE ANDAMAN AND NICOBAR ISLANDS

GS I –Geography
Context: The Indian government's renewed focus on developing the Andaman and Nicobar Islands(ANI) underscores their strategic significance in the Indo-Pacific region, prompting efforts to enhance infrastructure and security.

Strategic Importance of the A&N Islands:

- It is located 700 nautical miles southeast of the Indian mainland, the Andaman and Nicobar **Islands add 300,000 sq. km to India's exclusive economic zone,** with the potential for **undersea hydrocarbon and mineral deposits.**
- The islands' strategic location astride the **Malacca Strait**, makes them a crucial asset for **India's ability to monitor and project power in the Indo-Pacific region.**The Malacca Strait is a critical maritime choke point, with more than 90,000 merchant ships carrying about **30% of the world's traded goods passing through it annually.**
- The islands share maritime boundaries with Myanmar, Thailand, Indonesia, and Bangladesh, giving India substantial ocean space under the **United Nations Convention on the Law of the Sea (UNCLOS) in terms of exclusive economic zone and continental shelf.**
- The islands can serve as the **first line of defence against any attempt from the East** to undermine India's maritime security, especially in the face of China's growing influence in the Indo-Pacific region.
- **Port Blair** can become a regional hub for navies to collaborate on disaster relief, medical aid, counter-piracy, search and rescue, and other maritime security initiatives.

Issues/Challenges fordeveloping of the A&N Islands:

- The shift from **India's Look East policy to a stronger Act East policy**, along with the growing recognition of the significance of maritime power and the increasing capabilities of the Chinese PLA Navy, has underscored the need to develop Indian island territories, especially the Andaman and Nicobar group.
- **Lack of political prioritisation** until recently, with the realisation of the islands' strategic importance only now.
- Challenges of **distance from the mainland and difficulties in developing infrastructure.**
- Complex **environmental clearance procedures** and regulations on forest and tribal conservation.
- Coordination challenges due to the involvement of multiple ministries and agencies. The conflict between long-term strategic vision and immediate political gains.

Strategic infrastructure development that is required in A&N Islands:

- Ensuring comprehensive maritime domain awareness and surveillance over the islands.
- Bolstering deterrence capabilities against any naval misadventures from the East.
- Developing infrastructure to support **India's maritime economy, especially in the southern group of islands.**
- Improving transportation and connectivity to facilitate development and tourism. Developing **Galathea Bay transhipment port** on Great Nicobar Island.
- The plan to connect the A&N Islands to the mainland through **Submarine Optical Fibre Cable (OFC)** needs revitalisation. This will provide cheaper and better connectivity and access to **Digital India's benefits.**
- Enhancing transportation and connectivity for development and tourism.
- The **military must increase forces and deploy** the appropriate assets at the **Andaman Nicobar Command (ANC)** to maintain island security. This includes basing surveillance and fighter aircraft there, as well as conducting frequent detachments.
- Exploring partnerships with the **Quad and Indo-Pacific Oceans Initiative (IPOI)** for development initiatives.

Seeking infrastructure development concessions akin to those on India's northern borders.

STREET VENDORS IN INDIA

GS II –Welfare schemes for vulnerable sections of the population
GS I –Salient features of Indian Society

Context: The Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act completed a decade of its enactment. It was enacted on 1st May 2014.

Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act of 2014:

- It was implemented **to legalise the vending rights** of street vendors **(SVs)**.
- Its objective was to safeguard and regulate street vending in urban areas, with s**tate-level regulations** and programs overseen by **Urban Local Bodies (ULBs)** through the formulation of by-laws, planning, and enforcement.
- It outlines the **roles and responsibilities** of both vendors and various levels of government.
- It envisages accommodating all **existing vendors** in designated vending zones and issuing **Vending Certificates (VCs).**
- It establishes a **participatory governance framework** through the creation of Town Vending Committees (TVCs), where street vendor representatives must constitute **40%** of the members, with a sub-representation of **33%** for women SVs.These committees are responsible for ensuring the inclusion of all existing vendors in vending zones and theyinclude mechanisms for addressing grievances and disputes, proposing the establishment of a **Grievance Redressal Committee** chaired by a civil judge or judicial magistrate.
- It mandates that states/ULBs conduct surveys to identify SVs at least once **every five years.**

Government Initiatives for Street Vendors:

- **PM SVANidhi Scheme**launched by the **Ministry of Housing and Urban Affairs,** aims to provide affordable working capital loans to street vendors to restart their businesses or expand their existing ones. It also offers incentives for timely repayment.
- **National Urban Livelihood Mission (NULM)is a centrally sponsored scheme** that aims to reduce poverty and vulnerability of urban poor households by enabling them to access gainful self-employment and skilled wage employment opportunities.It includes provisions for **skill training, capacity building,** and access to credit for street vendors.
- **Urban Street Vendors (USV) Component under Deendayal Antyodaya Yojana-National Urban Livelihoods Mission (DAY-NULM)** focuses on street vendors.It provides support for setting up and upgrading vending infrastructure, organising vendors into **Self-Help Groups (SHGs)**, and facilitating access to credit and social security schemes.
- Under the **Street Vendors Act, Town Vending Committees** are formed at the municipal level to facilitate the implementation of the Act's provisions.These committees are responsible for identifying vending zones, issuing vending certificates, and addressing grievances of street vendors.

Challenges Faced by Street Vendors in India:

- Despite the Street Vendors Act which focuses on protection, street vendors endure increased harassment and eviction, often stemming from **outdated bureaucratic attitudes** to treat them as illegal migrants.
- Street vendor representatives in Town Vending Committees (TVCs) often lack influence, and the inclusion of women vendors remains superficial.
- The Act's alignment with urban governance frameworks is deficient, and Urban Local Bodies (ULBs) lack the necessary authority and capacity.
- Programs like the Smart Cities Mission prioritise infrastructure development over the integration of street vendors, undermining the Act's objectives.
- Conventional perceptions of **'world-class cities' marginalise** street vendors, impeding their acknowledgment as legitimate contributors to urban life.

Cases of '**rangadari tax' and 'hafta'** are common. In many cities, vendors have to part with substantial money to ply their trade.

Street Vendors in India:

- A street vendor is an individual who sells goods to the public **without a permanent built-up** structure for vending.
- They may operate from **stationary positions on pavements** or other public/private spaces, they may be mobile, carrying their merchandise on push carts or in baskets.
- In India, approximately **49.48 lakh street vendors** have been identified, with **Uttar Pradesh** having the **highest count at 8.49 lakh,** followed by Madhya Pradesh at 7.04 lakh. Conversely, Delhi has about 72,457 identified street vendors, while none have been identified in **Sikkim.**
- **Article 19(1)(g)** of the Indian Constitution grants citizens the fundamental right to practice any profession, trade, or business.

PAPER 2

CHINA'S INFLUENCE IN CENTRAL ASIA

GS II –Effect of policies and politics of developed and developing countries on India's interests.
Context: China hosted the "C+C5 summit", in the city of Xi'an (the first of its kind), with the leaders of five Central Asian countries.

Implications of the rising influence of China in CentralAsia for India:

- China's presence andleverage in India's neighboring region impact India's strategic interests and limit its influence in the region.Post the Taliban takeover, the Central Asian region has emerged as a geopolitical arena where great contestationsfor influence are unfolding between countries like Russia, China, the US, Turkey, Iran, Europe, the EU, and others.
- China is Central Asia's largest trading partner whileIndia's combined tradewith Central Asia is already only around $2 billion (much less than China's $32 billion).
- China's Belt and Road Initiative, which includes projects such as the China-Pakistan Economic Corridor(CPEC), could affect India's efforts to enhance its connectivity with Central Asia and limit its access to the region.China's influence may affect India's ability to foster closer ties with these countries through initiatives like theInternational North-South Transport Corridor (INSTC).
- The growing Chinese presence may impact theregional security dynamics and influence thebalance of power in the immediate neighborhood.India and the CARs share common concerns on the issue of rising threats from terrorism, extremism, anddrug trafficking which pose challenges to regional stability.
- The region is crucial for India's energy security. The growing Chinese influence might hinder India'sprospects for securing energy from the region.For example, Kazakhstan is the world's largest producer of uranium and Turkmenistan has huge reserves ofnatural gas.

Initiatives taken by India toincrease its imprint in the Central Asia Region:

- Through the renovation of Chabahar Port, faster development of the INSTC, becoming amember of the Ashgabat Agreement and TAPI pipeline.
- The Connect Central Asia (CAA) Policy of 2012 is a broad-based framework involving political, economic, security,and cultural connections.India recently hosted the first India-Central AsiaSummit.
- Commitment to Extended Neighborhood Policy (ENP)calls to diversify India's geopolitical partners anddiplomatic goals, and its willingness to engage its Central Asian partners on a multitude of fronts.
- India signed Strategic Partnership Agreements(SPA) with Kazakhstan, Tajikistan, andUzbekistan and a Civil nuclear agreement with Kazakhstan.India's International Technical and Economic Cooperation (ITEC) programme attracts young professionals forcapacity building.
- Multilateral organisationslike SCO and the Conference on Interaction and Confidence-Building Measures inIndia (CICA) have helped India to maintain regular channels of communication with the CARs.
- The $1b Line of Credit announced by India is aimed at High Impact Development Projects(HICDPS) for the socio-economic development of the region.

Challenges in India CAR relations:

- India does not share physicalborders with any of the Central Asian states. Also, India's antagonistic relations with Pakistanhave further complicated connectivity and tradeprospects for India.
- India's resource constraintsespecially financial can limit its ability to activelyengage with all Central Asian countries simultaneously.
- Political instability and security concerns in certain Central Asian countries, such asnarco-terrorism, and regional conflicts, can impactIndia's engagement.

CONSTITUTIONAL MORALITY

GS II –Indian Constitution

Context: The recent arrest of a serving chief minister on corruption charges raises legal, political, and constitutional concerns and poses questions about its consistency with constitutional morality, especially in a parliamentary democracy like India.

Constitutional Morality:

- Constitutional morality (CM) is a concept that refers to the **principles and values underlying a constitution that guide the actions** of both the government and the citizenry.
- The concept of constitutional morality was propounded by the British Classicist George Grote in the 19th century.He described **CM as a "paramount reverence for the forms of the Constitution" of the land.**
- In India, the term was first used by **Dr. B.R. Ambedkar.**

Pillars of Constitutional Morality:

- Upholds core values like justice, liberty, equality, fraternity, secularism, and individual dignity.
- Upholds the Rule of Law, ensuring everyone is subject to and accountable under the law.
- Upholds democratic principles, ensuring citizens have the right to participate in decision-making processes and hold elected representatives accountable.
- Respects and protects fundamental rights, including equality, freedom of speech, life, and personal liberty.
- Maintains a separation of powers between legislative, executive, and judicial branches.
- Establishes mechanisms for checks and balances to prevent power abuse and protect individual rights.
- Promotes constitutional interpretation that adapts to societal needs.
- Ensures ethical governance, transparency, accountability, and integrity in public service.

Judgments Upholding Constitutional Morality:

- **Kesavananda Bharati v. State of Kerala, 1973** established the **"basic structure doctrine,"** which essentially limits Parliament's power to amend the Constitution and ensures its core principles remain intact.This can be seen as an early instance of the court upholding the spirit of the Constitution.
- **SP Gupta Case (First Judges Case), 1982** labelled a constitutional breach as a severe violation of constitutional morality.
- **Naz Foundation v. Government of NCT of Delhi, 2009**decriminalised consensual same-sex relationships between adults.The court emphasised that **"constitutional morality" should prevail over societal perceptions** of morality, upholding individual rights.
- *Manoj Narula v. Union of India, 2014* stated that **"Constitutional Morality means to bow down to the norms of the Constitution** and not act in a manner which would become violative of the rule of law of action in an arbitrary manner.
- **Indian Young Lawyers Association v. State of Kerala (Sabarimala Case), 2018** struck down the practice of excluding women of a certain age group from the Sabarimala temple.It emphasised that "constitutional morality" includes principles of justice, equality, liberty, and fraternity, which outweighed religious customs restricting women's entry.
- **Navtej Singh Johar v. Union of India, 2018**read down **Section 377** of the Indian Penal Code, which criminalised homosexuality.

Challenges to Constitutional Morality in India

- Political interference undermines constitutional bodies' autonomy and impartiality.
- Criticisms exist over recent changes to the Election Commission of India and IT Rules 2023.
- Balancing judicial activism with restraint is a challenge.
- Excessive activism can encroach on executive and legislature domains.

Despite having a robust constitutional framework, ensuring effective enforcement and compliance remains a challenge.**Implementation gaps, delays in justice delivery,** and lack of awareness about constitutional rights among the general populace contribute to this challenge.

CUSTODIAL DEATH

GS II –ISSUES RELATING TO DEVELOPMENT

Context: The Supreme Court has emphasized the necessity of adopting a "more rigorous approach" when considering bail applications from police officers charged in cases of custodial deaths.

Custodial Death:

- Custodial death refers to a **death that occurs while a person is in the custody of law enforcement officials** or a correctional facility.
- It can occur due to **various causes such as the use of excessive force, neglect, or abuse** by the authorities.
- According to the **Law Commission of India**, the crime by a public servant against the arrested or the detained person who is in custody amounts to **custodial violence.**

Constitutional Framework Related to Custodial Death:

- **Article 21 of the Constitution of India** guarantees the **right to life** and personal liberty, which includes the **right to be free from torture** and other cruel, inhuman, or degrading treatment or punishment.
- **Article 20** grants protection against arbitrary and excessive punishment to an accused person, whether a citizen or foreigner or a legal person like a company or a corporation. It contains three provisions in that direction - **No ex-post-facto law (Article 20 (1)), No double jeopardy (Article 20 (2)), and No self-incrimination (Article 20 (3)).**
- **In Selvi v. State of Karnataka**, the court observed that the **state could not perform narco-analysis, polygraph, and brain-mapping tests on any individual without their consent.**

Legal ProtectionsAssociated with Custodial Death:

- **Section 24 Indian Evidence Act, of 1872** declares that all the confessions made by the **accused by succumbing to the threat** of investigating agencies would **not be admissible in a court of law.**This Section primarily works to prevent the accused from giving confessions using force against his will.
- **Sections 330 and 331 of the Indian Penal Code (IPC)** criminalise voluntarily causing hurt or grievous hurt to extort confession or information from any person.
- Section 41 of **the Criminal Procedure Code (CrPC)** was amended in 2009 to include safeguards so that:
  - Arrests and detentions for interrogation have reasonable grounds and documented procedures.
  - Arrests are made **transparent to family, friends, and the public,** and there is protection through legal representation.

International Conventions Against Custodial Torture

- International Human Rights Law of 1948 protects against torture and enforced disappearances.
- United Nations Charter of 1945 advocates for treating prisoners with dignity, recognizing their fundamental freedoms and human rights.
- Nelson Mandela Rules of 2015 treat prisoners with inherent dignity and prohibit torture.
- United Nations Convention Against Torture (UNCAT) aims to prevent cruel, inhuman, or degrading treatment or punishment worldwide.

Types of Custodial Death:

- **Death in Police Custody** can result from excessive force, torture, denial of medical care, or other forms of abuse or incidental cause.
- **Death in Judicial Custody** may occur due to overcrowding, poor hygiene, lack of medical facilities, inmate violence, or suicide.
- **Death in the Custody of the Army or Paramilitary Forces** can happen through torture or extrajudicial killings.

EXPANSION OF BRICS

GS II –Regional and Global Groupings and Agreements involving India and/or affecting India's interests.

Context: In the recent 15th BRICS summitheld in Johannesburg, South Africa, BRICS members agreed to induct six new member countries. They are Argentina, Egypt, Ethiopia, Iran, Saudi Arabia, and the UAE.

Significance of BRICS Expansion:

- The expansion signifies a growing alignment ofgeopolitical and economicagendas within the BRICS.It incorporates major global oil producers near crucialtrade chokepoints, such as the Suez Canal and the**Straitof Hormuz and Bab-al Mandab Strait.**
- Expanding BRICS opens up new horizons ofeconomic and people-to-people connections.India, Iran, and Russia are already **developing the International North-South Transport Corridor.**
- The inclusion of two African countries strengthens theongoing efforts to promote integration,development, and **growth through the African Continental FreeTrade Area.**
- The induction of 4 major Middle East Players**- Egypt,Iran, UAE, and Saudi gives BRICS** a major boost –as these countries have been at odds with each other.
- BRICS will greatly push forward the agenda for reform in the present system of multilateralism, whereWestern powers play a dominant role.

Challenges to BRICS Reforms and Expansion

- Rapid expansion could dilute the grouping's original purpose, especially if China dominates.
- Expansion is often seen as a political maneuver to advance China's international order.
- Balancing with other economic groups like Quad, Shanghai Cooperation Organization, and Indo-Pacific Economic Framework may be challenging.
- Tensions between countries may make BRICS more unwieldy to run, similar to SAARC.
- No clearly defined membership criteria have been made public.
- Sanctions against Iran could hinder intra-BRICS trade growth.
- Expanded BRICS could lead to "minilateralism," eroding global collective action.

India's Future in BRICS

- Balancing ties with Western powers, especially the US, as the expanded BRICS includes Iran.
- Balancing China's strategic influence with India's economic diplomacy and relations with Russia and Iran.
- Increasing India's influence and opportunity to collaborate on Global South issues.
- Pushing for institutionalizing intra-BRICS trade in member countries' national currencies.
- Inclusion of major oil-producing Middle Eastern powers in BRICS for energy diversification and better pricing.

Advocating a clear membership criterion for BRICSwill help ease India’s concernsabout who could be future members of the BRICS.

GRIEVANCE REDRESSAL ASSESSMENT & INDEX(GRAI)

GS II –Government policies and interventions

Context: Union Minister of State for Personnel, PublicGrievances and Pensions launched theGrievance Redressal Assessment & Index(GRAI) for the year 2022.

Grievance Redressal Mechanism (GRM):

- As per the 2nd ARC Report, GRM is one of the importantmechanisms that can be usefully employed to make theadministration citizen-centric.
- The basic principle of a grievance redressal system is that if the promised level of service delivery is not achieved orif a right of a citizen is not honored then the citizen should be able to take recourse to a mechanism to have thegrievance redressed.
- The two nodal agencies handling public grievances-
  - DARPG, Ministry of Personnel, Public Grievances and Pensions
  - Directorate of Public Grievances, CabinetSecretariat.

Issues with the existing public grievance system:

- Peopleare not aware that a system of redressal exists in many of the government departments.
- The departmenthasan internal system forthe redressal of grievances of its personnel, but it is notworkingsatisfactorily, and the non-settlement ofgrievances results in the filing ofpetitions in the courts on petty issues.
- Wide variations exist across ministries and other organisations concerning the framework,process, and capacity to handle grievances.
- GRM is not treated as mandatory like RTI in many departments.
- Public Grievance Cells often suffer from a shortage of staff and resources. Moreover, these cellshave not been adequately empowered.
- In severalcases, the complainant was asked to approach thestate government, and the grievance was disposedof and not forwarded to the concerned stategovernment.
- Slackness in administration, lowmorale of the services, inherent inertia, absence ofincentives, and lack of proper authority andaccountability.

Measures needed to strengthen GRM:

- Identifyareas susceptible to corruption and/or grievancegeneration and conduct a work audit of such areas.
- Wide publicity throughnational, regional, and local media as well as throughelectronic media.
- Throughrewarding good work awarding effectivesuggestions and punishing the deliberate negligence of civil servants.
- Recommendation of 2nd ARC:
  - Public grievance officers are on the lines of thePublic Information Officers under the RTI Act.
  - All grievance petitions received should besatisfactorily disposed of by these officerswithin 30 days.
  - Non-adherence to the time limit should invite financial penalties.
- Suggestions by Parliamentary Standing Committee on Personnel, Public Grievances, Law and Justice (2021):
  - CPGRAMS should play the role of a facilitator without impinging on the nature of federalism.
    All Ministries are to undertake a regular review of grievances raised in print and electronic media.

GRAI 2022:

- It was conceptualised anddesigned by the Department ofAdministrative Reforms and PublicGrievances (DARPG) under the Ministry ofPersonnel, Public Grievances & Pensions.
- These reports are part of the 10-step Centralised Public Grievance Redressal and Management System (CPGRAMS)reforms that were adopted by the DARPG to work on improving the quality of disposal of public grievances whilereducing the time it takes to address these issues.
- Its objective is to presentan organisation-wise comparativepicture and provide valuable insightsaboutstrengths and areas of improvement in grievance redressal mechanism (GRM).
- It is based on 12 Indicators spread across 4 key dimensions - Efficiency, Feedback, Domain, and Organisationalcommitment.
- 89 Central Ministries andDepartments wereassessed and ranked basedon a comprehensive index.
- The average disposal time ofCentral Ministries andDepartments hasimproved from 32 days in2021 to 27 days in 2022.

Centralized Public Grievances Redress and Monitoring System (CPGRAMS):

- It is an **online web-enabled system** developed by the **National Informatics Centre** (Ministry of Electronics & IT [MeitY]), in association with the Directorate of Public Grievances (DPG) and Department of Administrative Reforms and Public Grievances (DARPG).

INDIA AND GREECE

GS II –India's Bilateral Relations

Context: The Indian Prime Ministervisited Greece for thefirst time in four decades.

India and Greece Relations:

- Greece has historical linkages with India as Chanakya, in Chandragupta's Court records inArthashastramentioned the Yavan Ambassador in the Kings' court, namedMegasthenes.Trading between Mauryan Kings and Greece is evidenced bycoinage and writings. Gandhara art is believed to be a fusion ofIndian and Greek influences.
- Diplomatic relations were established between the twocountries in 1950. Greece opened its Embassy in Delhi in 1950 andIndia in Athens in 1978.
- Agreement on Science and Technology wasratified by both sides to encourage and support cooperation in thescience and technology field based on equality and mutual benefit.
- Greece supported India's membership at NSG (National Supplier Group) and bid for apermanent seat on the United Nations Security Council.
- India's relations with Greece and Armenia are crucialin the context of an emerging Turkey-Pakistan-Azerbaijan military axis (often referred to as the ThreeBrothers) and can counter increased Chinesepresence in theregion.
- Greece borders theMediterranean Sea, which covers threecontinents, i.e., Asia, Africa, and Europe, andthus can help enhance India's outreach tothe Mediterranean region.The Mediterranean region is vital forIndia's Indo-Pacific policy.
- India isexploring a multi-modal link betweenMumbai and the European mainland via theGreektranshipment port of Piraeus.This will allow much faster shipment ofIndian goods to markets in the entire Eurasian region.
- The Mediterranean region can be a potential source to meet India's increasing energy demand andenhance energy security given its vast resources of oil (1.7 billion barrels) and gas (112 trillion cubic feet).Greece's abundant sunshine and its robust windenergy capacity present substantial untappedopportunities forelectricity generation.
- With Greece expanding its defence expenditure fromUS$5 billion to 8.4 billion (2019- 2022),it can act as a potential market for India's defence exports.India and Greece have engaged in joint naval exercises in the Mediterranean Sea and multinational air forceexercise INIOCHOS-23.

Issues/Concerns associated with India- Greece Relations:

- The absence of high-level interaction in four decadeshas resulted in a lack of strongstrategic connections.
- Chinese state-owned shipping company COSCO has approx. 60% stake in the Piraeus port and thuscan hamper India's connectivity plans and tradeopportunities through the port.
- India's trade relations with Greece face challengescharacterised by limited businessopportunities, insufficient investment and capital inflow, a shortage of enterprises capable of bolstering exports, etc.

Way Forward:

- Greece and Cyprus are two EU member states thatcan act as a bridge between the EuropeanUnion and India, and thus, urgent steps are required tostrengthen bilateral ties.
- India can extend its Line of Credit, utilise its softpower and diaspora to counter China's presenceand provide an alternative to China's Debt Trap Policy.
- Collaborate on tourism promotion efforts, includingjoint marketing campaigns and improved airconnectivity.Greece is emerging as a popular tourist destination among Indians.
  Enhance cooperation on maritime security, including information sharing and joint navalexercises.

Highlights of the recent visit:

- India welcomedGreece into theInternational Solar Alliance andlooked forward to Greece'smembership of the Coalition forDisaster ResilientInfrastructure.
- Encourage joint efforts inpreserving and protecting ancientsites and strengthen cooperationwithin UNESCO.
  Both countries agreed on thefinalisation of the Mobility andMigration Partnership Agreement tofacilitateskilled migration.

MISLEADING ADVERTISEMENTS

GS II – GOVERNMENT POLICIES AND INTERVENTIONS

Context: In a move to safeguard consumers from misleading advertisements, the Supreme Court of India has issued directives for advertisers to submit self-declarations before promoting products in the media.

Misleading Advertisement:

- A misleading advertisement is defined under **Section 2 (28) of the Consumer Protection Act, 2019**, as any advertisement that:
  - Provides a false description of a product or service;
  - Offers false guarantees that mislead consumers;
  - constitutes an unfair trade practice through express representation;
  - Deliberately omits essential information about the product.

Key Directives from the Supreme Court to Regulate Misleading Advertisements:

- Advertisers must submit **self-declarations** before promoting products in the media.Advertisers are **now obligated to declare that their advertisements do not deceive or make untrue statements** about their products to prevent misleading consumers.
- Advertisers intending to run TV ads must upload declarations on the **Broadcast Seva portal**, which serves as a one-stop facility for stakeholders to request permissions, registrations, and licenses for broadcast-related activities from the **Ministry of Information and Broadcasting.**A similar portal for print advertisers is to be established.
- Social media influencers, celebrities, and public figures endorsing products **must act responsibly.**Endorsers should possess adequate knowledge about the products they promote to avoid deceptive advertising.
- **Ensuring consumer protection** by establishing a transparent process for consumers to report misleading advertisements and ensure they receive updates on complaint status and outcomes.

Regulation of Misleading Advertisements in India:

- The **Central Consumer Protection Authority (CCPA)** enforces the '**Guidelines for Prevention of Misleading Advertisements and Endorsements for Misleading Advertisements, 2022'**, which were issued per the powers conferred by the Consumer Protection Act, 2019. Its o**bjectives include:**
  - The guidelines seek to ensure that **consumers are not being fooled by unsubstantiated claims, exaggerated promises, misinformation, and false claims**.
  - Such advertisements violate various rights of consumers such as **the right to be informed, the right to choose and the right to be safeguarded against potentially unsafe products and services.**
  - The guidelines define "**bait advertising", "surrogate advertisement" and "free claim advertisements".**
  - They also lay down provisions to **protect children from exaggerated or unsubstantiated claims** in advertisements.
  - Advertisements targeting children are prohibited from featuring personalities from sports, music, or cinema for products that require a health warning or cannot be purchased by children.
  - Disclaimers in advertisements should not hide material information or attempt to correct misleading claims.
  - The guidelines also outline the **duties of manufacturers, service providers, advertisers, and advertising agencies** to bring more transparency and clarity to advertisements.This aims to help consumers make informed decisions based on facts.

- CCPA can **impose penalties of up to 10 lakh rupees** on manufacturers, advertisers, and endorsers for misleading advertisements.For subsequent violations, the penalty can be up to 50 lakh rupees.
- The Authority can also **prohibit the endorser of a misleading advertisement from making any endorsements** for up to 1 year, and for subsequent violations, the prohibition can extend up to 3 years.

- Deceptive advertising falls under **Section 53 of the Food Safety and Standards Act, 2006**, making it punishable. FSSAI mandates advertisements to be truthful, unambiguous and substantiated scientifically.
- **ASCI (Advertisement Standard Council of India)**is a **non-statutory tribunal** established as a self-regulated mechanism to introduce advertising ethics in India.It judges advertisements based on its **Code of Advertising Practice, also known as the ASCI code** which applies to advertisements seen in India, even if they are from abroad and directed at Indian consumers.
- **Consumer Protection Act of 1986 g**rants consumers the right to be informed about goods and services' quality, quantity, and price.Section 2(r) covers false advertisements under the definition of unfair trade practices. It also provides redressal against misleading advertisements.
- **The Cable Television Network Act of 1995 and the Cable Television Amendment Act of 2006prohibit**the transmission of advertisements that do not conform to the prescribed advertisement code. It ensures advertisements do not offend morality, decency, or religious sensitivities.
- The Cigarettes and Other Tobacco Products Actof 2003 prohibits direct and indirect advertisement of **tobacco products** in all forms of media.
- **The Drug and Magic Remedies Act of 1954 andthe Drugs and Cosmetics Act of 1940**regulate drug advertisements. Prohibits the use of test reports for advertising drugs.Penalties for its violations include fines and imprisonment.
- The **Prenatal Diagnostic Techniques (Regulation and Prevention of Misuse) Act of 1994 p**rohibits advertisements related to prenatal sex determination under Advertising harmful publications under the Young Persons (Harmful Publications) Act, 1956, is punishable.
- **The Indian Penal Code (IPC)**prohibits obscene, defamatory, or inciteful advertisements.Offenses related to inciting violence, terrorism, or crime are illegal and punishable under IPC provisions.

NO CONFIDENCE MOTION

GS II –FUNCTIONING OF PARLIAMENT

Context: The union government faced a no-confidence motion in the Monsoon session of the parliament.

No-Confidence Motion:

- It is a legislative resolution introduced in the Lok Sabha that allows the opposition to challengethe government's majority and ability to govern.
- In a parliamentary democracy, a government can be in power only if it commands a majority in the directly electedHouse. It implies that if a no-confidence motion is passed, the government of the day will have to resign.
- The motion of no-confidence can only be moved in Lok Sabha by any of its MPs.
- Article 75(3) of our Constitution embodies this rule by specifying that the Council of Ministers is collectivelyresponsible to the Lok Sabha.
- Rule 198 of Rules of Procedure and Conduct of Business in Lok Sabha provides a provision regarding the motionof no-confidence in the Council of Ministers.
- A motion of no-confidence need not set out any grounds on which it is based.Even when grounds are mentioned in the notice and read out in the House, they do not form part of the no-confidence motion.
- Unless leave is granted by the House to move the motion, no speech is permitted in support of the motionat the time of obtaining the leave of the House.
- After a no-confidence motion has been discussed and negatived by the House, another no-confidence motion cannot be moved in the same session.

Implications of the No-Confidence Motion:

- The passing of a no-confidence motion leads to the resignation of the Council of Ministers, whichmay lead to a political vacuum.

- The sudden removal of the political executive may lead to administrative confusion andpolicy uncertainty.
- Politically motivated no-confidence motions tend to affect the optimal time utilisation inParliament. For instance, up till the current motion, a total of 456 hours have been spent discussing no-confidencemotions in Parliament.
  If no other party is able to prove the majority, then the citizens have to bear the expense ofconducting general elections.

NON-MARKET ECONOMIES (NME)

GS II –BILATERAL GROUPINGS.

Context: Vietnam has urged the United States administration to promptly reclassify its status from "non-market economy" to "market economy".

Non-Market Economies (NME):

- In the United States, a **Non-Market Economy (NME)** refers to any foreign country that the **US Department of Commerce** determines **does not follow market-based cost or pricing structures.** Consequently, sales of goods in such countries may not accurately reflect their fair value.
- Countries in this list are Armenia, Azerbaijan, Belarus, China, Georgia, Kyrgyz Republic, Moldova, Russia, Tajikistan, Turkmenistan, Uzbekistan and Vietnam.

Criteria for getting a Non-Market Economy (NME):

- If the country's currency is **convertible**.
- If wage rates are determined by free bargaining between labour and management.
- If joint ventures or other **foreign investments** are allowed
- Whether the means of production are owned by the state.
- If the state controls the allocation of resources and price and output decisions.
- Other factors like **human rights.**

Vietnam's Arguments Regarding its Non-Market Economy (NME) Status:

- Vietnam's currency is **convertible** into other currencies transparently based on market principles.
- Wage rates result from **free bargaining** between labour and management.
- Foreign investment is **permitted**, and Vietnam has become an attractive destination for it.
- The government does not **own or control** the means of production significantly.
- The government does not have significant control over **resource allocation** or price/output decisions.
- Vietnam's economy operates on **market principles**, including legal frameworks, corporate governance, and diversified foreign relations.
- **Vietnam's Centre for WTO and International Trade** has said that the method used to calculate anti-dumping duties is flawed because it results in artificially high dumping margins, which do not accurately reflect the actual practices of Vietnamese companies.

US ApprehensionsRegarding the Issue:

- The US Commerce Department is currently reviewing Vietnam's status.
- The US steelmakers and the American Shrimp Processors Association have asked the US administration not to change Vietnam's status to a market economy.They cited Vietnam's restrictions on land ownership, weak labour laws, and lower shrimp duties that would hurt their members as reasons for their request. The change in Vietnam could benefit Chinese state firms invested in Vietnam by allowing them to bypass US tariffs more easily.

Non-Market Economies (NME)and World Trade Organisation (WTO):

- The **WTO does not explicitly recognize or endorse the NME status.** However, it allows members to use alternative methodologies to calculate normal values in antidumping investigations.

The **WTO Antidumping Agreement** provides flexibility for members to choose an appropriate methodology for NMEs. It does not prescribe a specific approach.

PRODUCTION-LINKED INCENTIVES (PLI) SCHEME

GS III –Growth and Development
GS II – Government policies and interventions
Context: The government has decided not to launch the Production-Linked Incentives (PLI) Scheme for new sectors and will assess the need for course correction for some sectors.

Production-Linked Incentives (PLI) Scheme:

- It was first introduced in March 2020, PLI isa special incentive scheme forproducers to promote manufacturingin multiple sectors.The incentives are linked to the performance of the organisation i.e., the government provides incentives forincremental sales.
- The PLI scheme is built on the foundation of multiplesectors with an incentive outlay of Rs. 1.97 lakh crore(about US$26 billion) to strengthen their productioncapabilities and help create global champions.

Benefits of the PLI Scheme:

- Since the incentives are directly proportional toproduction capacity/ incrementalturnover, it is expected to have a direct impact oncapacity expansion.
- PLI can significantly restructure India'sdomestic manufacturing, and push its share in theGDP to 25 percent.It is also expected to bring improvements inindustrial infrastructure, benefiting the overallsupply chain ecosystem.
- The implementation of the PLI Scheme will lead to a potential capital expenditure(capex) of Rs 2.5-3 lakh crore over the schemeperiod.It will account for 13-15% of average annualinvestment spending in key industrial sectors.
- Nearly 55% of the schemeis expected to be in green sectors such as electricvehicles, solar photovoltaics etc.
- PLI Schemes intend to plug the gap between thehighly skewed Indian import-export basket, which is mainly characterized by heavy imports of rawmaterials and finished goods.It will help in offsetting the effects of geopolitical conflicts and their consequent impact on food, fertilizer, andcrude oil prices.
- It fosters faster adoption of technology, e.g., in telecom and networking products, where timely intervention by thescheme will enable faster adoption of 4G and 5G products across India.
- The PLI scheme enables the smooth implementation ofregion-specificincentives to promote industrial development in underdeveloped or backward regions.

Issues/Challenges to the PLI Scheme:

- The manner inwhich incentives are to be awarded remainsambiguous. There are no set criteria or commonparameters for consideration for giving theseincentives.
- The efficacy ofmanufacturing depends on multiple factors like rawmaterials, the size of the domestic market, andthe relationship between upstream and downstreammanufacturers, etc.Thus, production subsidies to scale sector-specificmanufacturing will not work until other criticalfactors shaping the ecosystem are understood andfactored in.
- Withdrawing from these benefits at a later stage mayultimately lead to industrial inefficiencies andengendera decline in productivity both at the sectoral and firm levels.
- Beneficiaries under the scheme such asautomobiles, electronics, andtechnical textiles are largely constituted by big firms.This is notrepresentative of the actual configuration of the Indian industrial structure, which is largelycomposed of MSMEs.
- The lack of a centralized database that captures information like increases in production orexports makes the evaluation process difficult.This information ambiguity impacts transparency and can lead to malfeasance, further widening the fault linesand weakening the policy structure.

Way Forward:

- There should be dynamic collaborations betweenindustry players, research institutions,and academia to foster innovation, research, and development.
- For targeted implementation, a strong, technicallyevolved, institutionalmechanism accountable to thehighest levels of governance is necessary.
- To ensure streamlined and efficient implementationof the PLI scheme, steps likereducing bureaucratic hurdles and simplifying application processes can be considered.
- Regular monitoring and evaluation of the PLIscheme's outcomes can help identify areas forimprovement and make necessary adjustments toachieve the goal.
- Separate provisions can be created within the PLIscheme to incentivize Micro, Small, and MediumEnterprises (MSMEs).
- PLI scheme should be followed by measures to further improve the businessenvironment in the country through a transparent andpredictable policy framework.
  Alongside the PLI scheme, there should be a focus on supporting skill development programs andtechnologyupgrade initiatives.

ROAD SAFETY

GS II –GOVERNMENT POLICIES AND INTERVENTIONS

Context: The National Highways Authority of India (NHAI) has issued guidelinesfor rectification of accident-prone spots on the National Highwaysby implementing short-term measures.

Factors responsible for Road Accidents:

- Sometimes highways come uphaphazardly across the roads, and street designpermits speeding, which leads to road safetyconcerns. The problem is further aggravated due to inadequate signage, road markings or incorrect location, and poorconstruction of speedbreakers.
- Ongoing construction works astride roadsrestrictthe availability of restricted space to the road user.Improper road markings, lack of trafficcontrol, etc. at such sites furtherincrease the vulnerability
- Heavy rain, dense fog, and hail storms reduce visibility and make theroad surface slippery thus posing serious risksto the road users.
- Sidewalks notbeing available for pedestrian use who thenmove on the roads thereby becomingvulnerable to accidents.
- Old vehicles are relatively moreprone to breakdown and malfunction.It can result in the bursting of tyres, roadcollapse, wearing out of brakes, overturningof the vehicle, etc.
- Violation of traffic rules, non-use of safety devices (helmets and seat belts), triple riding, overloading, distracted driving, road rage, etc.
- Suboptimal training and testing, and lack of coordination among differentagencies.

Indian Initiatives Related to Road Safety:

- **Motor Vehicles Amendment Act of 2019** hikes the penalties for traffic violations, defective vehicles, juvenile driving, etc. It provides for a **Motor Vehicle Accident Fund,** which would provide compulsory insurance coverage to all road users in India for certain types of accidents. It also provides for a National Road Safety Board, to be created by the Central Government.
- **The Carriage by Road Act of 2007** provides for the regulation of common carriers, limiting their liability and declaration of the value of goods delivered to them to determine their liability for loss of, or damage to, such goods occasioned by the negligence or criminal acts of themselves, their servants or agents and for matters connected therewith or incidental thereto.
- **The Control of National Highways (Land and Traffic) Act of 2000** provides for the control of land within the National Highways, right of way, and traffic moving on the National Highways and also for removal of unauthorized occupation thereon.

- **National Highways Authority of India Act of 1998** provides for the constitution of an authority for the development, maintenance, and management of NHs and for matters connected therewith or incidental thereto.

Data:

- According to a report titled, **'Road Accidents in India-2022' published by** the Ministry of Road Transport and Highways, a total of 4,61,312 road accidents occurred in India, leading to 1,68,491 fatalities and 4,43,366 people injured in 2022.These figures represent an 11.9% year-on-year increase in accidents, a 9.4% rise in fatalities, and a substantial **15.3% surge in the number of people injured** compared to the previous year.

Global Initiatives Related to Road Safety:

- **The Brasilia Declaration on Road Safety (2015)was** signed at the second Global High-Level Conference on Road Safety held in Brazil. India is a signatory to the Declaration.The countries plan to achieve **Sustainable Development Goal** 3.6 i.e., to halve the number of global deaths and injuries from road traffic accidents by 2030.
- **Decade of Action for Road Safety 2021-2030 by t**he **UN General Assembly** adopted a resolution on "improving global road safety " with the ambitious target of preventing at least 50% of road traffic deaths and injuries by 2030.The Global Plan aligns with the **Stockholm Declaration**, by emphasizing the importance of a holistic approach to road safety.

The International Road Assessment Programme (iRAP) is a registered charity dedicated to saving lives through safer roads.

SHIFTING DIPLOMATIC POSTURE OF JAPAN

GS II –EFFECT OF POLICIES AND POLITICS OF DEVELOPED AND DEVELOPING COUNTRIES ON INDIA'S INTERESTS

Context: In the recent past of changing geopolitics, the world is witnessing a turning point as Japan, long a symbol of post-war pacifism, strengthens its military capabilities.

Japan's Shifting Diplomatic Posture:

- Japan has **significantly increased its defence budget,** ending the self-imposed cap of 1% of GDP.From 1960 to 2020, Japan's military spending remained at or **below 1% of GDP. Japan's** Prime Minister announced that Japan will **increase annual defence spending to 2% of GDP by 2027.**
- Japan is **acquiring new military capabilities**, including offensive weapons like cruise missiles, and easing restrictions on arms exports.
- Japan is working closely with the **US on joint military exercises** and exploring deeper integration of command structures.The key **Japan-US joint military exercises** are Keen Sword, Orient Shield, and Valiant Shield (a ballistic missile defence-focused exercise).The **Global Combat Air Programme (GCAP)** is a multinational initiative led by the **United Kingdom, Japan, and Italy** to jointly develop a sixth-generation stealth fighter by 2035.
- Japan has decided to **ease its strict defence export rules**, allowing it to **collaborate with Britain and Italy** to build a next-generation fighter jet for export under certain conditions.
- Japan is strengthening its relationships with other regional powers like **India andAustralia,** promoting a **"free and open Indo-Pacific" vision through the Quadrilateral Security Dialogue (QUAD) and Pacific Island Forum (PIF).**
- Japan's strong stance in support of Ukraine against Russia is seen as a signal of its commitment to upholding international norms and deterring similar aggression in Asia.
- Japan is attempting to **reconcile with South Korea,** a historical adversary, in an effort to create a more cohesive regional security architecture.

Factors that led Japan to Shift its Diplomatic Posture:

- China's growing military power and **assertive claims in the East China Sea,** particularly regarding disputed territories like the **Senkaku Islands**, have created a sense of urgency for Japan to strengthen its defences.
- North Korea's continued **development of nuclear weapons and ballistic missiles** remains a major security concern for Japan.

- The **perceived erosion of US commitment to Asian security** under the Trump administration, along with the **rise of isolationist tendencies in the US,** has pushed Japan to become **more self-reliant** in its defence.Examples include the **failure of the USA's Middle East policy to maintain peace.**
- A growing number of **conservative voices** in Japan advocate for a **more active security role** and argue that Japan, as a normal power has a responsibility to contribute to regional stability and defend its interests.
- Decades of relying solely on the US for security have led some to question the sustainability of this approach, especially in the face of a changing regional landscape.

Significance of Japan's Shifting Diplomatic Posture on India:

- Both India and Japan see **China as a strategic concern.** Japan's increased military capabilities and focus on the Indo-Pacific could strengthen the ability of both countries to deter Chinese aggression.**Both India and Japan** are members of **the Quad grouping, G20 and G-4, International Thermonuclear Experimental Reactor (ITER).**The **India-Japan Act East Forum** was established in 2017 which aims to provide a platform for India-Japan collaboration under the fabric of **India's "Act East Policy**" and **Japan's "Free and Open Indo-Pacific Strategy".**
- The new strategy emphasises cooperation with **like-minded countries like India.** This could lead to **more joint military exercises, technology sharing,** and potentially relaxed export restrictions on Japanese defence equipment for India.Japan is one of the few countries with whom India has a **2+2 ministerial dialogue.**India and Japan's defence forces also organise a series of bilateral exercises such as **JIMEX (naval), Malabar exercise (Naval Exercise), 'Veer Guardian'** and **SHINYUU Maitri** (Air Force), and **Dharma Guardian (Army).**
- The new **Japanese Official Development Assistance (ODA) Loan framework** for strategic purposes could provide India with much-needed funding for infrastructure projects in border areas with China. This would improve India's defence preparedness and connectivity.**India** has been the **largest recipient** of the **Japanese ODA Loan framework** for the past decades.**Delhi Metro** is one of the most successful examples of **Japanese cooperation** through the utilisation of ODA.India's **Western Dedicated Freight Corridor (DFC)** project is funded by a soft loan provided by the Japan International Cooperation Agency.
- A stronger Japan could be a more reliable economic partner for India, potentially leading to increased trade and investment.**Japan's bilateral trade** with India **totalled USD 20.57** billion during FY 2021-22. India was the 18th largest trading partner for Japan, and Japan was the 12th largest trading partner for India in 2020.

Issues/ChallengesDueto Japan's Shifting Diplomatic Posture on India:

- Both India and Japan are developing long-range strike capabilities. This could lead to an arms race in the region, potentially straining resources.Competition between Japan and India to **export defence equipment** in common nature of market and allies like Africa, Philippines, and South America might harm India's interests in the long run.
- It might become challenging for India to balance more **assertive powers** in competing blocks like the **Quad grouping** and **BRICS.**
- Ideological conflicts may arise in areas such as human rights, nuclear proliferation, and international interventions, where India's stance can differ from the Japanese stance.

COLLEGE AUTONOMY

GS II –Education

Context: The National Education Policy 2020 envisages a future where colleges evolve into autonomous institutions, characterised by innovation, self-governance, and academic freedom.**Granting autonomy to colleges is seen as pivotal for promoting innovation,** enhancing academic quality, and fostering institutional excellence.

Benefits of College Autonomy:

- Autonomous colleges have the freedom to design their **curriculum according to the specific needs and demands of their students, industries, and local communities.**This flexibility enables them to offer

innovative courses, update existing ones, and integrate interdisciplinary approaches, ensuring that graduates are well-prepared for the dynamic job market and societal challenges.

- Autonomous colleges are **better positioned to promote research culture among faculty and students.**Additionally, **autonomy facilitates collaborations with industries, other academic institutions,** and research organisations, fostering interdisciplinary research and knowledge exchange.
- The **decentralised governance structure allows for quicker responses** to emerging trends, opportunities, and challenges in higher education.It also **promotes transparency and inclusivity in decision-making processes**, involving various stakeholders such as faculty, students, alumni, and industry experts.
- Autonomy **creates an entrepreneurial mindset among college stakeholders by encouraging initiatives such as start-up incubators**, entrepreneurship development cells, and industry-academia partnerships.Colleges **can offer courses in entrepreneurship, provide mentorship to aspiring entrepreneurs**, and facilitate access to funding and networking opportunities, thereby contributing to economic growth and job creation.
- **Autonomous colleges prioritise the cultivation of critical thinking**, creativity, and problem-solving skills among students.

Significance of College Autonomy:

- The National Institutional Ranking Framework (NIRF) of 2023 highlights the positive impact of autonomy on academic excellence, with **a significant portion of top-ranked colleges being autonomous institutions.**Besides, **in the top 10 colleges of the NIRF Rankings of 2023 from the college category, five are autonomous colleges.**
- Having half the top spots occupied by autonomous colleges significantly strengthens the case for autonomy as a successful approach to achieving academic excellence.
- Higher education in India is witnessing a marked trend towards establishing autonomous colleges, **with the number soon expected to reach 1,000 across 24 States and Union Territories.States like Andhra Pradesh, Karnataka, Maharashtra, Tamil Nadu, and Telangana lead in establishing autonomous colleges**, indicating a nationwide interest in exploring autonomy's potential.

Challenges Faced by Autonomous Colleges:

- Despite being granted autonomy by the UGC, **some universities may impose restrictions on the extent of autonomy enjoyed by colleges.**This could include caps on syllabus changes, limiting colleges to modifying only a fraction of their curriculum.**Such limitations hinder colleges' ability to innovate and tailor their academic offerings** to meet evolving needs effectively.
- **Colleges may still find themselves entangled in bureaucratic processes** controlled by the university, limiting their ability to make independent decisions.
- While the UGC grants complete autonomy to colleges, **some universities may exhibit reluctance to cede control,** particularly in critical areas such as syllabus design, introduction of new courses, and assessment methods.**This reluctance may stem from a traditional hierarchical approach to governance within the university system,** where centralised control is favoured over decentralised decision-making.
- Autonomous colleges **may encounter challenges in implementing internal policies** and procedures without interference from external authorities.**This could include resistance from faculty or staff accustomed to the traditional hierarchical structure** or difficulty in enforcing decisions without the backing of university authorities.
- With autonomy comes the responsibility of maintaining and enhancing academic standards.However, **there is a risk that colleges may vary in their approach to quality assurance,** leading to inconsistencies in educational offerings and outcomes.
- Navigating the regulatory requirements while maintaining autonomy can be challenging, particularly for colleges with limited administrative capacity or expertise.**Collaboration with regulatory bodies and adherence to best practices can help colleges navigate these challenges.**

Autonomous colleges **may require support and capacity-building initiatives to effectively exercise autonomy.**Universities should provide training, workshops, and resources to college administrators, faculty, and staff on topics such as curriculum development, quality assurance, financial management, and governance.**This**

capacity-building support will enable colleges to leverage autonomy to drive innovation, excellence, and inclusivity in higher education.

PAPER 3

BALANCED FERTILIZATION

GS III –AGRICULTURE

Context: Balanced fertilization — discouraging farmers from applying too much urea, di-ammonium phosphate (DAP), or muriate of potash (MOP), which only have primary nutrients in high concentrations — is likely to be a key policy goal for the government taking over after the Lok Sabha elections.

Balanced Fertilization:

- It is a practice in agriculture that focuses on providing plants with the **optimal amounts of nutrients** that are needed for their healthy growth and development.
- Balanced fertilizationemphasizes supplying these essential nutrients in the correct ratios based on several factors:
  - Different soil types have varying levels of inherent nutrients. Testing the soil reveals its **nutrient profile, guiding fertilizer selection and application rates.**
  - Different crops have **specific nutrient needs at different stages of growth.** For example, legumes might require more nitrogen for nitrogen fixation, while fruits might benefit from additional potassium for better quality.

Benefits/Significance of Balanced Fertilization:

- By providing the **optimal mix of nutrients,** plants can grow to their full potential, leading to higher yields.
- Balanced nutrients contribute to stronger plants with **better resistance to pests and diseases**, ultimately improving the quality of the harvest.
- Overuse of **single-nutrient fertilizers** can disrupt soil health. Balanced fertilization helps maintain a **healthy soil ecosystem,** promoting long-term sustainability.
- Excessive fertilizer application can contribute to **nutrient runoff, polluting water** bodies. Balanced use minimizes this risk.
- By **preventing over-fertilization** and nutrient deficiencies, balanced fertilization can optimize resource utilization and reduce overall fertilizer costs.

Issues/Challenges related to Balanced Fertilization:

- **Urea**, a single-nutrient nitrogen fertilizer, is heavily subsidized by the government, making it c**heaper than other fertilizers like DAP** (diammonium phosphate) containing phosphorus and MOP (muriate of potash) containing potassium.This incentivizesthe overuse of urea and neglects other crucial nutrients.
- The current system for setting **fertilizer prices fails to consider market forces,** leading to imbalances. For example, the price of Muriate of Potash (MOP), a key source of potassium, is **too high for both** farmers using it directly and fertilizer companies including it in blends.This **discourages MOP use,** contributing to widespread potassium deficiencies in Indian farms.
- **The lack of adequate soil testing facilities** in the rural and remote areas of India makes it difficult for farmers to access them for balanced fertilization.Even if tests are done, **interpreting the results and translating them** into fertilizer recommendations requires proper training and mechanisms for both farmers and extension workers.
- Traditional practices and limited knowledge mostly hinderthe adoption of balanced **fertilization techniques.** It is due to the lack of precise fertilizer application techniques resulting in issues of over-fertilization and under-fertilization alongwith **a limited focus on micronutrients.**
- The **Nutrient-Based Subsidy (NBS) Scheme**, meant to encourage balanced use, failed because it didn't address urea pricing. Urea consumption continued to rise despite NBS.

Essential Nutrients for Plant Growth:

- **Primary Nutrients: Nitrogen (N), Phosphorus (P), and Potassium (K)** are the most crucial nutrients needed in larger quantities. They play **vital roles in plant** structure, energy production, and overall health.
- **Secondary Nutrients**: Sulphur (S), Calcium (Ca), and Magnesium (Mg) are also essential but **required in smaller amounts** compared to primary nutrients.
- **Micronutrients:** Trace elements like Iron (Fe), Zinc (Zn), Copper (Cu), Manganese (Mn), Boron (B), and Molybdenum (Mo) are needed in very small quantities but are still critical for specific plant functions.

CAPITAL EXPENDITURE (CAPEX) BY STATES

GS III –GOVERNMENT BUDGETING

Context: The Department of Expenditure, Ministry of Finance has approved capital investment proposals of Rs. 56,415 crores in 16 States in the current financial year. Approval has been given under the scheme entitled 'Special Assistance to States for Capital Investment 2023-24'.

Significance of Capex by States:

- Capital spending can induce high multiplier benefits,crowd in private investment, and ease criticalsupply constraints.Combined capex spending is more than that of the centre and thus they play a significant role.In FY22, statesspent ₹10.5 trillion on capexcompared to the centre's ₹8.4 trillion.
- Capital spendingcontributes to future economicgrowth through the creation of employment andcapacitydevelopment.
- A sustained improvement in the quality of publicexpenditure, with a focus on infrastructure,research and development, health, education, and other socialservices can play a conducive role in promotinginclusive and sustainable economic growth.
- The impact of capex in one State can have positiveexternalities on that of others.

Trends in Capital Expenditure (Capex) by States:

- The expenditure by states on subsidies rose at afaster pace even as their capitalspending slowed down in FY23.Their capex growth was flat in FY23 compared with 29% growth in FY22.
- The states' capital outlay-GDP ratio is expected to improve from 2.3% in 2021-22 to 2.9%in 2022-23.
- The state's debt-to-GSDP ratio was the highest inPunjab (48%), followed by Rajasthan andHaryana.
- According to a study by the Bank of Baroda, 25states have cumulatively achieved 76% of their capextarget in FY23.

Reasons/Issues for failure in increasing Capex by States:

- States are legally mandated underthe Fiscal Responsibility and Budget Management (FRBM)Act to keep their fiscal deficit under control.Unable to reduce their revenue expenditure onsalaries, pensions, subsidies, etc., they end up cuttingdown on capex to meet the deficit norms.
- States are also constrained by a lack ofprojects or skills in the absorption of high capitalexpenditure. There arestate-level political disruptions such as governmentinstability or elections which come in the way of decision-making about capex.The election cycle is also fraught with freebies and populist promises which results in wasteful expenditure.

Scheme for Special Assistance to States for Capital Expenditure:

- It was launched in FY 2020-21 in the wake of the **COVID-19 pandemic and has** been expanded and continued as a 'Scheme for Special Assistance to States for **Capital Investment 2023-24' with allocation of Rs 1.3-lakh crore.**
- The scheme is divided into the following parts. They are:
  - **Part Iis the largest with the allocation of Rs. 1 lakh crore**. This amount has been allocated amongst states in proportion to their share of central taxes and duties as per the award of the **15th Finance Commission**.

- **Part II provides incentives to states** for scrapping old vehicles and setting up automated vehicle testing facilities;
- Parts III and IV provide incentives to states for reforms in **urban planning** and urban finance;
- Part V provides funds for increasing the housing stock for police personnel and their families within the police stations in urban areas.
- Part VI of the scheme supports the vision of national integration, **Make in India, and One District One Product** by promoting cultural diversity and local products through Unity Mall projects.
  Under Part VII, Rs. 5,000 crore is provided as financial assistance to states for establishing libraries with digital infrastructure at the **Panchayat** and Ward level, primarily benefiting children and adolescents.

CARBON FARMING

GS III –AGRICULTURE

Context: In light of escalating environmental concerns and the pressing need for climate-resilient agricultural practices, carbon farming has emerged as a crucial mechanism within the agriculture sector globally.

Carbon Farming:

- Carbon farming represents a strategic approach aimed at maximising **carbon sequestration** and employing agricultural practices designed to enhance the absorption of carbon dioxide (**CO2)** from the atmosphere while facilitating its retention in both **plant biomass** and **soil organic matter.**
- This process requires **careful planning, monitoring, and adaptation to local conditions** to maximise its effectiveness in mitigating climate change.

Significance of Carbon Farming:

- Carbon farming is pivotal in the fight against **climate change** by sequestering carbon in soil and curbing greenhouse gas emissions.
- By nurturing healthy soil, carbon farming bolsters water retention, diminishes erosion, and boosts nutrient availability, resulting in amplified crop yields and agricultural productivity.Converting **organic waste into compost** can be used as a soil amendment to improve soil structure, fertility, and carbon content.
- Carbon farming fosters biodiversity by fostering intricate ecosystems in agricultural settings, attracting beneficial insects and pollinators that fortify crop health and lessen dependence on pesticides.
- Implementation of carbon farming practices opens avenues for farmers to tap into **carbon credit markets,** alongside potentially augmented yields from enriched soil, thus diversifying income streams and bolstering financial resilience.

Potential Opportunities for Carbon Farming in India:

- India's extensive agricultural base presents significant economic opportunities through the adoption of carbon farming practices with an estimated potential of **$63 billion** from approximately **170 million hectares** of arable land.
- Implementation of carbon credit systems can offer additional income streams to Indian farmers by recognising their contributions to environmental services.Agricultural soils in India have the potential to **sequester 3-8 billion tonnes of CO2 equivalent annually over 20-30 years,** thus providing opportunities for farmers to participate in carbon trading markets.
- Different regions of India offer varying degrees of suitability for carbon farming initiatives.The fertile plains of the Indo-Gangetic region and the **expansive Deccan Plateau** are particularly favourable for implementing carbon farming practices.

Challenges related to Carbon Farming:

- Soils with **poor structure or low organic matter** may have limited capacity for carbon storage and may require amendments or management practices to enhance fertility and carbon sequestration potential.

- Geographic factors like **elevation, slope, and proximity to water bodies** also impact land use options and agricultural productivity.**For example,** high-altitude regions may have limited crop options due to colder temperatures, while coastal areas may face challenges related to saltwater intrusion and soil salinity.
- The selection of crop varieties suitable for specific soil types, climates, and growing seasons is critical for optimising agricultural productivity and carbon sequestration potential, varieties that are well-adapted to local conditions and **resilient to pests, diseases, and extreme weather events** can enhance crop yields and contribute to soil health and carbon storage.However, limited availability of diverse crop varieties or lack of access to improved seeds may constrain farmers' ability to maximise carbon farming benefits.
- Adequate water is essential for **plant growth and photosynthesis,** which are fundamental processes for carbon sequestration.Arid regions face challenges in carbon farming due to inadequate water availability, which impedes plant growth and reduces the capacity for carbon sequestration.
- **Small-scale farmers** in developing nations like India often encounter financial barriers, struggling to afford the initial costs associated with implementing sustainable practices essential for carbon farming.
- The absence of robust policy frameworks and insufficient community engagement hinder the widespread adoption of carbon farming practices, impeding its potential impact on mitigating climate change.

CHOCOLATE INDUSTRY MELTDOWN

GS III –AGRICULTURE

Context: The chocolate industry is facing a crisis as the price of cocoa beans surges, reaching a record USD 12,000 per ton in April 2024.

Reasons Behind the Rising Prices of Cocoa:

- The immediate reason for the ongoing crisis is the **bad harvest season in West African countries Ghana and Ivory Coast**, where 60% of the world's cocoa beans come from.
- The development of **El Niño,** a weather pattern that refers to an abnormal warming of surface waters in the **equatorial Pacific Ocean**, led to **heavier-than-usual rainfalls in West Africa**, creating an ideal ground for the spread of **black pod disease**, which causes cocoa pods to rot on the branches of cocoa trees.
- **Climate change** is also a driving factor, with h**eat waves, droughts**, and heavy rainfalls further threatening cocoa production, posing long-term challenges for farmers and chocolate manufacturers alike.
- The underlying issue is that the big chocolate companies **do not pay enough to the cocoa farmers in West Africa,** who earn on average as little as less than USD 1.25 a day, which is well below the **United Nations absolute poverty line of USD 2.15 per day.**
- Farmers are **not able to invest in land to increase yield or build resilience against climate change** due to the lack of funds, leading to the use of slave and child labour, and the selling off of land to illegal gold miners.As a result, the farmers continue to live in poverty, unable to invest in their land or adopt sustainable practices, **leading to a decline in production and price surge.**
- Despite the **huge profits made by the chocolate companies, they have not done much to help raise farmers' income**, contributing to the long-term exploitation of farmers and potentially leading to **higher chocolate prices for consumers in the long run.**

Potential Consequences of the Ongoing Crisis:

- The **International Cocoa Organisation (ICCO)** predicts a global shortfall of about 374,000 tonnes for the 2023-2024 season, exacerbating the scarcity of cocoa beans and driving up chocolate prices.
- The scarcity of cocoa beans is likely to persist, leading to further exploitation of farmers and a rise in chocolate prices.
- Experts believe that the prominent chocolate companies have the room to redistribute wealth down the supply chain, but unless they do so, the situation is unlikely to improve.

International Cocoa Organisation (ICCO):

- ICCO is an **intergovernmental organisation established in 1973 under the United Nations**.
- It is based in **Abidjan, Ivory Coast.**

- ICCO was created to implement the **first International Cocoa Agreement negotiated in Geneva** at a **United Nations International Cocoa Conference**.

Cocoa Production in India:

- Cocoa is best cultivated in coconut and **areca nut** gardens in India, where **30-50% of sunlight can be intercepted by cocoa under the areca nut canopy**.
- In India, it is mainly cultivated in **Andhra Pradesh, Karnataka, Kerala, and Tamil Nadu** mainly as intercrop with Arecanut and Coconut.
- **National Horticulture Mission** provides a subsidy of Rs 20,000 per hectare to cocoa farmers in Andhra Pradesh for the first three years.
- **Central Plantation Crops Research Institute** Conducts systematic cocoa improvement programs with germplasm introductions.

CLIMATE MIGRATION

GS III –DISASTER MANAGEMENT

Context: The issue of climate migration has garnered significant attention, yet the world still lacks a comprehensive legal framework to protect individuals forced to flee their homes due to increasingly severe weather disasters.

Causes of Climate Migration:

- Reports by the **UN Office for the Coordination of Humanitarian Affairs (OCHA)** highlight that sudden-onset disasters like **floods**, **hurricanes**, and **earthquakes** often cause significant internal displacement.People flee to safer grounds within their countries, but returning home can be difficult due to **destroyed infrastructure** and livelihoods.
- The **UN Refugee Agency (UNHCR)** emphasises how disasters often disproportionately affect **vulnerable populations.**These populations, lacking resources or living in high-risk areas, are more likely to be displaced and struggle to recover.
- The **IOM** reports that slow-onset disasters like **droughts, desertification, and salinisation** degrade land and water resources.This makes it difficult for people to sustain their livelihoods, pushing them to migrate in search of better opportunities.
- Reports by the **Intergovernmental Panel on Climate Change (IPCC)** warn of **rising sea levels** threatening coastal communities. This can lead to permanent displacement as homes and farmland become submerged.
- The **UN Department of Economic and Social Affairs (UNDESA)** acknowledges that migration due to climate change is rarely caused by a single factor.**Poverty, political instability, and lack of social safety nets** often combine with disasters to force migration.
- **World Bank** highlights the challenges in accurately quantifying climate migration.This makes it difficult to develop effective policies to support displaced people and build resilience in vulnerable communities.

Issues/Challenges Faced by the Climate Migrants:

- The **International Labour Organization (ILO)** warns that climate migrants often **lose their skills** and assets due to displacement.This makes it difficult for them to find new jobs and rebuild their livelihoods in unfamiliar environments.
- The **UN Refugee Agency (UNHCR)** reports that climate migrants often end up in informal work sectors with low wages and poor working conditions.They may also be more vulnerable to exploitation due to their precarious situation.
- **The World Bank** highlights that climate migrants often struggle to access basic services like healthcare, education, and housing in their new locations.This can lead to social exclusion and marginalisation.
- The IOM emphasises the difficulties climate migrants face **adapting to new cultures and languages.**This can hinder their ability to integrate into new communities.

- Reports by the **UN Office of the High Commissioner for Human Rights (OHCHR)** point out that there's no clear legal framework to protect climate migrants.They don't qualify for refugee status under current international law.
- The Journal of Environmental Law claims that climate change-induced displacement can lead to **statelessness,** particularly for those who move across borders.In 2021, the **World Bank, in its Groundswell report**, estimated that by 2050, some **216 million people** worldwide would be **internally displaced** due to the impacts of climate change.
- **The WHO** highlights the **psychological distress and trauma** climate migrants experience due to displacement and loss.Access to mental health services is often limited, further exacerbating their struggles.
- Climate migrants may be exposed to **new health risks** in their new locations, such as infectious diseases or extreme weather events. This is especially concerning for **children and the elderly.**

Encouraging **labour migration among countries as an adaptation measure** for climate-displaced populations can help mitigate the impacts of climate change on vulnerable communities.

Climate Refugees:

- According to the **International Organization for Migration (IOM)**, "climate migration" refers to the movement of a person or group of **people who are predominantly forced to leave their homes due to sudden or gradual environmental changes caused by climate change.**This movement can be temporary or permanent and can occur within a country or across borders.
- This definition highlights that climate migrants are primarily those who have **little choice** but to leave their homes due to the impacts of climate change.

CRITICAL MINERALS

GS III –MOBILIZATION OF RESOURCES

Context: The Ministry of Mines released the first-ever report of the country on “Critical Minerals for India”.

Critical Minerals:

- They are elements that are the buildingblocks of essential modern-day technologies and are atrisk of supply chain disruptions.
- For example, Antimony,Beryllium, Bismuth, Cobalt, Copper, etc.
- For India, major import sources of Cobalt are China, the US, and Japan;Lithium (Chile, Russia, China); Nickel (Sweden, China), etc.
- These minerals are now used everywhere frommaking mobilephones, and computers to batteries, electric vehicles, and greentechnologies like solar panels and wind turbines.
- Based on their individual needs and strategicconsiderations, differentcountries create their own lists.

Significance of Critical Minerals for India:

- Minerals such as lithium, graphite, cobalt,titanium, etc. are essential for the advancement of high-techelectronics, telecommunications, transport, etc.Self-sufficiency in critical minerals ensures self-reliance andaddresses the vulnerability in its supply chain.
- Some critical minerals are important forclimate-friendly technologies like electric vehicles, solar panels wind turbines, etc.
- Critical minerals such as lithium, nickel, and siliconare vitalto manufacturing technologies and materials used in the aerospaceand defence sector.
- Critical minerals are necessary for India to achieveits geo-economic goals, energy security, renewableenergy goals, mineral security, and commitment toelectric vehicles by 2030.

Issues/Challenges/Concerns associated with Critical Minerals:

- Some critical mineral assets for the country'smanufacturing sectors (particularly greentechnologies) are not ready to be mined. There arealso some critical minerals of which there exist noknown resourceswithin the country.
- Geographical concentration of extraction orprocessing in a few locations may lead to supplychain disruption or vulnerabilities.For example, three-fifths of rare earth minerals, essential for clean energy, electronics, and defence technologies,are mined in China, while more than fourth fifths are processed there.
- The supply of critical minerals isdependent on trade agreements which can be driven bydomestic interests.For example, Indonesia (host to the world's largestnickel reserves), has banned export of raw nickel ore,as they seek to develop a domestic nickel processingindustry.
- India lacks periodicassessments for judging the criticality of the mineralsneeded. This affects the prices that are paid for suchminerals and long-term risk mitigation plans.

These minerals lacksubstitutes and have low end-of-life recycling rates.

Initiatives taken regarding Critical Minerals:

- India- Australia Critical Minerals Investment Partnership will build new supply chains underpinned by critical mineralsprocessed in Australia to help India's plans to lower emissions from its electricity network and become a global manufacturinghub, including for electric vehicles.
- India has joined the Minerals Security Partnership (MSP) which is a US-led collaboration of 14 countries that aims to catalysepublic and private investment in critical mineral supply chains globally.
- Mines and Minerals (Development and Regulation) Amendment Bill 2021ensures transparency in the auction process ofmines and enhances domestic and foreigninvestment in the mining sector.

India, Japan, and Australia unveiled Supply Chain Resilience Initiative (SCRI)to enhance the resilience of supply chainsin the Indo-Pacific Region and reliance on China.

DEEPFAKES IN ELECTIONS

GS III –Awareness in the fields of IT and Computers

Context: The emergence of deepfakes in our electoral process raises significant concerns as they undermine our ability to distinguish reality from fabrication.

Deepfakes:

- It refers to **synthetic media** created through **AI technology**, aiming to manipulate or generate visual and audio content to **deceive or mislead individuals.**
- The term **deepfake** was coined in **2017** by an anonymous **Reddit** user who identified as "Deepfakes."This individual utilized Google's open-source **deep-learning technology** to produce and share pornographic videos.
- The creation of deepfakes involves a technique known as **generative adversarial networks (GANs),** comprising two competing neural networks:
  - **The generatorproduces** fake images or videos that closely resemble reality, while
  - **The discriminator**to differentiate between authentic and fake content.
- Its creation necessitates a substantial amount of data, often sourced from the internet or social media without consent, including photos or videos of both the source and target individuals.
- It constitutes a component of deep synthesis, an umbrella term encompassing technologies such as deep learning and augmented reality, utilized to generate text, images, audio, and video to construct virtual scenarios.

Significance/Advantages of Deepfake in Elections:

- Deep learning algorithms enable political parties and candidates to analyze extensive **voter data**, encompassing **demographics,** social media engagement, and voting history.
- **Natural language processing (NLP)** algorithms enable campaigns to analyze and interpret vast amounts of textual data, including social media posts, news articles, and public forums, and targeting the voters for personal benefits.

- Utilizing deep-powered **predictive analytics such as AI cloud,** parties can forecast election outcomes by scrutinizing diverse factors such as **polling data, economic indicators**, and sentiment analysis from social media.
- AI algorithms continuously scan various data sources, including social media, **news outlets, and opinion polls,** to gauge public sentiment and identify emerging trends.
- **Deepfake-empowered AI chatbots** and virtual assistants engage with voters on social media platforms, addressing inquiries, disseminating information about candidates and policies, and even encouraging voter participation.
- AI-driven deepfake tools play a crucial role in **detecting and preventing electoral fraud**, including voter suppression, manipulation of electronic voting systems, and dissemination of disinformation.By analyzing data patterns and anomalies, AI algorithms contribute to upholding the integrity of elections.
- Governments and electoral authorities leverage AI and deep technologies to monitor and regulate political advertising, identify breaches of campaign finance laws, and ensure adherence to electoral regulations.AI-powered tools facilitate **transparency and accountability** in the electoral process.

Issues/Challenges related to Deepfakes in Elections:

- **Creating deepfake content**, and bombarding voters with highly personalized propaganda, leading to confusion and manipulation.
- Deepfake videos of opponents can be generated using AI, tarnishing their image influencing voter perceptions, and giving birth to the **Deep Fake Election concept.**
- Deepfake models, particularly **Generative Artificial Intelligence(AI)**, can manipulate democratic processes by spreading disinformation. This risk is exacerbated by social media platforms reducing their **fact-checking** and election integrity efforts.
- Deepfakes AI models, including AGI, are susceptible to inaccuracies and inconsistencies, raising concerns about their reliability.Instances of **Google AI models** misrepresenting individuals have highlighted the potential dangers of unchecked AI.
- The use of deepfake in elections raises ethical questions regarding **privacy, transparency, and fairness.**AI algorithms may perpetuate biases present in training data, leading to unfair treatment or **discrimination** against certain voter groups.
- Lack of transparency in AI decision-making processes can erode public trust in electoral outcomes.
- Unequal access to AI resources may disrupt the level playing field in elections, favouring parties with greater resources.
- Regulating deepfakes in electoral campaigns is challenging due to rapid technological advancements and the global nature of online platforms.Governments and election authorities struggle to keep pace with evolving AI techniques and may lack expertise in regulating AI-driven electoral activities.

International Practices to Combat Deepfakes:

- China emphasizes **obtaining consent** and verifying identities in the use of deepfake technologies, providers of such technologies are mandated to secure consent from depicted individuals and **authenticate user identity** and measures are in place to facilitate recourse for individuals adversely affected by deepfake content.
- Canada focuses on pre-emptively addressing the harms of deepfakes through **widespread public awareness campaigns** and prospective legislation and these campaigns aim to educate the populace about the risks associated with deepfake technology.

DIGITAL MONOPOLY

GS III –IT and Computers

Context: The Ministry of Electronics and Information Technology (MeitY) has expressed strong opposition to Google's delisting ofcertain applications from the Play Store.

Digital Monopoly:

- It refers to a situation where a single company orplatform holds significant control over theirrespective digital ecosystems and markets.
- Its examples include big tech companies like Google,Amazon, Facebook, Apple, and Microsoft.

Threats Posed by Digital Monopolies:

- Exclusionary behaviour hinders business users and consumers from switching to third-party serviceproviders. E.g., App stores mandating the use of their payment systems.
- A digital enterprise according tothe favourable treatment of itsproducts on itsplatform, thus creating a conflict of interest.
- Combining or bundling core or essential services withcomplementary offerings, thusforcing users to buy related services.
- Using personal data for consumer profiling to offertargeted online services and products, thus raisingdata privacy concerns.
- Predatory pricing strategies, or intentionally settingprices below cost price to excludecompetitors.
- Exclusive agreements with business users or sellers,thus preventing them from dealing with otherenterprises.E.g., Apple restricts installation of third-party applicationson iPhones.

India's Initiatives to Prevent Digital Monopolies:

- Consumer Protection (E-Commerce) Rules of 2020were introduced to govern the operations of e-commerce platforms anddigital marketplaces. These rules aim to prevent anti-competitive practices, ensure transparency, and protectconsumer interests.
- The Competition (Amendment) Act of 2023 introduced newmerger notification thresholds and deal value thresholds tocapture acquisitions of high-value, data-rich firms thatcan potentially lead to digital monopolies.
- The Competition Commission of India (CCI) also investigatesand takes action against anti-competitive practices bydigital platforms.
- The Digital Personal Data Protection Act of 2023emphasizes the importance of data to prevent theconcentration ofdata in the hands of a few dominant players.
- The proposed Digital India Act (DIA) proposes to regulate avast array of digital enterprises including social mediawebsites, artificial intelligence-based platforms, and e-commerce enterprises.
- Draft National Data Governance Framework Policyaims to create standardized data management andsecurityfor non-personal and anonymised data across all government bodies.

SUSTAINABLE & INCLUSIVE DEVELOPMENT OF NATURAL RUBBER SECTOR (SIDNRS)

GS III –Industrial Policy

Context: According to a recent report by Invest India,
an Investment Promotion and Facilitation Agency, India'sE-Commerce Sector is projected to reach USD 325 billion by 2030.

E-commerce:

- It is a short form for **Electronic Commerce. It** encompasses the buying and selling of goods and services over the Internet.
- It eliminates geographical barriers, allowing transactions to occur seamlessly across borders.
- It includes a wide range of activities, from online retailing to **digital payments,** and continues to evolve with advancements in technology and changes in consumer behaviour.

Driving factors behind developing E-commerce:

- The rise in smartphone usage has been a significant catalyst for e-commerce growth in India. It has democratised access to online platforms.**1.18 billion** people, representing over **80%** of India's population, will have access to smartphones **by 2026.**

- The **Unified Payments Interface (UPI)** has emerged as a significant player in digital transactions, facilitating transactions worth **USD 1.5 trillion in 2022.**
- **The affordability of cheap internet** plays a pivotal role in India's internet penetration.Now one gigabyte of data is priced at approximately **USD 0.17 (Rs 13.5)**, which gives incentive to a substantial number of the population to opt for online activities. India ranks 7th on the list of countries with the cheapest mobile data.
- The growth of e-commerce in India has been supported by the development of efficient logistics and supply chain networks.Government initiatives such as the **National Logistics Policy** streamline deliveries to the last mile, enhancing logistical efficiency and cost-effectiveness.
- India's growing middle-class population and increasing disposable incomes have fuelled the demand for e-commerce.According to the World Economic Forum, nearly **80% of households in 2030 will be middle-income** in India.
- E-commerce offers consumers the convenience of shopping from the comfort of their homes or on the go, saving time and effort. For e**xample, f**ood delivery platforms like **Zomato and Swiggy** have gained immense popularity due to the convenience they offer to consumers, allowing them to order meals without leaving their homes or offices.
- E-commerce platforms provide consumers with a vast array of product choices and competitive pricing options, making it easier to find desired products at affordable rates.This has been a significant draw for consumers, particularly in smaller cities and rural areas **where product availability and pricing can be limited.**
- Recent reports highlight the growing prominence of **Rural-Centric E-Commerce.**It expects a significant portion of demand to originate from **tier 2-4 towns** and **rural areas by 2026.**

Issues/Challenges faced by E-commerce:

- Cases of counterfeit and substandard goods being sold on popular e-commerce platforms have been reported in India.It can undermine consumer trust and lead to legal and financial consequences for e-commerce companies.
- Internet penetration remains relatively low in certain areas. **Postal addresses are not standardised,** affecting logistics.Due to a lack of supply chain integration, high delivery charges, and more time taken to deliver the product.
- Clear legislation is needed to regulate e-commerce practices domestically and internationally.The **rise of social commerce**, where consumers can make purchases directly through social media platforms, poses a potential challenge to the traditional regulatory framework.
- The e-commerce industry is susceptible to technological disruptions, such as the **emergence of new business models, advancements in artificial intelligence,** and cybersecurity threats like data breaches, hacking, and phishing attacks.Customers are sceptical of paying by credit card due to the increasing **threat of fraud** by hackers.

EMPLOYMENT TRENDS IN INDIA

GS III –Employment

Context: India has been witnessing a significant surge in employment in recent years, with the creation of over 80 million additional jobs between 2017-18 and 2022-23.

Key Trends in Employment Growth:

- Analysis using **National Sample Survey Office (NSSO)** data from 1983 to 2023 shows consistent growth in principal employment across all sub-periods.
- **Principal employment,** which measures those working for the bulk of the year, has grown consistently since 1983.Every sub-period under consideration has seen growth in principal employment, with no instances of jobless growth.
- The period from 2017-18 to 2022-23 saw the fastest increase, with about 80 million additional jobs, translating to an **annual growth rate of 3.3%.**Despite long-term deterioration since 2000, recent years have seen improvements in key labour market indicators such as **labour force participation rate,**

workforce participation rate, and unemployment rate. These improvements coincided with **economic distress periods, notably before and during the Covid-19 pandemic.**

- Employment growth has been well-distributed across rural and urban sectors, and various industries **(manufacturing, agriculture, construction, services).**
- The employment growth has been highest for women, at over 8% annually.Employment among those **aged 60 and above has also grown at around 4.5% annually.**There are various reasons for this trend, including **increasing distress, improved access to resources like water and energy**, and greater flexibility in care-related work.
- **The Employment Condition Index**is based on seven labour market outcome indicators, including the percentage of workers in regular formal work, casual labourers, self-employed workers below the poverty line, work participation rate, average monthly earnings of casual labourers, the unemployment rate of educated youth, and youth not in employment and education or training.The **"employment condition index"** has improved between 2004-05 and 2021-22.However, some states **(Bihar, Odisha, Jharkhand, UP) have remained at the bottom throughout this period**.

Evolution of Employment Quality:

- Around 50% of jobs in the formal sector are **informal.**Approximately **82% of the workforce is engaged in the informal sector.Nearly 90% are informally employed.**
- A large part of the employment growth (44 million) is in the form of own-account workers and unpaid family workers.This could be a result of government schemes like **Pradhan Mantri MUDRA Yojana (PMMY),** which have provided significant funding to this segment. **Self-employment is the primary source of employment**, constituting 55.8% in 2022.**Casual employment (hiring employees on a work-as-needed basis)** accounts for 22.7%, and regular employment for 21.5%.

Trends in Wages and Salaries:

- Aggregate **wages** and salaries have seen relative stagnation in recent years.From 2017-18 to 2022-23, the average annual growth of salaries and wages was **6.6% in nominal terms but only 1.2% after accounting for inflation.**
- While there is no apparent wage distress, there is also no significant improvement in living conditions.Possible reasons include the dampening of wages due to a large influx of workers and stagnating labour productivity.

Trends in Youth Employment:

- **Youthemployment and underemployment increased between 2000 and 2019** but declined during the pandemic years.However, unemployment among youths, especially those with **secondary-level or higher education, has intensified over time.**
- In 2022, the share of unemployed youths in the total unemployed population was 82.9%, and the share of educated youths among all unemployed people increased to 65.7% from 54.2% in 2000.
- The unemployment rate among educated youths was **six times greater for those with secondary education or higher (18.4%)** and nine times higher for graduates (29.1%) than for persons who could not read or write (3.4%) in 2022.This was **higher among educated young women (21.4%) than men (17.5%),** especially among female graduates (34.5%) compared to men (26.4%).

Issues/Concerns related to Employment in India:

- While the economy is growing, many new jobs are informal, lacking security, benefits, or minimum wage.
- Though the unemployment rate might not be high, youth **employment is often of poorer quality.**This means young people might be over-educated for the available jobs or find themselves in precarious situations like the **gig economy**.
- Women's participation in the workforce hasn't grown as expected. Many end up in unpaid family work or low-paying self-employment instead of formal jobs.
- The **education system might not be aligned with the current job market needs.**

- A significant portion of the Indian workforce remains in the informal sector.This translates to **lower tax revenue for the government and limited social security benefits for workers.**
- As with many countries, automation poses a threat to certain sectors in India. This can lead to job displacement in industries like manufacturing.The rise of **artificial intelligence (AI)** may impact employment, particularly in the outsourcing industry in India, where some back-office tasks could be taken over by AI.
- Many of the workforce relies on informal or casual employment. This makes them **highly vulnerable to economic downturns or external shocks**, as seen during the Covid-19 pandemic.
  There is a significant demand for government jobs due to the lack of job creation in the private sector.This situation underscores the appeal of stable employment provided by the government.

Principal Employment:

- It **refers to the main job worked for the majority of the year,** while subsidiary employment is **typically part-time, of shorter duration, and in addition to the main job.**

FIXED DOSE COMBINATION (FDCS)

GS III –Health

Context: The Ministry of Health and Family Welfare prohibited the manufacture, sale, ordistribution of 14 fixed-dosecombination (FDC) medicines.

Fixed Dose Combinations (FDCs):

- It refers to products containing one or moreactive ingredients used for a particular indication.
- An Active Pharmaceutical Ingredient (API) is thebiologically active component of a drug product(tablet, capsule,cream, injectable) that produces the intended effects.
- If it is combined for the first time, it would fall underthe definition of a new drug.
- New Drugs require prior approval from the DrugsController General of India (DCGI) before beinglicensed by the State licensing authorities (SLAs).
- DCGI heads the Central Drugs Standard ControlOrganisation (CDSCO) which is responsible forensuring quality drug supply across the country.
- CDSCO works under the Ministry of Health & FamilyWelfare.

Advantages/Merits of FDCs:

- Advantages over the single entity preparations such asincreased efficacy, and/or a reduced incidence ofadverse effects,reduced cost due to proportional mixing of variousdrugs which reduces the cost of production, and simpler logistics of distribution relevant tosituations of limited resources.
- FDCs have shown tobe particularly useful in the treatment of infectiousdiseases like HIV, malaria, and tuberculosiswhere giving multiple antimicrobial agentsisthe norm.FDCs are also of use in chronic conditionsespecially when multiple disorders oftenco-exist.
- FDCs canreduce the ‘pill burden’ for customers andhence improve adherence.

Disadvantages/Demerits of FDCs:

- The mismatch between the two componentssuch as one drug havingadditive/antagonistic effectsleads toreduced efficacy or enhanced toxicity.
- Drugs having non-compatibility when mixed for FDCs may leadto decreased shelf life of FDCs.
- Diseases such asTuberculosis (TB) treatment in theabsence of Direct Observed Therapy Short course (DOTS) and therapeutic drug monitoring questionsoften get raisedabout FDCs effectiveness.
- The irrational FDCs with sub-standard doses ofantimicrobials, fail to offer adequatetherapeutic response to successfully kill the bugs and end up rather helping them develop resistance to the FDCs.

Issues in the Regulation of FDCs in India:

- To evade drug price control, some companiesreformulate individual drugs into an FDC.
- Compromised quality as New FDCs after 4 years getproduced by other manufacturers by getting licenses fromState licensing authorities (SLAs) without checking laxityconcerning pharmacological studies.
- Lack of coherence among stakeholders as SLAs have beenissuing manufacturing licenses for FDCs without priorclearance from DCGI.
- Poor capacity of State Drug Control Organisations (SDCO)such as infrastructure facilities, number and quality of druginspectors, etc.
- The adverse drug reaction reporting system is poor in India.
- Lack of data as India does not have the exact database ofcurrently available FDCs in the market, their sales turnover,and use patterns.

Steps taken by India for regulation of FDCs:

- Drugs & Cosmetics (Amendment) Act, 2008 providesstringent penalties for the manufacturer of spurious andadulterated drugs and certain offences have also beenmade cognizable and non-bailable.
- Testing capacities of Central Drugs Testing Laboratoriesunder CDSCO are being constantly strengthened toexpedite testing of drug samples in the country
- Drugs and Cosmetics Rules 1945 amended in 2017 provides that the applicant shall submit the result ofa bioequivalence study along with the application for grantof manufacturing License of oral dosage form of drugs.

Central Drugs Standard ControlOrganisation (CDSCO):

It is a statutory bodyestablishedasthe Central Drug Authority fordischargingfunctions assigned to the CentralGovernment under the Drugs and CosmeticsAct of 1940.

FLOODS IN DUBAI

GS III –Disaster Management

Context: Recently a storm hit Oman and UAE, with the UAE experiencing its heaviest rainfall in 75 years, raising questions about the role of cloud seeding in this unusual weather event.

Why has Dubai experienced heavy rainfall?

- A low-pressure system in the upper atmosphere, combined with low pressure at the surface, acted like a pressure squeeze on the air.
- **Heavy thunderstorm** intensified squeeze, along with the contrast between warmer ground-level temperatures and colder temperatures higher up, created the conditions for the powerful thunderstorm.
- Rising global temperatures, driven by human-led climate change, are leading to more extreme weather events worldwide, including intense rainfall.
- **A mesoscale convective system**is a series of medium-sized thunderstorms caused by massive thunderclouds, formed as heat draws moisture up into the atmosphere which can create large amounts of rain.
- In a recent study, Sharjahthe capital of the third-largest emirate in the U.A.E., found that the city's rapid growth over the past half century had made it vulnerable to flooding at far lower levels of rain than before.
- There were speculations about cloud seeding contributing to the rainfall, however, the metrology department clarified that no cloud seeding operations were dispatched before or during the turbulent weather episode.

Consequences of heavy floods in Dubai:

- *Al Ain*, a city in UAE bordering Oman recorded 10 inches of rainfall, the largest ever in a 24-hour period since records started in 1949.
- Heavy downpours have caused flooded homes, power outages, and emergency responses, Dubai Airport, the world's second busiest, was brought to a standstill following the rainfall.

The roads were collapsed, residential communities were hit by heavy flooding and many householders reported leaks from roofs, doors and windows.

Cloud Seeding:

- It is a weather modification technique that improves a cloud's ability to produce rain or snow, it can be done from ground-based generators or aircraft.
- Salts like silver iodide, potassium iodide, sodium chloride, or dry ice (solid carbon dioxide)are released into the targeted clouds. These seeding particles facilitate the formation of larger water droplets thereby augmenting rainfall.

GENE-DRIVE TECHNOLOGY

GS III –Biotechnology

Context: Genetically modified mosquitoes have been used in outdoor but controlled conditions in India, Brazil, and Panama whichshowed promising drops in mosquito populations, around 90% during the trials.

Gene-Drive Technology (GDT):

- GDT is a type of genetic engineering technique thatmodifies genes to alter the typical rules of Mendelianinheritance.
- Mendelian inheritance refers to certain patterns ofhow traits are passed from parents to offspring.
- A gene drive consists of three key components:
  - The gene to be spread;
  - The Cas9 enzyme tocut DNA; and
  - CRISPR is aprogrammable DNAsequence thatdetermines wherethe enzyme shouldcut.

Outcomes:

- The genetic material that encodes for these threeelements gets inserted into an animal's DNA, in placeof the naturally occurring gene.
- It increases the likelihood that a particular suite ofgenes will be passed onto the next generation,allowing thegenes to rapidly spread through a population and override natural selection.

Applications of GDT:

- GDT can be used to exterminate insects such asmosquitoes that can spread malaria, dengue, and theZika virus.
- Gene drives can also be used to control invasivespecies such as rodents.

Concerns/Issues related to GDT:

- Long-term disruption in an ecosystem like a disturbance inthe balance of food-web, potentially leading tounintended ecological consequences.
- Gene drive could get out of control and spreaduncontrollably across international borders.
- It could potentially be used for bioterrorism or othermalicious purposes.
- Deciding which species to target and the irreversiblenature of GDT raises complex ethical and moralissues.

CRISPR(ClusteredRegularlyInterspacedShortPalindromicRepeats):

- They are repetitive DNA sequencesdeveloped as a method ofrepurposing bacterial immunesystems tomake breaks in DNA atprecise locations.

Cas9:

It is an enzyme that acts as amolecular scissor that cuts the DNAat a specific location in the genome.

ILLEGAL IMMIGRATION IS A THREAT TO INTERNAL SECURITY

GS III –Internal Security

Context: Violence between Manipur's Kuki tribe and the majority Meitei community continued in several parts of Manipur.

Illegal Immigration is a threat to InternalSecurity:

- It is alleged thatamong the illegal migrants there are alsomilitants especially groups such asNSCN/KUFLA, who enter India to carryout terrorist activities.
- In recentdecades, trafficking of women andhuman smuggling has become quiterampant across borders.
- The Commission onIntegration and Cohesion found thattension usually exists with the presenceof high levels of migration combine withother forms of social exclusion likepoverty, poor housing, etc.
- Immigration has increasedpressure on the part of the government, as thegovernment has to increase the expenditure oneducation and health facilities for the immigrants.
- Most illegal immigrants have got theirnames enlisted in the voting list illegally, therebyclaiming themselves as citizens of the state.
- The influx of immigrants created a crisis of identityamong the indigenous people of the region.
- Large areas of forestland were encroached upon bythe immigrants for settlement andcultivation.This creates conflict in resource utilization with the local communities.

Laws in India to tackle Illegal Migration:

- Under the Foreigners Act of 1946, the centralgovernment can deport illegal foreign nationals.
- Under the Passport (Entry into India) Act of 1920, the state governments have been entrusted toremove an illegal foreigner by force.
- The Citizenship Act of 1955provides for the acquisitionand determination of Indian citizenship.

Way Forward:

- A considerable portion of the problem of illegalimmigration lies in the fact that we have no sucheffective refugee policy. The central governmentshould frame a holistic refugee policy.
- India has to make a diplomatic effort to getneighboring countries to cooperate as illegalmigrationcannot be solved unless the origin country cooperates.
- Government should introduce identity cards for theborderland people who frequently used tocross borders for different reasons.
- There is a vast area of the border that is notprotected by fencing. The central government shouldcomplete the border fencing.
- Regional forums like BIMSTEC can be used to discuss issues like illegal migration from neighboringcountries and garner support and coordination from the members.

Border guards like BSF, and ITBP should bestrengthened and a second line of defencecan be created by the State government through the state police force.

Ethnic composition of Manipur:

- There are 33 recognized tribes ofManipurthat either fall under the Nagas or the Kukis.
- Despite the socio-political and linguisticdifferences among them, all the ethnictribesoriginate from the same Mongoloid group.
- The tribals make up 40 % of Manipur'spopulation and live largely in the hills.

The Meiteis make up 53 % of thepopulationand live in the Imphal valley.

INDIAN POULTRY INDUSTRY

GS III –Economics of Animal-Rearing

Context: The H5N1 outbreak has highlighted critical vulnerabilities in the industrial livestock sector, underlining the imperative for a comprehensive reassessment of animal welfare within India's environmental and legal frameworks.

Issues faced by the Indian Poultry Industry:

- Regular outbreaks of **Avian Influenza (bird flu) disrupt production,** lead to the culling of birds, and create market panic, impacting consumption.
- **Newcastle Disease (ND)**is another highly contagious viral disease that affects poultry health and productivity.
- Inadequate **biosecurity measures** in farms and live bird markets facilitate the spread of diseases.Chickens in industrial settings are often **confined in wired cages, known as 'battery cages,'** in high densities, leading to overcrowding and stress.This practice leads to **poor air quality,** waste accumulation, and **greenhouse gas emissions**, contributing to environmental pollution and degradation.
- Volatile prices of **crucial poultry feed ingredients**, such as **corn** and **soybean meal**, not only affect production costs but also exacerbate **import dependence** due to their significant reliance on imports.
- Rumours and misinformation surrounding poultry products during disease outbreaks can drastically **reduce consumption**, affecting overall market stability.
- **Limited cold chain infrastructure** leads to spoilage and wastage, especially during peak production periods.
- A fragmented supply chain with **multiple intermediaries** raises transaction costs and lowers farmer profits, while poor transportation infrastructure hampers product movement, affecting delivery times and freshness.
- Multiple overlapping regulations across different levels of government create **confusion and compliance challenges** for poultry farmers.
- Small and medium-scale poultry farmers often struggle to **access formal credit,** hindering growth and modernization.
- Finding and retaining skilled labour can be difficult for poultry farms, impacting operational efficiency.
- Poultry farming can contribute to **water pollution** and air quality issues if waste management practices are inadequate.The rising demand for **protein has led to increased antibiotic** use in poultry farming, raising concerns about antibiotic resistance and public health risks.
- Ensuring proper animal welfare standards across the industry remains a challenge.
- Poultry farmers often face challenges in exiting the industry due to **contract farming arrangements**, accumulated debts, and specialised skills required for the sector.

Initiatives for the Poultry Industry:

- The **Department of Animal Husbandry and Dairying** is implementing the **Poultry Venture Capital Fund (PVCF)**under **"Entrepreneurship Development and Employment Generation" (EDEG)** of the **National Livestock Mission.**
- Different programmes under the **National Livestock Mission (NLM)** in which financial assistance is provided to States/Union Territories to implement **Rural Backyard Poultry Development (RBPD)** and Innovative Poultry Productivity Project (IPPP).
- **Assistance to States for Control of Animal Diseases (ASCAD) Scheme**under "**Livestock Health and Disease Control" (LH&DC)** which covers the vaccination of economically important poultry diseases viz., Ranikhet Disease, Infectious Bursal Disease, Fowl Pox, etc., including control and containment of emergent and exotic diseases **like Avian Influenza.**

Investing in **robust cold chain infrastructure** to minimise spoilage during transportation and storage is a global best practice.India can prioritise developing efficient cold chain networks, connecting remote production areas to major consumption centres.

Status of Poultry Birds in India:

- According to the **20th Livestock Census**, there are 851.8 million poultry birds in India. About 30% of this is **'backyard poultry**' or small and marginal farmers.

- Chickens, turkeys, ducks, geese, etc., are reared in poultry farms for meat and eggs.
- Tamil Nadu, Andhra Pradesh, Telangana, West Bengal, Maharashtra, Karnataka, Assam and Kerala have the highest poultry populations.

LAND SUBSIDENCE IN THE CHENAB VALLEY

GS III –Disaster Management

Context: There were reports of land subsidence in different parts of the Chenab Valley, especially in the Ramban, Kishtwar, and Doda districts, which led to the destruction of several houses.

Land Subsidence:

- According to the **National Oceanic and Atmospheric Administration (NOAA)**, l**and subsidence** is sinking off the ground because of underground material movement.
- It can happen for many reasons, man-made or natural, such as the removal of water, oil, or natural resources, along with mining activities. **Earthquakes**, **soil erosion,** and soil compaction are also some of the well-known causes of subsidence.
- It can happen over very large areas like whole states or provinces, or very small areas.

Causes of Land Subsidence:

- Over-extraction of resources like water, natural gas, and oil leads **to decreased pore pressure** and increased effective stress, causing ground subsidence.Over 80% of the world's extracted water is used for irrigation and agricultural purposes, contributing to ground subsidence.
- Exploitation of **underground solid mineral** resources leads to the **formation** of large empty spaces **underground (goaf),** which can lead to the ground sinking or subsiding.Mining activities, such as coal mining, can lead to the formation of goaf areas, which contribute to ground subsidence.
- The construction of tall buildings and heavy infrastructure can exert significant pressure on the ground, leading to soil deformation and subsidence over time.
- Continuous **low load and soil creep can cause slow deformation** of the foundation, contributing to ground subsidence.

Reasons for Land Subsidence in the Chenab Region:

- The region has the presence of **soft sedimentary deposits** and **alluvial soils,** which contribute to land subsidence.These materials are **prone to compaction** under the weight of overlying structures and the influence of external forces such as groundwater extraction.
- **Urbanisation** and unplanned construction in hilly regions put immense pressure on the land.The **Himalayan foothills** have witnessed rapid development, leading to land subsidence.
- The construction of **hydroelectric stations** can alter the natural flow of water and impact the stability of the land.
- Inadequate drainage systems in the **Chenab region** can worsen land subsidence through waterlogging, increased groundwater levels, **soil erosion**, saltwater intrusion, and infrastructure damage.
- Scattered **rocks** in the area are covered with old landslide debris comprising boulders, gneissic rocks, and loose soil, with a low bearing capacity.These gneissic rocks are highly weathered and have a low cohesive value with a tendency of high pore pressure when saturated with water, especially during **monsoons.**

Taking measures to address climate change, such as reducing **greenhouse gas** emissions and promoting sustainable practices, can slow down glacial melting and mitigate associated subsidence.

Note:

- **Soil creep is the slow, gradual movement of soil** downhill due to gravity and can contribute to ground subsidence over time.

MARINE CLOUD BRIGHTENING (MCB)

GS III –Environmental Pollution

Context: Scientists are testing a **geoengineering technique called marine cloud brightening that** involves using machines **to inject tiny saltwater particles into marine stratocumulus clouds**, aiming to increase their reflectivity and cool the Earth.

Marine Cloud Brightening (MCB):

- **It** is a scientific initiative that explores how **altering atmospheric particles (aerosols)** can **impact cloud reflectivity.**
- By releasing tiny **aerosol particles** into the atmosphere, researchers aim to enhance cloud brightness, leading to **increased sunlight reflection.**
- Aerosols of the **right size and concentration** could significantly **increase the reflectivity** of specific types of clouds.
- This phenomenon is visible in satellite images of clouds brightened by **ship emissions** (known as "ship tracks").

Goals of the Marine Cloud Brightening Program:

- It helps in better understanding of the present-day effects of **pollution aerosols** on clouds.
- Investigate whether aerosol particles made from **sea salt** could be used to intentionally reduce near-term climate warming while **greenhouse gas** concentrations are brought down to safer levels.
- It aims to understand the **benefits, risks, and efficacy of the intentional use of aerosols** to reduce warming through different implementations of marine cloud brightening.

Concerns/Challenges associated with Marine Cloud Brightening (MCB):

- MCB involves the **large-scale spraying of seawater** into the atmosphere at significant altitudes, which presents **engineering complexities** in terms of design, cost, maintenance, and operation of the spraying devices.
- Alterations in cloud patterns and precipitation due to MCB **could affect regional climate** and hydrological cycles, potentially causing **unintended consequences like droughts or floods.**Changes in clouds **over broad regions affect the circulation of the atmosphere, weather, and precipitation.**Both marine cloud brightening **(MCB) and pollution aerosols can change clouds**, which in turn **affects regions** both nearby and far from where the brightening occurs.
- MCB raises **ethical dilemmas** about human intervention in natural processes and the governance and decision-making processes surrounding its implementation.

MCB might lead to **complacency among policymakers** and the public, diminishing their commitment to reducing greenhouse gas emissions and adapting to climate change.

MINING IN THE ARAVALLI RANGE

GS III –Environmental pollution and degradation

Context: The Supreme Court has stopped the issuance of new mining licenses and renewals for existing ones in the Aravalli ranges and hills based on a report by the
Forest Survey of India (FSI).

The Aravalli Range:

- The Aravallis are **one of the oldest fold residual mountains of the world** comprised primarily of **folded rock strata.** This formation resulted from the convergence of tectonic plates during the **Proterozoic Era (2500-541 million years ago)**
- The Forest Survey of India (FSI) report, **defined the Aravallis** to include the **hills and a uniform 100-meter-wide buffer zone** around the downsides of the hills.
- They have an **elevation of 300m to 900m.** The mountains are divided into two main ranges – the **Sambhar Sirohi Range** and the **Sambhar Khetri Range** in Rajasthan.
- **Guru Shikhar Peak on Mount Abu** is the **highest peak** in the **Aravalli Range** (1,722 m).
- Major **tribal communities** include the **Bhil, Bhil-Meena, Meena, Garasia,** and others.

- The **Supreme Court in 2009** ordered a **complete ban on mining** in the Aravalli hills of Faridabad, Gurgaon, and Nuh districts of Haryana.

Significance of the Aravalli Range:

- It provides **habitat to 300 native plant species, 120 bird species**, and many exclusive animals like the jackal and mongoose.
- It acts as a **barrier between the fertile plains** in the east and the **Thar desert** in the west.
- The **excessive mining** in the Aravalli Range is linked to the spread of the Thar Desert.The **presence of loess,** a desert windblown sediment, **found in Mathura and Agra** suggests that the **desert is expanding** due to the weakened ecological barrier created by the degraded Aravalli Hills.
- It plays a significant role in shaping the **climate of northwest India**. During the monsoon season, these **mountains** act as a **climatic barrier**, guiding **moisture-laden south-westerlywinds** towards Shimla and Nainital.This, in turn, helps **nourish the sub-Himalayan rivers** and contributes to the rainfall that replenishes the vast north Indian plains.
- In the **winter months**, it **protects the fertile alluvial river valleys** from the cold westerly winds from Central Asia.

Issues/Concerns Related to the Mining in the Aravalli Range:

- **Mining activities** break and destroy **Aravalli ecosystems, displacing wildlife** like leopards, hyenas, and various bird species.This **disrupts food chains** and **ecological balance.**
- Mining in **Rajasthan's ecologically sensitive areas** has threatened the habitat of the **Great Indian Bustard, a critically endangered** bird species.
- The Aravallis acts as a natural **water reservoir.** Mining **disrupts natural water** flow and table recharge, leading to **water scarcity downstream**, and impacting agriculture and human settlements.A 2018 research paper noted a decline in spring recharge due to mining in Haryana.
- Mining activities **generate dust** and **release harmful pollutants** like silica, affecting air quality and causing respiratory problems in nearby communities.
- Mining **removes vegetation cover,** exposing the soil to erosion.Wind and rain wash away fertile topsoil, leading to desertification.A study by the **Centre for Science and Environment (CSE)** revealed a 37% decline in forest cover in the Aravalli region of Haryana between 2001 and 2016, likely linked to **mining activities.**

The government should **support communities dependent on mining** by creating alternative **livelihood opportunities** in sustainable sectors.

Green Walls:

- They are **vertical structures** that have different types of plants or other greenery attached to them.
- These can help in isolating the desertified land areas.

Green Muffler:

- It is a measure adopted to **control noise pollution**, by planting green plants.

MPOX

GS III –Biotechnology
Context: A study revealed a novel **adaptation mechanism** of the **Mpox virus** enhancing its ability to infect humans amidst recent outbreaks.The name was changed from **"monkeypox" to "mpox"** to avoid stigma towards monkeys and reflect the virus's direct human infectivity.
Mpox:

- It is also known as monkeypox and is a **DNA virus**. It belongs to the **family Poxviridae**, which consists of large, **double-stranded DNA** viruses.

- The virus was first identified in monkeys in 1958 but has since been found to **infect humans as well.**
- **It** is primarily transmitted to humans **from animals,** particularly **rodents and primates**, through direct contact or contaminated objects.
- **Its** infection in humans typically presents with **fever, headache, muscle aches,** and a characteristic rash that progresses from macules to papules to vesicles and pustules.
- While a vaccine for mpox exists, its **availability and effectiveness** are **limited,** highlighting the need for improved prevention and control measures.
- Mpox gained global attention during a widespread outbreak that affected over 100,000 people in more than 118 countries in 2022-2023.
- The outbreak was characterised by high **human-to-human transmission**, particularly through **close contact and the sexual route.**
- The **World Health Organization (WHO)** declared the mpox outbreak a **public health emergency**, leading to coordinated efforts to control its spread.

Genomic Features:

- Mpox genomes are divided into **clades I and II,** with evidence suggesting **clade I has higher mortality.**The **2022 outbreak** involved a new lineage, **clade IIb,** better adapted for human-to-human transmission.
- Researchers found evidence of a **distinct lineage of clade I associated with human-to-human transmission**, suggesting a recent **zoonotic spillover event.**
- Mpox viruses can undergo **genomic accordion** through gene duplication or deletion to adapt to different hosts and environments.

A study published in Nature Communications sequenced the genome of the mpox virus from the 2022 outbreak, revealing that certain sections **strongly influence** human-to-human transmission and are the virus's genomic accordions.

MULTILATERAL DEVELOPMENT BANKS

GS III –Infrastructure

Context: The UN Secretary-General, highlighted that reforming multilateral development banks (MDBs) will be a major focus at this year's Summit of the Future, occurring during the UN General Assembly in September 2024.

Multilateral Development Banks (MDBs):

- They are **international financial institutions** that provide financing and professional advice for economic and social development projects in developing countries.
- They are formed and capitalised by **multiple countries** through pooled resources and shared representation on their boards.
- They originated in the aftermath of **World War II** to rebuild war-ravaged nations and stabilise the global financial system.
- Unlike commercial banks, MDBs do not seek to maximise profits for their shareholders.Instead, they prioritise development goals, such as ending extreme poverty and reducing economic inequality.
- They often lend at low or no interest or provide grants to fund projects in **infrastructure, energy, education, environmental sustainability,** and other areas that promote development.
- **Major MDBs include the World Bank Group, the Asian Development Bank**, the **African Development Bank, the European Bank for Reconstruction and Development, and the Inter-American Development Bank.**

Issues/Challenges faced by MDBs:

- MDBs are often constrained by the amount of capital they have available to lend. This can **limit their ability to finance large-scale development projects,** especially in the face of growing needs.

- The world faces new and complex challenges like **climate change, pandemics, and technological disruption.**MDBs have not fully adapted to these growing challenges in their strategies and approaches to effectively address these issues.
- The current voting structure in some MDBs gives more power to developed countries.Developing nations are **pushing for a greater say in decision-making** to reflect their needs and priorities.
- Concerns exist about the **transparency of MDB decision-making processes** and the need for stronger accountability mechanisms to prevent corruption and mismanagement.For example, the **United States holds a considerable 15.85%** of the voting power in the World Bank, granting it substantial influence over the institution's decisions.
- One-size-fits-all lending conditions from MDBs, such as **uniform interest rates or repayment schedules,** are challenging for countries of the global south due to their diverse economic structures and **financial capabilities.**

Way Forward:

- MDBs can play a crucial role in mobilising resources for climate change mitigation and adaptation projects in developing countries.This could involve creating **dedicated climate finance facilities, offering green bonds,** and developing innovative risk-sharing instruments for renewable energy projects.
- Encouraging MDBs to facilitate knowledge exchange between developing countries.This could involve **connecting countries facing similar challenges** and fostering collaboration on successful development strategies.
- As middle-income countries develop, creating clear pathways for them to **"graduate" from concessional loans** to market-rate financing from private sources.This frees up MDB resources for low-income countries that still need significant support.

Strengthening safeguards to ensure MDB-funded projects **avoid negative social or environmental impacts** and promote **inclusive sustainable development.**

India and MDBs:

- India is a member of **four of the five constituents** of the World Bank Group viz., the **International Bank for Reconstruction and Development (IBRD), International Development Association (IDA), International Finance Corporation (IFC), and Multilateral Investment Guarantee Agency (MIGA).**India is not a member of **ICSID (International Centre for Settlement of Investment Disputes).**World Bank assistance in India started in **1948** when funding for the **Agricultural Machinery Project** was approved.
- India is a founding member of the **Asian Development Bank (ADB)**and the bank's fourth-largest shareholder.Since commencing operations in 1986, ADB has aligned its operations in the country to the government's developing priorities.This approach will be pursued through the forthcoming country partnership strategy, 2023–2027.
- India is a founding member of the Asian Infrastructure Investment Bank (AIIB) with the **second-highest voting share after China.**It is headquartered in Beijing.
- India, the founding member of the **New Development Bank (NDB),** is the second-largest recipient of NDB's financial support at USD 7.5 billion after China.It was established in 2015 by the **BRICS countries (Brazil, Russia, India, China and South Africa).**

NABARD'S CLIMATE STRATEGY 2030

GS III –Environmental Conservation

Context: The National Bank for Agriculture and Rural Development (NABARD) unveiled its Climate Strategy 2030 document which aims to address India's need for green financing.

NABARD's Climate Strategy 2030:

- **It** is structured around four key pillars:
  - **Accelerating Green Lending by f**ocusing on increasing green financing across various sectors.
  - **Market-making role by p**laying a broader role in creating a conducive market environment for green finance.

- **Internal Green Transformation by i**mplementing sustainable practices within NABARD's operations.
- **Strategic Resource Mobilisation by m**obilising resources effectively to support green initiatives.
- This strategy is designed to tackle the financial gap between the required investment for **sustainable initiatives** and the current inflow of green finance.
- India requires approximately **USD 170 billion annually by 2030**, aiming for a cumulative total of over USD 2.5 trillion.However, current green finance inflows are critically insufficient, with only about **USD 49 billion garnered as of 2019-20.**
- The majority of funds in India are earmarked for mitigation efforts, with only **USD 5 billion allocated towards adaptation and resilience.**This reflects minimal private sector engagement in these areas due to challenges in bankability and commercial viability.

Green Financing:

- Green financing refers to the mobilisation of financial resources to support investments that have a **positive environmental impact.**
- These investments can range from **renewable energy projects** and **energy efficiency initiatives** to sustainable infrastructure development and **climate-smart agriculture.**

Significance of Green Financing:

- The traditional financial system often prioritises short-term profits over long-term environmental sustainability. Green financing aims to bridge this gap by:
- **Facilitating the Transition to a Low-carbon Economy b**y channelling funds towards renewable energy and clean technologies, green financing helps reduce reliance on fossil fuels and mitigate greenhouse gas emissions.
- **Promoting Climate Adaptation and Resilience by making i**nvestments in green infrastructure like **flood defences and early warning systems** can help communities adapt to the changing climate and reduce the impact of natural disasters.
- **Unlocking New Economic Opportunities by** shiftingtowards a green economy creates new markets for clean technologies and sustainable practices, stimulating innovation and job creation.

Issues/Challenges Related to Green Financing:

- Green projects often require **higher initial investments** compared to conventional projects, which can deter investors despite long-term cost savings and environmental benefits.
- Green projects often have **longer payback periods** and may not align with short-term investment horizons or financial targets of investors and financial institutions.
- The absence of globally accepted standards for **green investment** leads to ambiguity and inconsistency in evaluating their environmental impact and financial performance.Also, without clear and standardised criteria, there is a risk of **greenwashing**, where **investments are misrepresented as environmentally friendly** without delivering substantial sustainability benefits.

By implementing a system where financial products earn **"green impact scores"** based on their environmental footprint. This allows consumers to make informed choices and prioritise green options.

PATIENT SAFETY CHARTER

GS III –Health
Context: The World Health Organization (WHO)launched a first-ever Patient Safety Rights Charter at the Global Ministerial Summit on Patient Safety.

Key Features of the Patient Safety Rights Charter:

- It outlines **the core rights of all patients in the context of the safety of health care** and seeks to assist governments and other stakeholders to ensure that the voices of patients are heard and their right to safe health care is protected.

- The Charter covers **10 patient safety rights** crucial to **mitigate risks** and prevent unintentional harm, which include the
  - Timely, effective, and appropriate care
  - Safe health care processes and practices
  - Qualified and competent health workers
  - Safe medical products and their safe and rational use
  - Safe and secure healthcare facilities
  - Dignity, respect, non-discrimination, privacy and confidentiality
  - Information, education, and supported decision-makingaccess to medical records
  - To be heard and fair resolution
  - Patient and family engagement.

Need for Patient Safety Charter:

- **Approximately 1 in 10 patients** encounter harm during healthcare procedures, resulting in over 3 million annual deaths attributed to unsafe care, as reported by the **Organisation for Economic Co-operation and Development (OECD)**.According to the OECD, **investing in patient safety positively impacts** health outcomes, reduces costs related to patient harm, improves system efficiency, and helps reassure communities and restore their trust in healthcare systems.
- The **majority of patient harm is avoidable,** underscoring the vital role of engaging patients, families, and caregivers in minimising harm.Patient harm often results from **poorly designed healthcare systems**, not isolated incidents.
- A **2023 survey of WHO Member States** revealed gaps in implementing the **Global Patient Safety Action Plan 2021-2030,** emphasising the need for patient representation and addressing income-based disparities in implementation.Interim results of the survey showed **only 13% of responding countries have a patient representative** on the governing board or an equivalent mechanism in the majority of their hospitals.

Patient safety is a **critical global priority** and essential for achieving the **Sustainable Development Goals (SDG)-3**: "Good Health and well-being".

Patient Safety:

Patient safety encompasses efforts to prevent unintended harm during healthcare provision, a critical aspect of global healthcare.

PUBLIC STOCKHOLDING

GS III –World Trade Organization (WTO)
Context: At the 13th ministerial conference of the World Trade Organization (WTO), India asserted a robust position concerning the permanent solution for public stockholding for food security.

Public Stockholding:

- It refers to the practice of governments **purchasing, storing, and eventually distributing food grains.**
- India along with many other countries uses this system to ensure food security for its population.

Significance/Advantages of Public Stockholding:

- Public stockpiles ensure a **buffer against potential food shortages** caused by factors like droughts, crop failures, or market disruptions.This helps maintain food availability for the population, especially during emergencies.
- By releasing stocks when prices rise due to short supply, governments can **moderate price fluctuations** and prevent sharp spikes that can burden consumers, particularly low-income households.
- Governments can buy grains at predetermined **minimum support prices**, offering some income security for farmers. This can incentivise production and maintain agricultural output.
- Stockpiled food can be used for **social welfare programs,** providing subsidised food to vulnerable populations and those facing food insecurity.

Issues/Disadvantagesof Public Stockholding:

- Maintaining large stockpiles can be expensive for governments. Storage and maintenance costs can **strain public finances and divert resources** from other development priorities.
- Subsidised food grains from public stockpiles may **undercut market prices,** discouraging private sector investment in agriculture and potentially impacting overall production efficiency.
- Improper storage leads to **spoilage and wastage of food grains,** causing economic losses and undermining the program's effectiveness.
- The management of public stockpiles is **vulnerable to corruption and mismanagement**, leading to inefficiencies and leakages within the system.

Subsidised stockpiling practices can create friction in international trade.Some countries argue such practices **distort fair market competition** and disadvantage exporters from other nations.For instance, **Thailand recently accused India** of using its public stockpiles of rice, meant for domestic food security, to gain an unfair advantage in the export market.

Key points highlighted by India at the 13th ministerial conference of the World Trade Organization (WTO):

- India has called for a shift in the WTO's focus, urging it to move beyond catering solely to the trade interests of agricultural exporters.Instead, the organisation should prioritise **addressing fundamental concerns such as food security and sustaining livelihoods.**
- India argues that public stockholding programs are essential for developing countries to ensure **food security for their populations,** especially for vulnerable sections of society.The current WTO rules allow some leeway for developing countries regarding **public stockholding** programs.However, these provisions are temporary and **India seeks a permanent** solution that acknowledges their development needs.

India stressed the necessity of **creating a level playing field in international agriculture trade**, especially for low-income or resource-poor farmers across the globe. This aligns with the broader goal of promoting fairness and equity in trade practices.India pointed out the glaring **disparities in domestic support** provided by countries to their farmers.Notably, subsidies in some developed countries are reported to be up to 200 times higher than those in developing nations.

RECIPE FOR A LIVABLE PLANET REPORT

GS III –Environmental Conservation

Context: The World Bank released a Recipe for a Livable Planet Report stating that annual investments of USD 260 billion are necessary to cut agri-food emissions in half by 2030 and achieve net zero by 2050.

Key Highlights of the Report:

- **"Recipe for a Livable Planet"** provides a **global strategic framework** for reducing the agri-food system's impact on climate change.It outlines how the **world's food production** can significantly **lower greenhouse gas (GHG) emissions** while continuing to ensure global food security.
- The **global agri-food system** can decrease nearly a third of the world's GHG emissions through **feasible and accessible measures.**These measures will enhance food security, increase the climate resilience of the food system, and **protect vulnerable communities** during this transition.
- Agri-food contributes roughly **one-third of global GHG emissions**, more than all of the **world's heat and electricity emissions** combined.
- About three-quarters of these emissions originate from developing countries, necessitating **targeted mitigation actions as per the specific needs of the region.**
- Addressing emissions from the entire **food value chain**, including land use changes, is critical as over half of the emissions stem from beyond the farm level.

Opportunities Highlighted by the Report:

- The agri-food sector offers significant, cost-effective opportunities for climate action, including drawing carbon from the atmosphere through enhanced land management.

- The financial outlay required to halve agri-food emissions by 2030 would yield substantial returns, greatly outweighing the costs with beneficial impacts on health, the economy, and the environment.
- **High-incomecountries**should **reduce their agri-food energy demands,** support lower-income countries through **funding and technology transfer,** and modify consumer diets away from high-emission foods.
- **Middle-incomecountries**can achieve significant emissions reductions through better land use management and agricultural practices.
- **Low-incomecountries f**ocus on **sustainable growth** without the burden of high-emission infrastructures, leveraging strategies like agroforestry to boost productivity and resilience.
- Enhance private sector investment in agri-food mitigation, repurpose subsidies, and implement policies favouring low-emission technologies.
- Use digital technologies for better emissions data and invest in innovations to transform the agri-food system, ensuring inclusive stakeholder participation for a just transition.

Key Highlights Related to India in the Report:

- The report identifies India as one of the **top 3 countries in terms of total annual agri-food system** emissions, along with **China, and Brazil.**
- The report notes that in countries like India, around 80% of the technical mitigation potential in agriculture could be achieved by **adopting cost-saving measures alone.**This represents a major opportunity for India to reduce emissions while also improving agricultural productivity and incomes.
- Key mitigation options for India include **better livestock feeding** (**Harit Dhara, a nti-methanogenic feed**) and breeding, fertiliser management, and better water management in water-intensive crops.
- A **marginal abatement cost curve** for India's agriculture sector shows these are some of the most cost-effective interventions India can pursue to cut agri-food emissions **substantially by 2030**.
- India needs to **curb methane emissions** from **agricultural production**.Adopting practices like **intermittent irrigation** and promoting varieties that emit less methane provide mitigation opportunities.
- India has high rates of food loss and waste. As per the **Food Waste Index Report 2021**, Indian households generate **50 kg of food waste per capita per year**.**Reducing food loss and waste can** provide another high-impact, cost-effective avenue for India.

India will need **international financial and technical support** to realise its agri-food mitigation potential.

SUNDARBANS

GS III –Environmental Pollution and Conservation
Context: A study conducted by prominentenvironmental scientists has warned about the substantial threat of air pollution to the Sundarbans, an essential
mangrove ecosystem in West Bengal.

Sundarbans:

- Sundarbans hosts the largest mangrove forests in the world, lying on the **delta of the Ganges, Brahmaputra**, **and Meghna rivers** on the **Bay of Bengal.**
- The mangrove ecosystem is an **ecotone** between the land and the sea in the tropical and subtropical regions.
- Due to the ecotonic effect, this zone fosters a rich blend of habitats, from freshwater swamps and intertidal mangroves to saline forests and open water.It is a sanctuary for a wide variety of species from different habitats, including rare and globally threatened wildlife such as the E**stuarine Crocodile**, Water Monitor Lizard, **Gangetic dolphin**, and O**live Ridley Turtle.**
- **40% of Sundarbanslie in India** and the rest in Bangladesh.It was designated a **UNESCO World Heritage site** in 1987 (India) and 1997 (Bangladesh).
- Sundarbans Wetland in India was recognised as the '**Wetland of International Importance**' under the **Ramsar Convention** in January 2019.
- **Royal Bengal Tigers** in the Sundarbans are top predators that control prey populations, preventing overgrazing and maintaining the delicate balance of the ecosystem.**Protecting tigers** also safeguards a vast habitat for other plant and animal species, contributing to a **healthy forest ecosystem** in the Sundarbans.

- In 2011, **India and Bangladesh signed an MoU on the Conservation of the Sundarbans,** recognising the need to monitor and conserve the Sundarbans.

Issues/Challenges Faced by the Sundarbans:

- As a consequence of climate change, **rising sea levels** threaten to inundate low-lying mangroves. This **saltwater intrusion disrupts their delicate** balance and makes them more vulnerable to **storm surges during cyclones.**
- Climate change is also linked to more frequent and intense storms. These cyclones can batter mangroves, causing **physical damage** and disrupting sediment patterns crucial for their survival.
- The conversion of mangrove forests for agriculture like cash crops (palm oil) or food production (rice paddies) destroys their habitat.This not only reduces the area available for these ecosystems but also fragments existing ones, impacting biodiversity.
- Mangroves provide crucial services like **shoreline protection and nursery grounds for fish**. Deforestation disrupts these services, impacting coastal communities and fisheries.
- The loss of mangrove habitats due to **climate change** is leading to the loss of species in the near-threatened or endangered category.Settlement mangroves used to be safe havens for diverse molluscs and crustaceans, but they are disappearing due to polluted discharges and breeding activities of these species.
- Pollutants, enriched with **black carbon** or soot particles, from nearby urban areas and the entire **Indo-Gangetic Plain region** are worsening the air quality of the Sundarbans, impacting its ecosystem.These air pollutants significantly affect the ecology and biogeochemistry of the Sundarbans mangrove ecosystem.

Way Forward:

- Cultivating **native grass species** like wild rice, *Myriostachyawightiana,* biscuit grass, and salt couch grass can stabilise streambanks and prevent erosion, instead of introducing non-local species like vetiver, which are not **salt-tolerant.**
- Promoting soil-tolerant paddy varieties and **organic agriculture** practices can increase agricultural productivity and income for farmers while minimising environmental impact.Implementing **rainwater harvesting** and **watershed development initiatives** will further enhance agricultural production.
- Using natural processes and microorganisms for wastewater treatment, such as lactic acid bacteria and photosynthetic bacteria, can support water quality and ecosystem health.
- The India-Bangladesh Joint Working Group (JWG) should be transformed into a **high-powered board of interdisciplinary experts** to plan and implement climate resilience for the Sundarbans and the communities dependent on it.
- The corrective measures include solar energy promotion,electric transportation, subsidised LPG, regulated tourism, closing down of pollutant factories, regulation of brick kilns and land use, and **strengthening of coastal regulations.**

A multi-layered approach to multi-engagement and multidimensional planning can be followed by the ministries of tourism, **disaster management**, agriculture, fisheries, and rural development.

UN FINANCING FOR SUSTAINABLE DEVELOPMENT REPORT 2024

GS III –Inclusive Growth

Context: The UN Financing for Sustainable Development Report 2024released by the United Nations (UN) stated that more investment is needed if the 17
Sustainable Development Goals (SDGs), agreed upon by all United Nations members back in 2015, are to be achieved by 2030.

Key Highlights of the UN Financing for Sustainable Development Report 2024:

- Rising geopolitical tensions, climate disasters, and a global cost-of-living crisis have hit billions of people globally, which has halted the progress on healthcare, education, and other development targets.
- Debt services in the **Least developed countries (LDC)** will increase from USD 26 billion annually in 2022 to **USD 40 billion annually** between 2023 and 2025.Stronger and more frequent disasters, caused by the ongoing climate crisis, account for more than **half of the debt upsurge** in vulnerable countries.

- The poorest countries now spend 12% of their revenues on interest payments, 4 times more than they spent a decade ago.Roughly **40%** of the global population lives in countries where governments spend more on interest payments than on education or health.
- In least-developed **countries,** development fundingis slowing down.Due to several reasons such as **low domestic revenue growth** due to tax evasion and avoidance, a falling rate of corporate tax (which was 28.2% in 2000 to 21.1% in 2023), due to **globalisation** and tax competition, etc.
- According to the **Financing for Sustainable Development Report: Financing for Development at a Crossroads Report** 2024, around **USD 4.2 trillion** in investments is needed to close the development financing gap.This number was **USD 2.5 trillion before the Covid-19** pandemic began.

Suggestions made by the UN Financing for Sustainable Development Report 2024:

- The international financial system, which was set up at the **1944 Bretton Woods Conference**, is no longer fit for purpose.A **massive surge of financing** and a **reform of the international financial architecture** can help to achieve SDG goals by 2030.
- To achieve the SDGs **global cooperation, targeted financing, and, crucially, the political will** is the need of the hour.

Progress made by India in Achieving SDGs:



- India ranks **112th out of 166 countries** (from **121st in 2022**) in terms of progress towards the Sustainable Development Goals (SDGs) in the United Nations SDG Index and Dashboards Report 2023.
- India has successfully lifted millions of people out of poverty, **reducing the poverty rate from 45% in 1993 to around 21% in 2011.**Around **415 million** people exited poverty within just 15 years between 2005 and 2021 in India, according to the latest **Global Multidimensional Poverty Index (MPI) 2023**
- The prevalence of **undernourishment** in India has decreased from 18.2% in 2004-2006 to **14.5% in 2016-2018.**However, India is still home to a **quarter of all undernourished people worldwide,** making the country a key focus for tackling hunger on a global scale.
- India has made significant improvements in **maternal and child health**, with a reduction in maternal mortality ratio from **384 in 2000 to 103 in 2020 (UN MMEIG 2020 report).**The under-five mortality rate has also declined from 89 per 1,000 live births in 1990 to **34 in 2019.**
- According to the Education Ministry, the literacy rate in rural India stands at 67.77%, while in urban India, it is 84.11%The **ASER 2023** data shows that in the surveyed rural districts, more than **85%** of youth (age 14-18) are currently enrolled in some kind of educational institution.

The **female labour force participation rate** in India increased from 23.3% in 2017-18 to 37.0% in 2022-23 **(PLFS-5).**

UNITED NATIONS FORUM ON FORESTS (UNFF)-19

GS III –Environmental Conservation

Context: India participated in the 19th Session of the United Nations Forum on Forests (UNFF) at the UN Headquarters in New York.

Key highlights of the UNFF19:

- **India presented its revised National Forest Policy** emphasising forest fire prevention and management through recommendations and technological solutions.
- According to UNFF, about 100 million hectares of forest, or **3% of the world's total forest area are affected by fires each year.**
- India proposes operationalising the **Global Fire Management Hub,** a collaborative effort by the **UNEP** and the **FAO** to share knowledge and experiences in mitigating forest fires.

- India suggests establishing **universally accepted standards like the Model Forests Act** for forest certification programs for consistent and responsible forest management practices worldwide.
- The forum **reviewed** the **UN's strategic plan for forests (2017-2030)** and progress made in achieving global forest goals like **enhancing forests' economic, social, and environmental benefits** and securing finances.
- A UN report highlighted **concerns about the "Climatisation" of forests, driven by market-oriented approaches for carbon sequestration,** reducing forests' ecological and social values roles to mere carbon sinks.
- **Indonesia** presented its **Forest and Other Land Use Net Sink 2030 strategy,** and **Malaysia** committed to keeping **at least 50%** of its **territory under tree cover.**

India's Initiatives Highlightedin Forest Management at UNFF19:

- India highlighted its success in using technology to manage forest fires.Examples include **real-time fire monitoring** via remote sensing, online fire reporting through **web portals**, and utilising ecological methods for restoration.Based on the forest inventory records**, 54.40% of forests** in India are exposed to **occasional fires, 7.49% to moderately frequent fires**, and **2.40% to high incidence levels.**
- Globally, **India ranks third** in the **net gain of average annual forest area** between 2010 and 2020.
- India completed its **50 years of Project Tiger** and **30 years of Project Elephant**, demonstrating its commitment to species conservation and habitat protection.
- India introduced the **'Green Credit Program'** to incentivise tree plantation and restoration of degraded forest lands to strengthen climate action initiatives.
- In 2023, India hosted the **country-led Initiative under UNFF** at Dehradun, attended by representatives from 40 countries and 20 international organisations, focusing on forest fire management and forest certification.

Indian Forest Policy:

- **The National Forest Policy of 1894prioritised timber production** and custodial management. It introduced f**orest classification** with an emphasis on protecting commercially valuable areas.
- **The National Forest Policy of 1952focused on national priorities** like land-use management and environmental protection. It emphasised securing resources like timber, pasture, and fuelwood for national development.
- **The National Forest Policy of 1988 p**rioritised **environmental stability, biodiversity conservation**, and soil and water security. It advocated for large-scale afforestation and social forestry programs.

Draft National Forest Policy of 2018 proposed **revisions to address modern issues like climate change** and human-wildlife conflict. It focuses on mitigating climate change and promoting public-private partnerships for forest restoration.

United Nations Forum on Forests(UNFF):

- The UNFF is **an intergovernmental policy forum** that promotes **"management, conservation and sustainable development of all types of forests.**
- It was **established in 2000** by the **UN Economic and Social Council and has universal membership.**

UNIVERSAL HEALTH COVERAGE

GS III –Health
Context: The Tracking Universal Health Coverage: 2023 Global Monitoring report was jointly released by the World Health Organization and World Bank.

Universal Health Coverage (UHC):

- UHC means that all people have access to a fullrange of quality health services without financial hardship. It covers the full continuum of essentialhealth services, from health promotion to prevention, treatment, rehabilitation, and palliative care.

Importance of Universal Health Coverage:

- It maximizes human capital by making investments in health, which is crucial for both educational attainment in children and workforce productivity in adults, which promotes economic growth.
- A well-functioning health system with equitably distributed healthcare infrastructure increases the demand for skilled jobs.
- It helps in achieving Sustainable Development Goal 3 (Good Health and Well-being).
- It helps to reduce social barriers to care and, the distribution of health resources between different population groups and empowers patients in decision-making.

Challenges in achieving Universal Health Coverage:

- As per, the Economic Survey2022-23, central and state governments' budgetedexpenditure on healthcare increased marginally from1.6% in FY21 to 2.1 % of GDP in FY23 and is among thelowest in the world.
- As per NITI Aayog, at least30% of the population is devoid of any financialprotection for health due to existing gaps in coverageand overlap between schemes.
- India's High out-of-pocket expenditure (OOPE) asa percent of current health spending is 47.1% in 2019-20,significantly above the global average of 18%.
- Rural communities in India sufferfrom significant access to healthcare such as a lack ofqualified medical professionals, physical limitations likedistance, lack of established healthcare infrastructure,andinability to pay for necessary medical treatment.As per the Lancet publication (2023), the provision of core health services lacks uniformity across state-rundistrict hospitals.
- India faces a shortage of healthcare professionals,including doctors, nurses, andparamedical staff, which impacts the delivery of healthcare services.

Initiatives taken to achieve Universal Health Coverage:

- Ayushman Bharat Yojana aims to undertakeinterventions to holistically address health at the primary,secondary, and tertiary levels.
- Ayushman Bharat Digital Missionwill improveequitable access to quality healthcare by encouragingthe use of technologies such as telemedicine and enablingnational portability of health services.
- The National Health Policy of 2017aims toachieve universal health coverage and deliver quality healthcare services to all at affordable cost.
- Intensified Mission Indra Dhanush focuses onreaching zero-dose children aged between 0 and 5years and pregnant women who might have missed anyvaccine doses in the national immunization schedule.

Central government should adopt a collaborativemechanism allowing for flexible policy-making, and differential financing from the centre to the state.This will also allow states to better meet the diverserequirements and develop health plans that areconsistentwith the needs and requirements of populations.

Key findings of the Report:

- The UHC service coverage indexincreased from 45 to 68between 2000 and 2021.
- The world is off track to makesignificant progress towardsthe UHC (SDG target 3.8) by 2030 asimprovements tohealth services coverage have stagnated since 2015.
- The proportion of the populationnot covered by essentialhealthservices decreased by about 15%between 2000 and 2021, withminimal progress made after 2015.The COVID-19 pandemic disruptedessential services in 92% ofcountries at the height of thepandemic in2021. In 2022, 84% of countries still reported disruptions.
- The proportion of the populationthat facedcatastrophic levels ofOOP health spendinghas increased.
- Financial hardship is moreprevalent amongpoorer householdsmostly due to higher ratesof impoverishing health spending ratherthancatastrophic health spending.

VULTURE SPECIES IN INDIA

GS III –Environmental Conservation

Context: Studies have found that even vultures in protected areas are not safe from toxic drugs like Diclofenac. Scientists investigated the feeding habits of vultures in India by analyzing the DNA in vulture faecal samples from vulture nests and roosts across six states between 2018 and 2022.

Vultures Species in India:

- It is **one of the 22 species** of large scavenger birds that live predominantly in the **tropics and subtropics.**
- They act an important function as nature's garbage collectors and help to keep the environment clean of waste.They also play a valuable role in keeping **wildlife** diseases in check.
- **India is home to 9 species of Vulture** namely the Oriental white-backed, Long-billed, Slender-billed, Himalayan, Red-headed, Egyptian, Bearded, Cinereous, and the Eurasian Griffon.
- A **significant decline** in vulture populations has been observed in South Asian nations, **particularly India, Pakistan, and Nepal.**This decline is primarily attributed to the **widespread use of diclofenac**, a veterinary drug, during the late 1990s and early 2000s.This resulted in a **population decline exceeding 97%** in some regions, creating a consequential ecological crisis.

Role of Vultures in Ecosystem:

- Vultures efficiently consume carrion (dead animals), **preventing carcasses from accumulating and rotting.**This **helps to decompose organic matter** and return nutrients to the soil, which benefits plant growth and the overall health of the ecosystem.
- Vultures have **incredibly strong stomachs with very acidic digestive juices.** This powerful acid **can kill bacteria and viruses** that can cause diseases like anthrax, rabies, and botulism, thus, acting as **a true "dead-end" for pathogens.**
- Vultures are **sensitive to changes** in their environment. A decline in vulture populations can be **an indicator of a wider ecological problem,** such as pollution or a shortage of food sources.

Reasons behind the decline in Vulture Populations:

- The widespread use of veterinary drugs like **diclofenac, ketoprofen, and aceclofenac** in the **late 20th century** has had devastating consequences for vulture populations.These drugs, commonly **used to treat pain and inflammation in livestock**, are **toxic to vultures** when they feed on carcasses of treated animals.
- Vultures are scavengers, often consuming carcasses contaminated with **pesticides or other toxins.** Vultures feeding on carcasses of animals hunted **with lead ammunition** can suffer fatal **lead poisoning.**This "secondary poisoning" poses a significant threat, further declining their populations.
- **Urbanisation, deforestation, and agricultural expansion** have led to habitat loss, destroying vulture nesting sites, roosting areas, and food sources. The lack of suitable habitat hinders their survival.
- Vultures are vulnerable to collisions with power lines, wind turbines, and other man-made structures, leading to injuries or fatalities and contributing to population decline.
- In some areas, vultures are **targeted due to cultural beliefs** or **illegal wildlife trade**, adding to their struggle to survive.

Diseases like **avian pox and avian flu** can also have a detrimental impact on vulture populations, leading to further decline.

Vulture Conservation Efforts Taken by India:

- Recognising the devastating impact of **diclofenac**, India **banned its veterinary use in 2006.**This was a critical step in protecting vultures from kidney failure caused by ingesting carcasses of treated livestock.
- The **Ministry for Environment, Forests and Climate Change** launched a **Vulture Action Plan 2020-25** for the conservation of vultures in the country.It will **ensure minimum use** of Diclofenac and **prevent the poisoning** of the principal food of vultures, the cattle carcasses.
- In **August 2023**, India further **banned the use of ketoprofen and aceclofenac for veterinary purposes**, acknowledging their potential threat to vultures.

- India established a network of **Vulture Conservation Breeding Centres (VCBCs)**,the **first being set up in Pinjore, Haryana in 2001.**These centres **focus on the captive breeding of endangered vulture species,** providing a safe environment to raise healthy populations for reintroduction into the wild.
- In a proactive effort to conserve the declining vulture population in **Jharkhand**, a **'Vulture Restaurant'** has been established in **Koderma district**. The initiative aims to address the adverse impact of livestock drugs, particularly diclofenac, on vultures.

WATER TRADING

GS III –Environmental Conservation

Context:NITI Aayog released a document on water trading mechanisms to promote the reuse of treated wastewater.

Water Trading:

- It is a water market mechanism that considers water as a commodity rather than a public good.
- It can be traded amongst users according to their needs.
- A water company, rather than creating its own water source, can buy the water from a third party andsell it to the users.
- Enabling elements for Water Trading:
  - Ownership of water - should be indisputable.
  - Water entitlements- could be transferred.
  - Water Reuse Certificates (WRCs) - as tradable permits.

Need for reuseof treated wastewater through water trading:

- Only about 40% of urban wastewater is treated in India, and even that is not beingreused.
- Wide temporal and spatial variations in the distribution of water. For example, India receives more than 80 % of the rainfall during four months of the year.
- The National Water Policy of 2002 & 2012 has proposed a private hand in watermanagement and has emphasised public-private partnership.
- Wastewater, including agricultural runoff, is the leading cause of water pollution.It is intricately linked to human and ecosystem health, with over 80% of global wastewater estimated to enterwater bodies untreated. (UN World Water Report 2023)
- Around 80% of people living under water stress lived in Asia, in particular, northeast China,as well as India. (UN World Water Report 2023)
- Around89% of groundwater extracted is used for irrigation, andcrops such as paddy and sugarcane consume more than60% of irrigation water. India is also a large virtual net export of waterbecause of agricultural products export.

Challenges/Issues for reuse of treated wastewater through water trading:

- Treating the wastewater and supplying it to the farm fields or the industrial units will involve certaincosts that may not be competitive with the cost of fresh water.As fresh water is available at a substantially low or sometimes even at no cost.
- Since the process of wastewater generation is continuous, the treatment process also should be continuous.However, the demand may not remain constant throughout the year.For example, the crops may not require irrigation for a certain period; demands in industrial units may also vary.This would require either storage of treated wastewater or discharging the treated water to other freshwatersources.
- To make water/treated wastewater a tradable commodity, it is necessary to havea demand.
- Maintenance of the plant, disruption in the sewage network, quality problems in inlet sewage or treatedsewage, disruption in the distribution network, etc. could interrupt supply.
- In India, water is neither considered a commodity nor possesses any significanteconomic value.This is mainly because of the due consideration of the essentiality of water as a fundamental human right.

The water neutrality concept advocates that the total water demand should be thesame after a new development is built as it was before. Promoting water reuse can aid in achieving water neutrality.

Note:

- Under the WRC trading system, water users across sectors, including municipal/residential, industrial, andagricultural sectors, shall be registered and get individual targets for water reuse.
- Water users having WRCs may sell them in the open market, allowing other water users to bridge the gapbetween their wastewater reuse targets.

PAPER 4

ETHICS OF NUDGE

GS IV –Ethics

Context: The Haryana government launched the Pran Vayu Devta PensionScheme, offering pensions to trees aged 75 years or older located onthe property of the state's residents. The scheme aims at nudgingthe residents towards conserving and protecting aging trees and theenvironment.

Nudge:

- A nudge is simply an intervention that gently steersindividuals toward a desired action. It is an aspect ofthe choicearchitecture that alters people's behaviourwithout forbidding any options or significantly changingtheir economicincentives.
- Nudges are rooted in behavioural sciences,acknowledging that individuals may not always makerational decisions.
- Instead, people's choices are influenced by cognitivebiases, heuristics, and emotions. Nudges leverage thisunderstanding to guide people toward better decisionswithout limiting their choices. E.g., nudge to keep yoursurroundings clean via Swacch Bharat Mission (SBM).

Nature of Ethical Nudging:

- Nudges should be transparent, not covert or hidden,and in the interests of the people who are beingnudged and consistent with their values.
- The formation and expression of preferences is animportant aspect of individual autonomy and thenudger should consider preferences when setting the goals of a nudging campaign.For example, economicallyunderprivileged groups may benefit from nudges thathelp them avoid unnecessarybank fees or high-interest loans more than those designed to increase savings.
- The goal should be to have a positive impact on individualsand society and promote better decision-making.Nudging should not exploit individuals' vulnerabilities,such as cognitive limitations, emotional states, orsocioeconomic factors.
- Nudges should be culturally sensitive andconsider the diversity of values, norms, and beliefs within apopulation.Individuals should have the option to opt out of nudges.

Significance of Nudges:

- Nudge Units within government agencies have proventhat simple nudges can reduce lawviolations without using heavy-handed penalties.
- When applied thoughtfully, nudges can be moreeffective than mandates, financial incentives, orawareness campaigns alone.
- Unlike traditional interventions that impose restrictionsor regulations, nudges maintainindividuals' freedom of choice.
- Implementing nudges can be relatively simple, and low-cost interventions compared to moretraditional approaches.
- Nudges are often arrived at after empirical researchand evidence, lending them significantcredibility and legitimacy.
- Nudges can be tailored to accommodate diversepreferences, values, and cultural contexts, allowing forthecustomization of interventions.

Key Ethical Concerns with Nudge:

- Nudges can be used by many different actors such as individuals, governments, supermarkets, or other corporations. The potential impact of nudges depends upon the nudgers' motives and who benefits from thenudges.

- Nudges workwith humanshortcomings, especially uncertainty, inaction, and impatience; thus, nudging may exploit people's irrationality.For instance, auto-enrolment in savings schemes takes advantage of an individual's inaction.
- Concerns of manipulation and lack of transparency areraised with nudges operating on asubconscious level.
- A nudge does not stimulate reflection and so is lessunlikely to create the knowledge, dissonance,orevaluation required to change people's beliefs, attitudes, and behaviour in the long term.

In some cases, a nudge may produce reactance(negative reaction due to perception ofchoice restriction) or boomerang effect (reversal from follow-up of intended outcome).

MEDIA TRIAL AND ETHICS

GS IV –Media Ethics

Context: The Supreme Court (SC) directedthe Union government to prepare guidelines on how the police should brief the media about ongoing investigations inorder to prevent media trials.

Media Trial:

- It is generally considered as the process in whichmedia publishes their own versions of facts innewspapers, newswebsites, and news channels inorder to sensationalize the case.This function of media is seen as an invasion of the domain of the judiciary by conducting indirect trials parallel withthe courts.
- In some cases, it has also been noticed that irresponsible media briefings by police officers during ongoinginvestigations enable the process of media trials.E.g., statements made by the accusedunder Section 161 [before the police] and 164 [beforea judicial magistrate]of the Criminal Procedure Code were released to the media.

Ethical Issues Involved in the Media Trial:

- It undermines the principle of innocent untilproven guilty which advocates that every accused ispresumed innocent untilproven guilty by the law. E.g. Aarushi–Hemraj murder case.
- It also leads to contempt of court (disrespect or disobedience toward acourt by entering into their domain).
- When sub-judice cases (cases under trial) arebeing discussed by experts in the media, their opinionscan affect the perception ofjudges toward the accused/victim. E.g. Jasleen Kaur harassment case.
- It threatens the Right to Privacy as the identity/personalinformation of the accused and victim is revealedwhich cannegatively affect the public image of an individual. E.g. in Sushant Singh Rajput case.
- It violates key principles of Media Ethics such astruth and accountability. It goes againstthe idea of responsible journalism.

Issues/Challenges in Controlling Media Trials:

- In India, organisations like the NewsBroadcastingand Digital Standards Authority (NBSA) andBroadcasting ContentComplaints Council have beenset up as internal self-regulatory mechanisms fortelevision news and entertainment, respectively.Thismechanism has not been very effective as thesebodies lack statutorybacking.
- Under Article 19 of the Constitution, media enjoysthefreedom of expression, but it is at times misusedwhen engaging in media trials.Also, for regulatingagencies it becomes difficult to decide whatconstitutes a Media Trial as there is no cleardefinition or guidelines. E.g., in cases of investigativejournalism.
- With the rise in internet penetration, social mediahasemerged as the new platform for media trials.E.g., sharing of sensationalizedor fake posts to alter public opinion.

Code of Journalistic Conduct:

It defines the boundaries of the media and the propercode of conduct to be followed by Journalists.

PATRONAGE APPOINTMENTS AND ETHICS

GS IV –Ethics in Public Administration

Context: Patronage appointments in the public sector raise several ethical issues that can affect the efficiency and accountability of the civil service.

Patronage Appointments:

- It refers to the appointments that are made based on the recommendation or source(personal connections) of an influential person, political leader, public servant, etc. This has not only affected the peopleinvolved by has affected the overall Governance of India.

Ethical Issues involved in Patronage Appointments:

- When appointees get theirposition throughpatronage, then they are likely to bemore accountable to thatperson rather than to thesystem.
- •Ideally, positions should be awarded based onindividuals' abilities and qualifications. Bypassingmeritocracy indirectly compromises the integrity of thesystem and can result in ineffective or incompetentindividualsoccupying important roles.
- In the light of appointments through patronage, qualified and capableindividuals may feel their efforts and qualifications are not given equal consideration.
- It erodes public trust in the fairness and impartialityof the system and people in power. It leads to aloss of confidence in the administration as a whole.The associated nepotism/favouritism further erodes the trust of the individual in the system.

Impacts of Patronage Appointments:

- It can be used as a means to reward political allies, secure support, or engage inquid pro quo arrangements. It promotes the diversion of resources away from public welfare toward personal gain.
- Incompetent and inexperienced officeholdersappointed through patronagestruggle to implement policies effectively. This can result in poor policy outcomes, delays, or hindrances in the overallgovernance process.
- This can lead to a loss of motivation and talent withinthe public service asindividuals feel undervalued and unappreciated.
- Citizens may perceive the system as corrupt, nepotistic, and lacking in transparency. This canweaken the legitimacy of the government andundermine its ability to govern effectively.
- Growth and development are dependent ongovernance. Ineffective officials whoare appointed through a patronage system may not be able to effect optimal utilization of resources.

Way Forward:

- Ensure that institutions have the authority, resources, and mandate to conduct fair andtransparent selection processes based on merit.
- Reduce political influence in the appointmentprocess by establishing an independentand non-partisan body responsible for overseeingappointments in public offices.
- Emphasize the importance of qualifications, experience, and competence in therecruitment and selection process.
- Promote the internalization of concepts of publicservice and integrity toensure ethical decision-making.

Foster collaboration with civil society organizations, professional associations, and academic institutions topromote accountability, monitor the appointmentprocess, and advocatefor reforms.

Practice Questions

Q1.) With reference to the recharge wells, consider the following statements:

1. Recharge wells are structures designed to directly recharge groundwater.
2. They play a crucial role in replenishing aquifers and maintaining sustainable water availability.
3. In urban areas, recharge wells help manage stormwater runoff, reduce flooding, and maintain stable water tables.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q2.) Consider the following statements:
Statement-I:
Carbon farming refers to a set of practices aimed at reducing greenhouse gas emissions from agriculture and land use.
Statement-II:
Carbon farming aims at storing carbon in the soil, crop roots, wood, and leaves.
Which one of the following is correct in respect of the above statements?

a. Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
b. Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
c. Statement-I is correct but Statement-II is incorrect
d. Statement-I is incorrect but Statement-II is correct

Q3.)With reference to the West Nile Fever (WNV), consider the following statements:

1. West Nile Fever is an infectious disease caused by the West Nile virus.
2. The principal vector for transmission is the Culex species of mosquitoes.

Which of the statements given above is/are correct?

a. 1 only
b. 2 only
c. Both 1 and 2
d. Neither 1 nor 2

Q4.) Which among the following flower was recently banned from being part of temple offering due to its poisonous nature :

a. Parijata or Indian Magnolia
b. Nerium Oleander
c. Safed Aak Or Crown Flower
d. Cobra saffron or Nagavriksha

Q5.)The word Zero Day Vulnerability recently seen in news is related to:

a. An imminent nuclear attack
b. Use of Chemical weapon
c. Release of Biological Weapon
d. Cybersecurity

Q6.) Consider the following countries:

1. Canada
2. China
3. France
4. India
5. Japan
6. United States

How many of the above-mentioned countries are the members of the Group of Seven (G7)?

a. Only two
b. Only three
c. Only four
d. Only five

Q7.) With reference to the Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act, consider the following statements:

1. The Act mandates a survey of street vendors to identify and protect their livelihoods.
2. Street vendors receive a certificate of vending, which establishes their legitimacy.
3. The Act provides a mechanism for resolving grievances and disputes among street vendors.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q8.)Consider the following statements about the Biohacking:

1. Biohacking refers to the practice of using science and technology to optimize and enhance the human body's performance and well-being.
2. Currently, there are no laws in India that specifically address biohacking

Which of the statements given above is/are not correct?

a. 1 only
b. 2 only
c. Both 1 and 2
d. **Neither 1 nor 2**

Q9.) Supreme Court of India has recognized the right to die with dignity as a fundamental aspect of the right to life under

a. Article 14
b. Article 20
c. Article 19
d. **Article 21**

Q10.) Rafah Border Crossing, recently seen in news is the sole crossing point between

a. Russia and the Crimea
b. Egypt and the Gaza Strip
c. Armenia and the Nagorno-Karabakh
d. Sudan with the Tigray Region

Q11.)With reference to sand mining, consider the following statements:

1. Sand mining involves the extraction of sand from various sources, including rivers, beaches, and seabeds.
2. Unregulated Sand extraction destroys river and beach habitats critical for various aquatic species.
3. Illegal sand mining in India is a crime under the Indian Penal Code, 1860.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q12.) Consider the following statements about the Oxytocin:

1. Oxytocin is a hormone produced in the brain's hypothalamus and secreted by the pituitary gland.
2. Oxytocin helps contract the uterus during labour, facilitating childbirth.
3. In the dairy sector livestock is injected with the hormone to increase milk production.
4. The World Health Organization (WHO) recommends carbetocin as a safe and effective alternative to oxytocin.

How many of the statements given above are correct?

a. Only one
b. Only two
c. Only three
d. **All four**

Q13.) With reference to the Central Drugs Standard Control Organisation (CDSCO), consider the following statements:

1. It is the Central Drug Authority responsible for discharging functions assigned to the Central Government under the Drugs and Cosmetics Act of 1940.
2. It operates under the Ministry of Health & Family Welfare.
3. The Union Minister of Health and Family Welfare is the head of the CDSCO department.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three

d. None

Q14.)Consider the following statements about Drip Pricing:

1. It is a pricing technique used by firms where they initially advertise only part of a product's price.
2. It makes comparison shopping easy.
3. An airline ticket advertised without including baggage fees is a form of drip pricing

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q15.)With reference to the Chabahar port, consider the following statements:

1. It is the Iranian port that is the closest to India.
2. It is located at the mouth of Gulf of Oman.
3. Chabahar is envisaged as a crucial link in the eastern route of the International North-South Transport Corridor (INSTC), facilitating the movement of goods between India and Central Asian countries

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q16.)55 Cancri e recently heard in news is

a. An exoplanet with atmosphere
b. A malware infecting power grid
c. A new covid variant
d. A new type of cancer

Q17.) The transplantation of an organ or tissue from one individual to another of the same species with a different genotype is referred to as:

a. Allotransplantation
b. Xenotransplantation
c. Isotransplantation
d. Autotransplantation

Q18.)Consider the following Countries

1. Ethipoia
2. Slovakia
3. Uzbekistan
4. Armenia
5. Belarus

How many of the above given countries are landlocked?

a. Only two
b. Only three
c. Only Four
d. **All**

Q19.)Consider the following statements about mercury pollution :

1. Mercury is emitted into the air naturally from volcanoes, the weathering of rocks and forest fires.
2. Medical sphygmomanometers contain mercury.
3. Children, infants and fetuses are at the highest risk due to mercury pollution.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q20.) Consider the following statements about Polling stations during Lok Sabha elections.

1. Private buildings cannot be used as polling stations under any circumstance.
2. Under Section 15 of the Representation of the People Act, 1951 (RPA), the district election officer (DEO) is assigned the responsibility to set up and publish the list of polling stations in their district.
3. Candidates can set up their booths beyond 100 m of the polling station.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q21.)Consider the following statements about the Consumer Protection Act 2019:

1. The Act applies to all goods and services, covering all sectors whether they are private, public, or cooperative.

2. The act provides for an Alternate Dispute Resolution Mechanism through mediation.
3. It has no provision covering misleading advertisements.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q22.) The word FLiRT recently seen in news is :

a. A new coronavirus variant.
b. A cyber attack through Social media platforms
c. An early warning system for fire hazards
d. A type of Suicide drone

Q23.) Consider the following statements about India Volatility Index (VIX)

1. India VIX is a volatility index computed by the SEBI based on the order book of NIFTY Options.
2. It is a measure of the market's expectation of volatility over the long term.
3. India VIX typically subsides ahead of events like elections that could have a major impact on the market's trajectory.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q24.) Consider the following countries

1. Liechtenstein
2. Uzbekistan
3. Kyrgyzstan
4. Ethiopia

Which among the above countries are "doubly landlocked":

a. 1 only
b. 2 and 3
c. 1 and 2
d. 3 and 4

Q25.)Consider the following statements about Competition Commission of India:

1. It derives it powers from Monopolistic and Restrictive Trade Practices Act (MRTP).
2. It is a statutory body under the Ministry of Corporate Affairs.
3. It consists of a Chairperson and not more than 6 Members appointed by the Central Government.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q26.) Project Udbhav, recently heard in news is

a. a visionary initiative by the Indian Army that seeks to integrate age-old wisdom with contemporary military pedagogy.
b. an initiative by Competition Commission of India to regulate global digital giants.
c. an initiative by ministry of finance to control tax evasion
d. an initiative by education ministry to establish world class universities

Q27.)Consider the following statements about Malaria:

1. Malaria is a mosquito-borne infectious disease caused by parasites of the genus Plasmodium
2. It is spread exclusively through bites of infected male Anopheles mosquitoes.

Which among the statements given above is/are incorrect?

a. 1 Only
b. 2 Only
c. Both 1 and 2
d. Neither 1 nor 2

Q28.) Humboldt glacier, recently heard in news, was located in

a. Venezuela

b. Australia
c. Malaysia
d. China

Q29.)Consider the following statements about National Pharmaceutical Pricing Authority of India (NPPA);

1. The NPPA monitors the prices of scheduled as well as non-scheduled medicines under the Drugs (Prices Control) Order.
2. NPPA was constituted by the Government of India in 1997 as an attached office of the Department of Pharmaceuticals.
3. NPPA, under its Central Sector Scheme named Consumer Awareness, Publicity and Price Monitoring (CAPPM) has set up Price Monitoring and Resource Unit (PMRU) in various states.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q29.) Consider the following countries:

1. Poland
2. Lithuania
3. Norway
4. Finland
5. Azerbaijan
6. Mongolia
7. Slovakia

How many of the above mentioned countries share land border with Russia?

a. Three only
b. Four only
c. Five only
d. Six only

Q30.)Consider the following statements about Green Credits Program

1. Indian Council of Forestry Research and Education (ICFRE), is in charge of administering the programme.
2. Green credits earned through the program are tradable.

Which among the statements given above is/are correct?

a. 1 Only
b. 2 Only
c. Both 1 and 2
d. Neither 1 nor 2

Q31.)Consider the following statements about INDIAN COUNCIL OF MEDICAL RESEARCH (ICMR):

1. The Governing Body of ICMR is presided over by the Union Health Minister.
2. ICMR established the Clinical Trials Registry - India, which is India's national registry for clinical trials.

Which among the statements given above is/are correct?

a. 1 Only
b. 2 Only
c. Both 1 and 2
d. Neither 1 nor 2

Q32.) Consider the following countries

1. Iran
2. India
3. Ethiopia
4. Brazil
5. Russia

Which among the above countries are part of BRICS:

a. Only two
b. Only three
c. Only four
d. **All**

Q33.)Consider the following statements about Costal Regulation Zone (CRZ):

1. CRZ-I A constitutes ecologically sensitive areas and geomorphological features which play a role in the maintaining the integrity of the coast like coral reefs and mangroves.
2. Even eco tourism activities are prohibited in CRZ-I A.
3. Clearance for projects or activities located in CRZ-I and CRZ-IV are to be dealt with by the Ministry of Environment, Forests & Climate Change.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q34.) Consider the following statements about International Solar Alliance (ISA)

1. Its headquarter is located in India.
2. All member states of the United Nations are now eligible to join the ISA.

Which of the statements give above is/are correct?

a. 1 only
b. 2 only
c. Both 1 and 2
d. Neither 1 nor 2

Q35.)Consider the following statements about International Court of Justice

1. It is also known as the world court.
2. It is the only one of the six principal organs of the UN that is not located in New York City.
3. All members of the UN are automatically parties to the ICJ statute

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q36.)Which of the following organisation published the Climate Inequality Report 2023

a. World Inequality Lab
b. World Wide Fund for Nature
c. IUCN
d. UNEP

Q37.) Consider the following rivers

1. Gandak
2. Gomti
3. Koshi
4. Yamauna

How many of the rivers mentioned above are left bank tributaries of Ganga?

a. Only one
b. Only two
c. Only three
d. All four

Q38.)Consider the following

1. Taiwan
2. Hawaii
3. Hainan Island

How many of the above are located in pacific ocean?

a. Only one
b. Only two
c. All three
d. None

Q39.) Consider the following statements about G7

1. The G-7 does not have a formal constitution or a fixed headquarters.

2. China has never been a member of G7.
3. European Union (EU) is a "non-enumerated member".

Which of the statements give above is/are correct?

a. 1 only
b. 2 and 3
c. 1 and 3
d. All of the above

Q40.)Consider the following statements about tropical cyclones

1. They are important on Earth for transferring heat and energy between the equator and the poles.
2. Tropical cyclones are difficult to form over a region within 5 degrees of latitude from the Equator because the Coriolis force there is too small.
3. As soon as its source of warm moist air begins to ebb or is abruptly cut off, a tropical cyclone begins to weaken.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q41.) Consider the following statements regarding Earth's heat budget:

1. The heat budget of Earth is the balance between the energy Earth receives from the Sun and the energy it radiates back into space.
2. Albedo refers to the fraction of solar energy reflected back into space by Earth's surface and atmosphere.
3. Greenhouse gases such as carbon dioxide and methane play no role in Earth's heat budget.

Which of the statements given above is/are correct?

a. 1 and 2 only
b. 2 and 3 only
c. 1 and 3 only
d. 1, 2, and 3

Q42.) Bering strait separates

a. Alaska and Chukchi Peninsula of the Russian Far East
b. Alaska and North western Canada
c. Alaska and rest of United States
d. None of the above

Q43.)Consider the following statements

1. Banks can mobilise funds through issuing certificates of deposit (CDs).
2. A certificate of deposit is fully taxable under the Income Tax Act.
3. Banks are allowed to grant loans against CDs, unless specifically barred by the Reserve Bank.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q44.) Anopheles stephensi ,recently heard in news is

a. A highly invasive mosquito which caused malaria outbreaks in Africa.
b. Mosquitoes which transmits Kala Azar
c. Mosquitos Responsible for Dengue outbreaks in Northern America
d. None of the above

Q45.)Consider the following statements with respect to Trade Deficit

1. A rising trade deficitcan cause the country's currency to appreciate.
2. Reducing a trade deficit involves a combination of increasing exports and decreasing imports.

3. A trade deficit is also referred to as a negative balance of trade.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q46.)Consider the following statements about International Criminal Court

1. Russia& Israel are not members of the ICC.
2. It was established in 2002 pursuant to the multilateral Rome Statute.
3. It is also known as the world court.

How many of the statements given above are correct?

a. Only one
b. Only two
c. All three
d. None

Q47.) Consider the following statements about Sensex and NIFTY

1. The Sensex is reconstituted biannually.
2. Sensex constitutes 30 companies trading on the National Stock Exchange of India (NSE).

Which among the statements given above is/are correct?

a. 1 only
b. 2 only
c. Both 1 & 2
d. Neither 1 nor 2

Q48.)Consider the following statements:

1. Indian Renewable Energy Development Agency (IREDA) is a 'Navratna' Government of India Enterprise.
2. Navratna companies have greater financial flexibility and can undertake strategic projects without seeking explicit government approval.

Which of the statements given above are correct?

a. 1 only
b. 2 only
c. Both 1 and 2
d. Neither 1 nor 2

Q49.) Consider the following pairs:

Fabrics with GI Tag	State
1. Chanderi Fabric -	Madhya Pradesh
2. Kanjeevaram Silk-	Tamil Nadu
3. Ajrakh fabric -	Kerala

How many of the pairs given above are correctly matched:

a. Only one
b. Only two
c. All three
d. None

Q50.)Consider the following pairs:

Minerals		Ores
1. Tin	-	Cassiterite
2. Tantalum	-	Bauxite
3. Tungsten	-	wolframite

How many pairs given above are correctly matched?

a. Only one
b. Only two
c. All three
d. None

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